

Complete Notes on **ICT Project Management**

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20th July 2022

Chapter 1

Introduction (4 Marks)

1.1 Project

- Set of organized activities designed to fulfill specific objectives within a limited amount of time and limited resources
- Every project has a definite beginning and an end.
- Project are planned, executed, and controlled
- The term "**project**" refers to a particular job that is unique, non-routine, temporary, and has a clear objective.
- A project plan is a schedule of activities indicating the start and stop for each activity.

According to Cleland and King

A project is a combination of human and non human resources pulled together in a temporary organization to achieve a specific purpose.

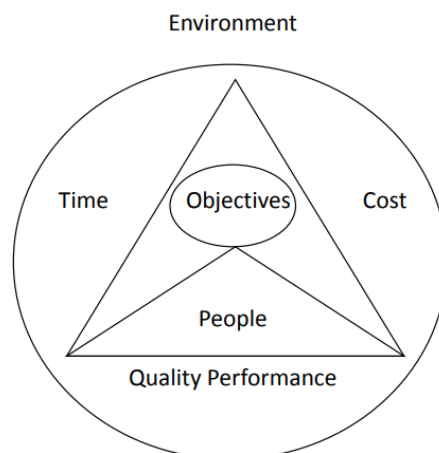
According to Management Institute of USA

A project is a temporary effort done to create a unique product or service.

1.1.1 Activity

- Must have clear start and stop i.e Known duration
- Some activities may not start until other activities are completed
- The start and stop of activity must be visible and easy to measure

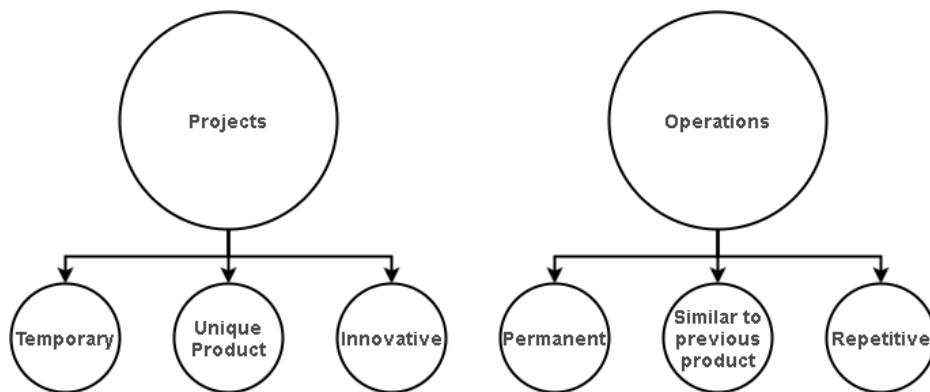
1.1.2 Model of Project



1.1.3 Engineering Project

- The engineering project is a specific kind of technological system that is included within the context of technological systems in general.
- In many Countries, Engineering projects are specifically defined by rules and regulations.
- Such projects should be carried out by registered engineers and/or registered engineering companies.

Project vs Operational Work



SN	Project	Operational Work
1	A project is an activity carried out to create a special product that is temporary in nature.	Operations are ongoing execution of activities which occur after a product is made to produce the same result or a repetitive service.
2	Temporary in Nature	Permanent in nature
3	The budget is defined for Projects.	The budget is not Defined for operations..
4	Unique product is created.	Same product is produced to keep the system running.
5	It has more risk as it is done for first time	It has less risk as such products have already been made before
6	Management of project is called Project Management	Management of operation is called Business Process Management.

1.2 Project Objectives

- Any project must have objectives. Project without objectives does not exist.
- First steps in a project's life is to define its objectives.
- End goals: A project ends when an objective has been achieved.
- How should the project objective be? => SMART objectives

1.2.1 How to define objectives of a project?

- defined uniquely and must be specific
- Focus in the other member of the project team about what the project is about
- Create team commitment and agreement about the project objectives.
- Ensure that you involve all interested parties in achieving a successful project output.

1.2.2 SMART Objectives of Project

- **S: Specific**
 - clearly defined, not ambiguous (fuzzy)
- **M: Measurable**
 - Project objectives and goals must be measurable in terms of its benefits and achievements.
 - Should answer
 - How much?
 - How many?
 - How will I know when it is accomplished?
- **A: Achievable**
 - A goal is said to be achievable when its accomplishment is within reach.
 - The goal must be something, we have the skill and abilities to achieve it.
 - Must be achievable and agree by all the team members
- **R: Realistic**
 - Should have practical range of achievement
 - The goal should meet available resources, experiences, knowledge and time.
- **T: Time bound**
 - If there is not enough time to complete the project, it won't be completed.
 - Project must be time specific.

Bad Example of SMART: I want to build a software.

Good Example: I want to build a software regarding “**Music Streaming**” within **2 years** since i have all the **needed resources** and **aggrement of our team members**.

1.3 Project Management

It is the use of methods, techniques, skills, knowledge, and experience to power activities of planned projects. Project management involves project planning and monitoring and include such items as:

1) Project Planning

- a) Definition of work requirements
- b) Definition of quantity of work
- c) Definition of resources needed

2) Project Monitoring

- a) Tracking progress
- b) Comparing actual to predicted
- c) Analyzing impact
- d) Making adjustments

1.3.1 Model of Project Management (Triple constraints)



fig: Quality Triangle



fig: Scope Triangle



fig: ProjectTriangle

Quality/Scope /Project triangle shows the trade inherent in any project.

1) Time constraint

- a) project's schedule for completion
- b) deadlines for each phase of the project

2) Scope constraint :

- a) The tasks required to fulfill the project's goals
- b) Project's specific goals, deliverables, features, and functions
- c) Other tasks required to complete the project.

3) Cost constraint

- a) refers to Project's budget
- b) financial resources needed to complete the scope of project on time
- c) Include cost for materials, labor, vendors, quality control, etc

Relations

- 1) **Time and Scope** : You can reduce your project scope to also reduce your project duration if you're running behind schedule.
- 2) **Scope and Cost**: By reducing the project scope, you'll need to execute fewer tasks, which means lower costs.
- 3) **Time and Cost** : In some projects, time and cost can be directly related. For example, the costs of renting equipment or labor are directly proportional to the time you need them for.

1.3.2 Challenges in IT Projects

- Communication and Connectivity Problems
- Staff Turnover and Visibility
- Information Security and Privacy
- Political & Cultural Risks
- Environmental & Infrastructure Risks
- Brain Drain & Loss of Institutional Knowledge
- Legal Requirements

1.4 Project Characteristics

1) Specific Objective

- A project clearly defines objectives, on achievement of which a project succeeds. Objective of the project is specific.

2) Temporary (Life Span)

- A project cannot continue endlessly. It has a beginning and end from its birth to death.

3) Non-routine and Non-repetitive:

- A project is non routine and non repetitive in nature.

4) Constraints

- A project operates within constraints of time, cost and quality.

5) Uniqueness

- No two projects are exactly similar. There are a complex set of activities involved within a project which may not be similar.

6) Flexibility

- Risks and changes are unavoidable so a project needs to address these issues for which a project needs to be flexible.

7) Resource Integration

- Every project uses resources such as man, machine, money and minutes. So, integration of these resources is necessary for their efficient use.

8) Team Work

- A manager leads the team to accomplish the goal of the project.

9) Planning and Control

- Each project has an effective planning and control system in order to efficient and effective completion of the project.

1.5 Project Classification

Project can be classified based on

1. Sponsorship of Project

- Customer
- Organization
- Contractor
- Government
- Donor

2. Nature of Projects

- Individual
- Staff
- Special
- Complex

3. Orientation of Project

- Product Oriented
- Process Oriented

4. Speed of Project

- Normal
- Crash
- Disaster

5. Funding Source of Project

- Private sector project
- Government sector
- Grant Project
- Loan Project

6. Size of Project

- Mega : 5 - 10 years
- Major : 3 - 5 years
- Medium : 1 - 3 years
- Small : small duration

7. Technique of Project

- Labor Intensive
- Capital Intensive

8 Number of Key Purposes

- Single Purpose
- Multi Purpose

9. Type of Relationship

- Independent
- Dependent
- Mutually Exclusive

1.6 Some Examples of Projects

1) Construction Projects

- a) Warship
- b) Customer Call center
- c) IT system

2) Research Projects

- a) Business modeling
- b) Developing model of nepali economy
- c) Developing new species of crops
- d) Code breaking

3) Reengineering Projects

- a) Taking sterling into euro
- b) Renumbering Nepal telephone system
- c) Designing and installation of intranet

4) Procurement Projects

- a) Outsourcing a specific construction or research projects
- b) Outsourcing complete business function

5) Business Implementation Projects

- a) Developing a new business process to repackage and exploit existing assets
- b) Installing ecommerce

1.7 Types of IT Projects

1) System Integration

- It brings together a range of diverse ICT products and services.
- Eg: building a nationwide WAN

2) Facilities management

- These are very close to system integration.
- But, aim is to take a full responsibility and accountability but not take the ownership for establishing and operating customer IT facility
- Eg: Establishing automated security and disaster control center

3) Software Application Development

- Eg: ERP Application Development

4) Services Consulting

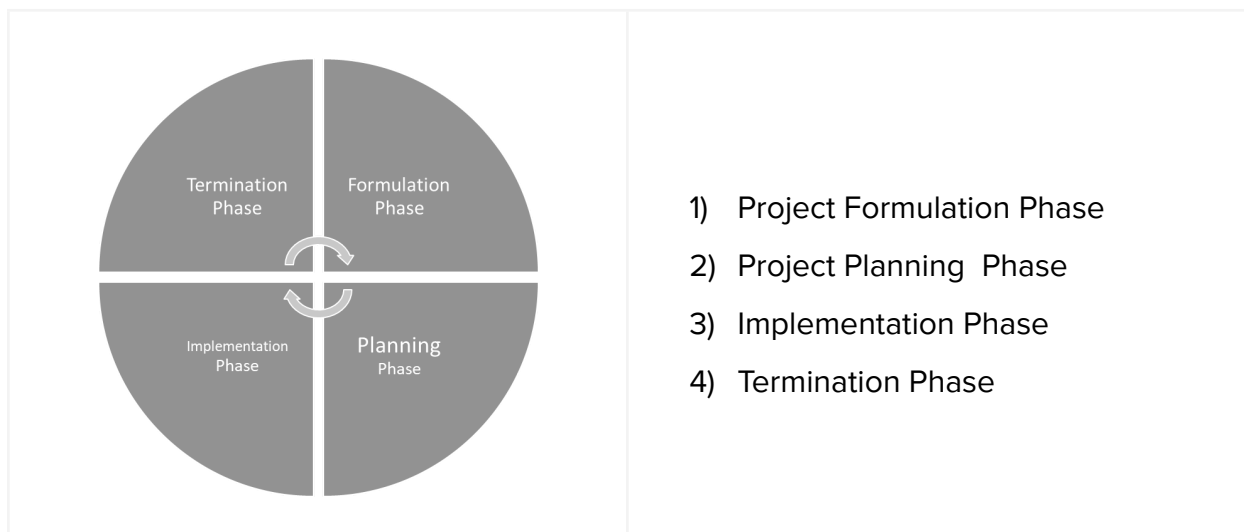
- Delivers specific consulting service to customer
- This may or may not result in delivery of It product or services to that customers
- Eg: Counseling and training on different ICT issues

1.8 Project Life Cycle

The project life cycle refers to a logical sequence of activities to accomplish the project's goal or objectives. Each project has its own life i.e. beginning and end.

1.8.1 Phases of life cycle

Generally the life cycle of a project has four phases. Project development process is always a cycle because each stage of analysis is built on the one before it and triggers a new stage of study.



Project Formulation Phase

- Identify and set evaluation criteria
- Includes
 - 1) **Project Identification :**
 - identify creative ideas based on the environment, internal and external resources.
 - 2) **Project Evaluation :**
 - formulates the project to identify the constraints of the project
 - objective is determined and project proposal is developed after pre feasibility study and prototype design of the project

Project Planning Phase

- preparation of a detailed plan with time, cost and quality.
- Includes

1) Feasibility Study :

- It is a measure of the software product in terms of how beneficial product development will be for the organization from a practical point of view.
- perform technical, economic, market, management and environmental analysis

2) Project Appraisal/ Estimation:

- Based on the results of the feasibility study, the project's success is evaluated.
- Proposed project is compared with similar product
- Project requirements and implementation plans are developed.

Implementation Phase

- During this stage, project activities are implemented and controlled.
- A proper team is formed.
- Project managers monitor and supervise the project activities.
- Inspection and Testing is done from time to time.
 - Time to time evaluation and the review of the project's progress is done.

Termination Phase

- Once the team has completed project work, they enter the termination phase.
- Project is evaluated and later handed over to customers.
- They provide project results, release project resources, and evaluate the project's success during this phase.
- Includes

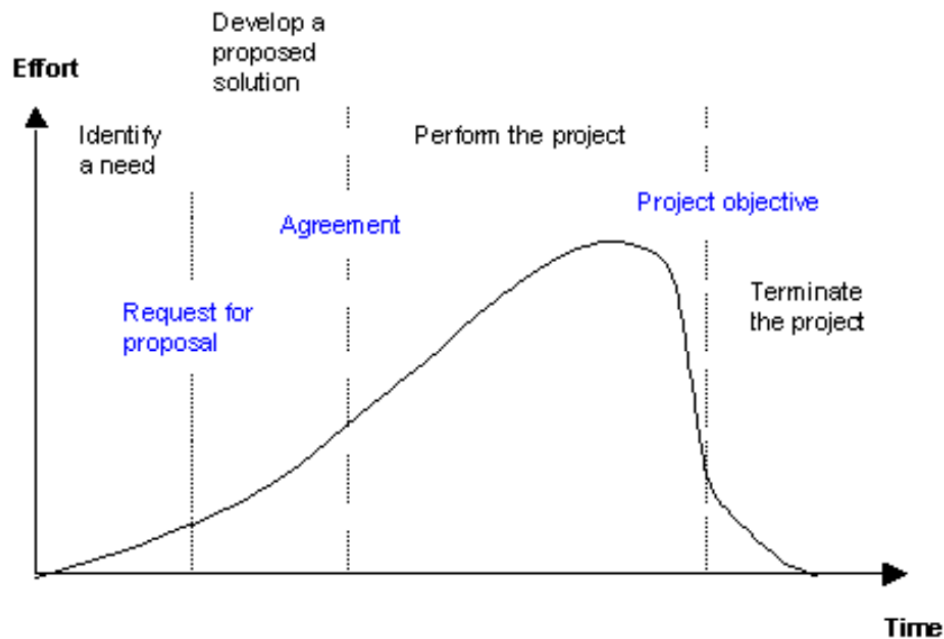
1) Project Operation and Evaluation

- Review the project outcome
- Evaluation of projects results
- Go to major preparation phase if required

2) Project Handover

- Handed over to clients

1.8.2 Efforts and Tasks during Project Life Cycle Phases



1.8.3 Differences between project and general management

SN	General Management	Project Management
1	It mainly focuses on management of general activities.	It mainly focuses on management of projects and related tasks.
2	It is a continuous process.	It is a temporary (one time) process.
3	Nature is Repetitive and regular	Nature is Non-repetitive and unique
4	Single State	Move from one state to another i.e. results in change
5	No real start and end points	Definite Start and End points
6	General managers only work within organizational premises.	Project managers not only work within organizational premises, but they also can work outside organizational premises.
7	General managers have unlimited authority over their staff.	Project manager does not have any authority over their staff.
8	Role of manager is permanent or long term.	Role of manager and team membership is temporary.

1.9 Feasibility Study

- It is a measure of the software product in terms of how beneficial product development will be for the organization from a practical point of view.
- It is a study to evaluate the feasibility of a proposed project or system.

1.9.1 Types of feasibility Study

1) Technical feasibility

- In Technical Feasibility, current resources of both hardware software along with required technology are analyzed to develop the project.
- also analyzes technical skills and capabilities of technical team,

2) Economic feasibility

- In the Economic Feasibility study, the cost and benefit of the project is analyzed.
- It is analyzed whether the project will be beneficial in terms of finance for the organization or not.

3) Operational feasibility

- degree of providing service to requirements is analyzed.
- how easy the product will be to operate and maintenance after deployment is also analyzed.

4) Legal feasibility

- In the legal Feasibility study, the project is analyzed from a legality point of view.
- Analyzing data protection acts such as copyright, license, legal implementation of project

1.9.2 Schematic Diagram of Feasibility study

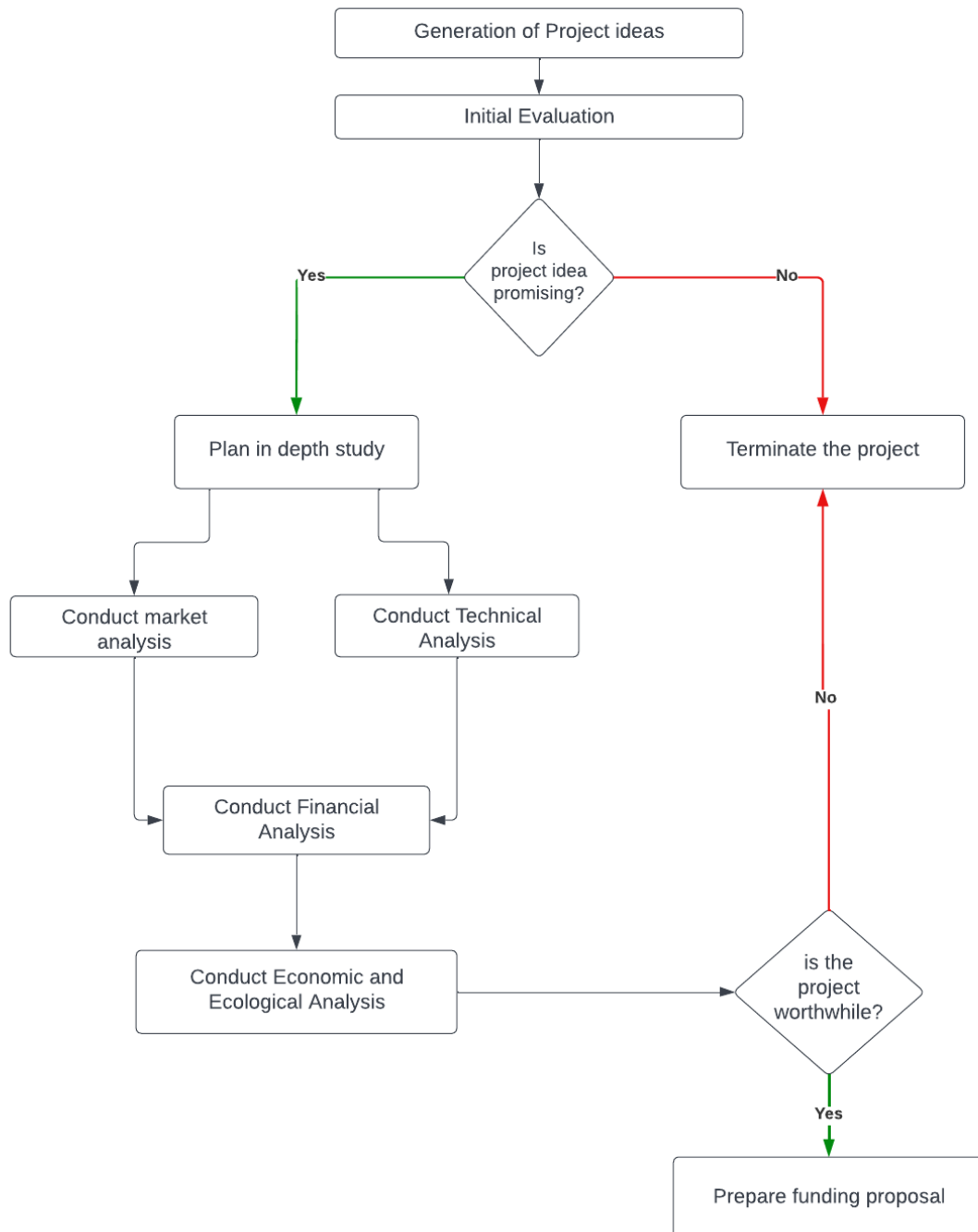


fig: systematic digram of feasibiity study

1.10 IT System Development

1) Communication

- Requirement Analysis
- Feasibility Analysis
- Project Proposal Development

2) Planning

- a) Scheduling
- b) Cost Estimation
- c) Quality Management
- d) Change Management
- e) Risk Management

3) Modeling

- a) Analysis
- b) Design

4) Construction

- a) Coding
- b) Testing

5) Deployment

- a) Maintenance
- b) Feedback

1.10.1 Elements of project proposal

The elements of a project proposal are a document which explains

- the overall planning of the project,
- why the project exists in a particular situation,
- what kind of problems the project attempts to address,
- what makes the project successful, and
- What are the required resources?

Some of the key elements of project proposal are:-

- 1) Reasons for the Proposal
- 2) Project Planning Matrix : Presentation of goal, purpose, inputs and outputs of the project
- 3) Solution of the Problem
- 4) Project Implementation Plan: Estimates, Cost Benefit Analysis, Resources, Schedule, Risk & Contingency Plan, Milestones and Deliverables.
- 5) Plan for technical and administrative support
- 6) Budget or Financial Plan
- 7) Relevant Past Experiences

Chapter 2: PMBOK

Project Management Body of Knowledge (7 Marks)

2.1 PMBOK

- PMBOK is a collection of Processes and Knowledge Areas generally accepted as best practice within Project Management.
- PMBOK is sum of knowledge within the profession of project management
 - widely used proven traditional practices
 - limitedly used innovative and advanced practices
- Guidelines for managing individual projects
- Following PMBOK is a good practices which are applicable to most project most of the time
- PMBOK recognizes FIVE Process Groups and NINE Knowledge areas for every type of project.

2.1.1 Process Groups

Processes are described in terms of inputs, tools & techniques and outputs.

- **Inputs:** Documents, Plans, Designs, Artifacts, etc.
- **Tools and Techniques:** Mechanisms Applied to Inputs
- **Output:** Documents, Products, Service, Knowledge.

Five Process Groups

1) initiating processes

- The PMBOK defines the project initiation phase as “the process of formally recognizing that a new project exists or that an existing project should continue into its next phase.”
- All projects start with an idea for a product, service or another desirable outcome.
- The initiating process group then determines the nature and scope of the project.
- The first project document is the project charter.
- The charter answers the question, "What are we trying to do?"

2) planning processes

- After initiating, the project is planned to an appropriate level of detail.
- The major goal is to successfully manage risk throughout project execution by planning time, cost, and resources in an efficient manner.

3) executing processes

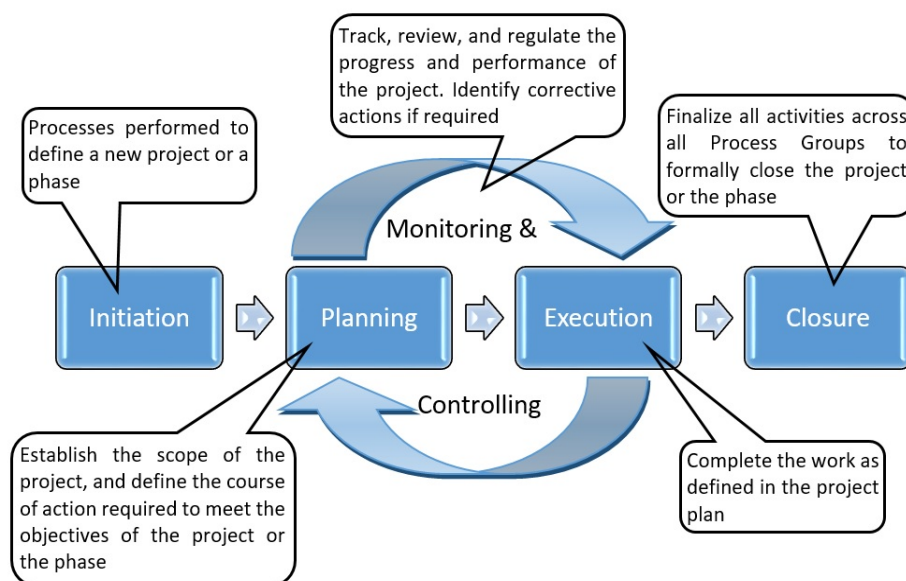
- The procedures required to finish the tasks defined in the project management plan are referred to as "executing."
- It involves achieving the project's goals.
- Coordinating people and resources, completing the project operations, and integrating them all are all part of the execution process.

4) monitoring and controlling processes

- Track, review and monitor the progress and the performance of the project
- Potential problems can be identified quickly for the team to take corrective action.
- The monitoring process group ends once the project has achieved its goals and objectives, as detailed in the project contract.

5) closing processes

- Project closing is an important part of project management.
- In order to close a project, all tasks associated with each process group must be completed, the project team must be disbanded, and the project must be approved by the client.



2.1.2 Knowledge Areas

Each Knowledge Area contains some of or all of the Project Management Processes.

- 1) Project Integration Management
- 2) Project Scope Management
- 3) Project Time Management
- 4) Project Cost Management
- 5) Project Quality Management
- 6) Project Human Resource Management
- 7) Project Communications Management
- 8) Project Risk Management
- 9) Project Procurement Management

2.1.3 Basics of PMBOK

- Much of PMBOK is unique to Project Management, like CPM, PERT, WBS, etc.
- Some areas overlap with other management disciplines.
- General Management includes Planning, Organizing, Staffing, Coordinating, Leading and Controlling operations of an organization.
- Financial Forecasting, Organization Behavior and Planning techniques are also similar

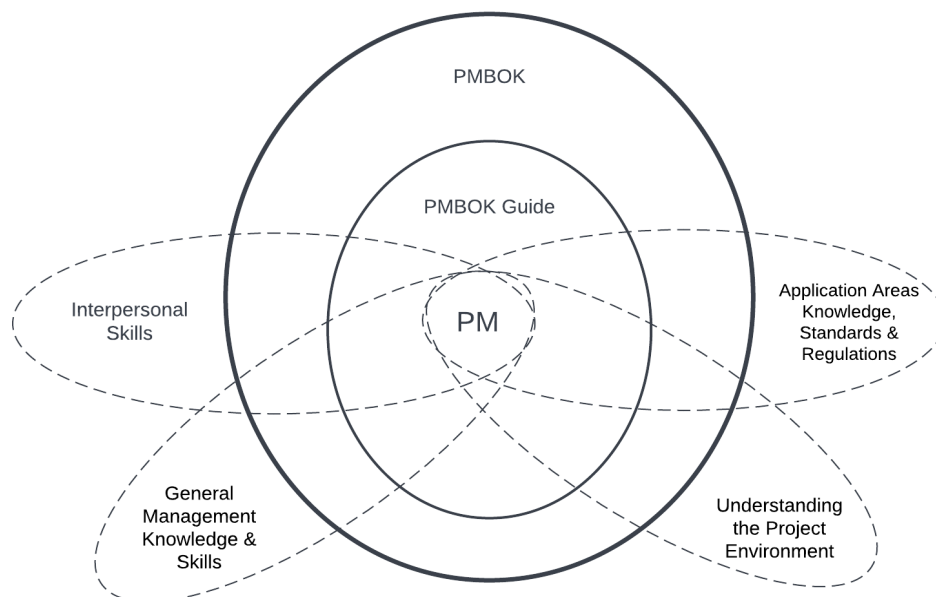


fig: Areas of Expertise needed by Project Team/ Manager
According to PMI

2.2 Understanding of Project Environment

- Projects are Environment Specific so, Projects must continually adapt to environmental changes.
- Environment consists of forces that influence the project's ability to achieve its objectives.
- Environmental influences on projects occur through:
 - Complexity
 - Uncertainty
 - Competition for Resources
 - Flexibility
 - Rapid Technological Changes

2.2.1 Project Environment - Classification

- 1) Internal Environment**
- 2) Task Environment**
- 3) External Environment**

2.2.2 Internal Environments affecting ICT Project

The internal environment of an organization refers to events, factors, people, systems, structures and conditions inside an organization. They are generally under the direct control of the company. They provide strengths and weaknesses to the project.

The forces in the internal environment consists of:

1) Project Objectives

Objectives are the desired outcomes or end results of the project. Project activities must be conducted within project objectives. A project must focus on the objective of projects. Once the goal has been reached, it vanishes.

2) Constraints

A project must operate within the constraints of time, cost and quality. They describe the scope and boundaries of the project.

3) Structure

A project is temporary. Project must function within the boundary of its structure.

4) Resources

A project consists of human and non-human resources. It usually has its own budget.

How does the internal environment affect?

1) Staff

Employees are an important part of the internal environment of an organization.

Managers must be able to manage lower-level employees and, at the same time, supervise the other factors of the internal environment.

Even when everyone is capable and talented, politics and internal conflicts can destroy a good organization from within.

2) Budget

Even a company's financial situation in business can determine whether it will survive or not. When cash resources are too limited, it affects the number of people we are hiring, the quality of equipment and the type and amount of advertising we are buying.

If you have enough money instead, you have much more flexibility to grow and expand the business, or to prepare for an economic downturn.

3) Corporate Culture

The values, attitudes, and priorities that staff members consistently follow make up the internal business culture.

A corporation where competition between employees is encouraged would undoubtedly result in a different and more toxic environment than the corporation where cooperation and teamwork are valued.

2.2.3 Task Environments affecting ICT Project

- This project environment affects the organization 's ability to reach the goals.
- Includes no. of stakeholders such as client, contractor, consultants, suppliers, government, competitors, and financiers.
- Here, Interest and Impacts are Interrelated

2.2.4 External Environments affecting ICT Project

The external environment is composed of factors that occur outside the organization but which can cause internal changes. They are beyond the company's control.

Customers, competition, economy, technology, political and social conditions, and resources are common external factors that influence the organization.

Even if the external environment occurs outside an organization, it can have a significant influence on its current operations, growth and long-term sustainability.

Ignoring external forces can be a damaging mistake for managers. As such, it is necessary that managers continue to monitor and adapt to the external environment.

1) Economy

In a bad economy, even a well managed organization may not be able to survive.

It is not possible to control the economy, but understanding it can help identify threats and opportunities. Some of the economic factors are:

- Inflation rate
- Interest rate
- Unemployment rate, foreign exchange rate

2) Competition

Unless the organization is a monopoly, you will always have to deal with the competition.

When you launch a business, you typically have to compete with more established and experienced companies in the same industry. On the other hand, when a company has established itself, it will have to compete with competing companies that are fighting for the same market share.

3) Politics

Changes in government policy can have a huge impact on an activity. Some of the political factors are:

- Government policies
- Taxes, laws
- Government Stability

A classic example is the tobacco industry.

Since the 1950s, cigarette manufacturers have been asked to put warning labels on their products and have lost the right to advertise on television. Smokers have less and less places where they are allowed to smoke. Therefore, the percentage of people who smoke is vanishing, with a corresponding effect on the sector's revenues.

4) Customers and Suppliers

Customers and suppliers are typically the most important people a corporation must interact with after its staff.

Suppliers have a huge impact on costs. The value of a specific provider is determined by the uniqueness of his goods or services and, hence, by the potential of a negotiation.

The power of customers depends on the fact that they are free to choose between a specific organization and its competition.

2.2.5 Success Criteria for Projects

1) Technical Tangible Quantitative

- a) Deadlines
- b) Performance specifications
- c) Quality Standards
- d) Cost requirements
- e) Resource Constraints

2) People Intangible Quantitative

- a) Commitment
- b) Cooperative attitude
- c) Positive image and total project focus
- d) Risk evaluation

2.2.6 Obstacles to Project Success

- 1) Poor project planning and direction
- 2) Insufficient communication
- 3) Lack of change, risk, financial, and performance management
- 4) Failure to coordinate with stakeholders
- 5) Ineffective involvement in management
- 6) Lack of skilled team members
- 7) Poor or missing technology and tools

2.3 General Management Skill

❖ Planning

- Determination of objectives.
- Establish planning premises and constraints.
- Decide the planning period.
- Collection, classification and processing of information.
- Deciding alternative courses of action.
- Evaluation of alternatives.
- Selection of best plan.
- Alternative plans to help master plan and Controlling plans.

❖ Organizing

❖ Staffing

❖ Directing

❖ Motivating

❖ Controlling

❖ Coordinating

❖ Communicating

2.3.1 Effective and Ineffective Project Managers

SN	Effective Project Managers	Ineffective Project Managers
1	Lead By Example	Set Bad Examples
2	Visionaries	Confused
3	Have Technical Skills	Lack Technical Expertise
4	Ability to quickly make decisions	Indecisive
5	Good Communicators	Poor Communicators
6	Good Motivators	Poor Motivators/ Demotivates
7	Stand up to Top Level Management	Complaining about Top Level Management
8	Support Team Members	Lacks Team Spirit
9	Encourages New Ideas	Conservative

2.3.2 Essential Interpersonal and Managerial Skills

The Project Manager must be a people manager. They must have soft skills such as negotiations, communication and interpersonal skills.

Essential Interpersonal and Managerial Skills are:

1) Energized and Initiators

- Project manager should not lack energy and fitness.
- S/he must be able to work continuously under considerable pressure and odd conditions.
- S/he must have the ability to take necessary initiative to see a positive result as and when required.

2) Communication

- must be able to express ideas in written and oral forms
- Ensure simplicity, no complexity and great feedback if necessary

3) Influencing

- Ability to get people to do what they want otherwise.

4) Leadership

- Impart Vision, Gain Consensus for Goals, Establish Direction, Inspire, Motivate, Self Assured

5) Motivator

- Energize people to achieve high level of Performance to Overcome Barriers to change

6) Negotiation

- Ability to resolve conflicts.
- Make possible agreement
- Understand the best solution to the Problem

7) Problem Solver

- Able to deal with Problems.
- Have Problem Solving Attitude.
- Have Problem Analysis

8) Perspective Nature

- Ability to look beyond the team.
- See how Project & Team fit into Organization.

9) Result Oriented

- Not just complete work for work's sake but to achieve the Project Objectives

10) Global Illiteracies

- Ability to work in a Cross Cultural Environment.
- Understand Cross Cultural Issues.

11) Problem Solving using Problem trees

- Examine the Problem
- Analyze the Problem
- Develop Problem Tree

2.4 Problem Solving using Problem Trees

This helps in analyzing an existing situation by identifying the major problems and their main causal relationships.

Steps for Developing a Problem Tree

- 1) Identify major problems existing within the stated problem areas.
- 2) Analyze their interrelationship and common issues. Determine the core problem among the major problems.
- 3) Write the causes of the major problems.
- 4) Write effects caused by the core problem.
- 5) Form a diagram of a problem tree showing causes, effects and problems.
- 6) Review the diagram as a whole. Verify its validity and completeness.
 - a) Repeat from step 2 if required.
 - b) Iteration helps!

Brainstorming

Brainstorming is a method of generating ideas and sharing knowledge to solve a particular commercial or technical problem, in which participants are encouraged to think without interruption.

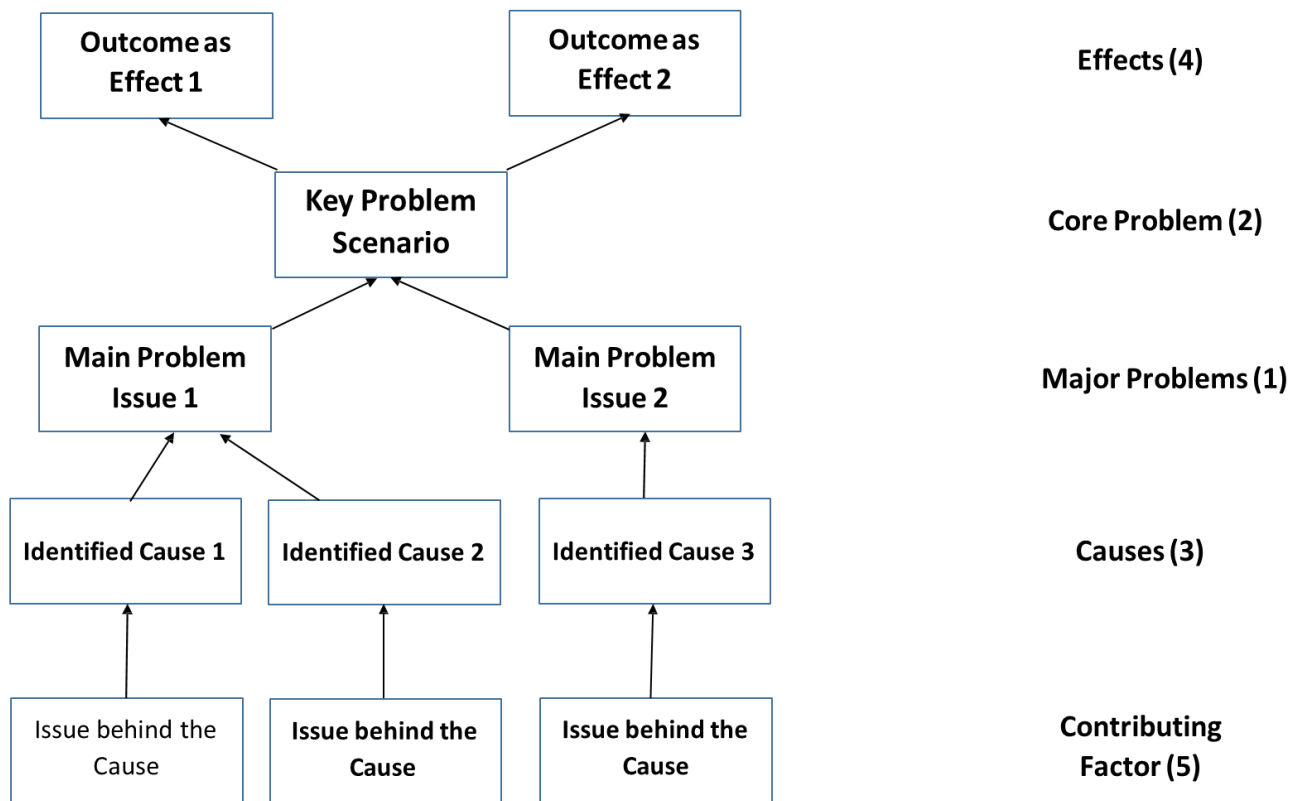
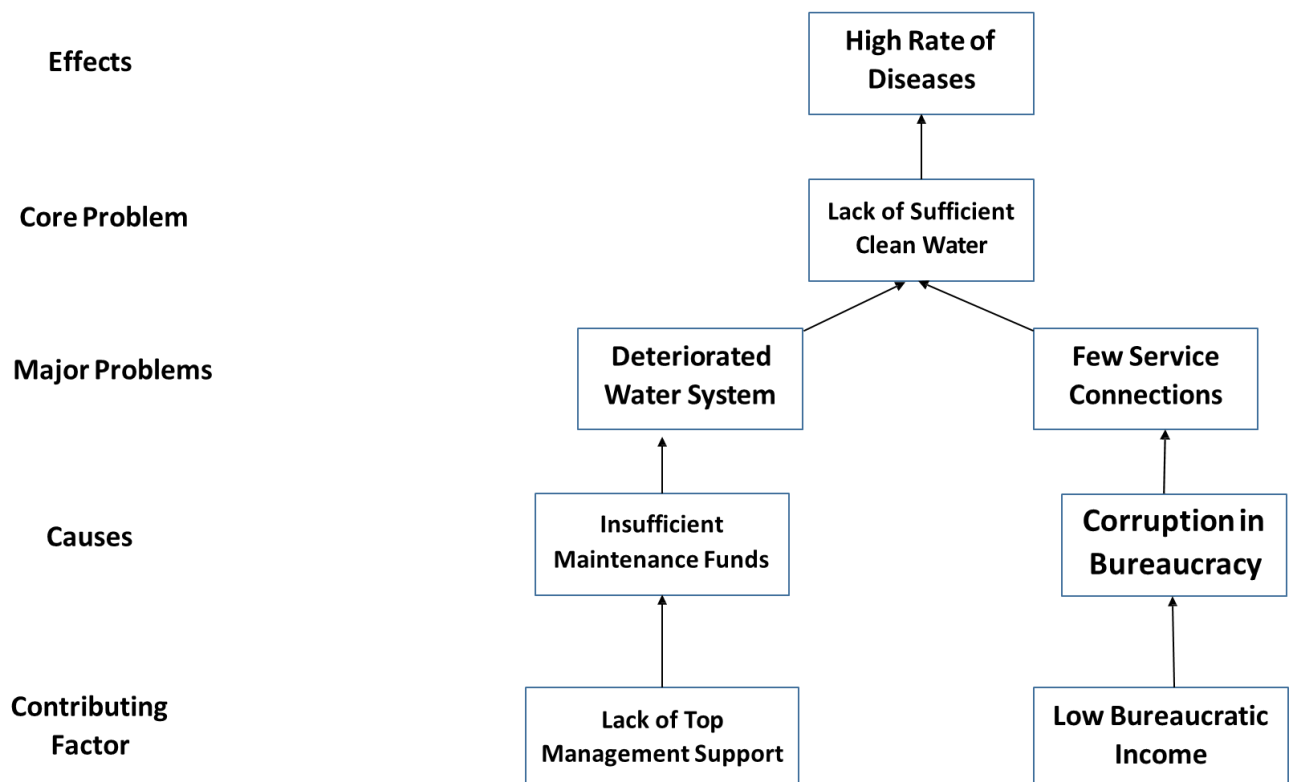


fig: Problem Solving problem tree schematic diagram

Example:



Chapter 3: Portfolio and Project Management Institutes

(PMI) Framework (4 Marks)

3.1 Portfolio

- Collection of projects, programs, and processes that are managed together to meet objective of an organization
- Project defines end goals/objectives but portfolio represents strategic planning commitment 

3.2 Project Portfolio Management (PPM)

- Project portfolio management (PPM) is the analysis and optimization of the costs, resources, technologies and processes for all the projects and programs within a portfolio.
- carried out by portfolio managers or project management office (PMO)
- goal: balance risk and reward
- The key focus of PPM is to make sure that all the outcomes in the portfolio support the strategic goals and business objectives of the organization.

3.2.1 Steps of the Project Portfolio Management Process

1) Define Business Objectives

2) Collect Project Ideas for Your Portfolio

3) Select the Best Projects for Your Portfolio

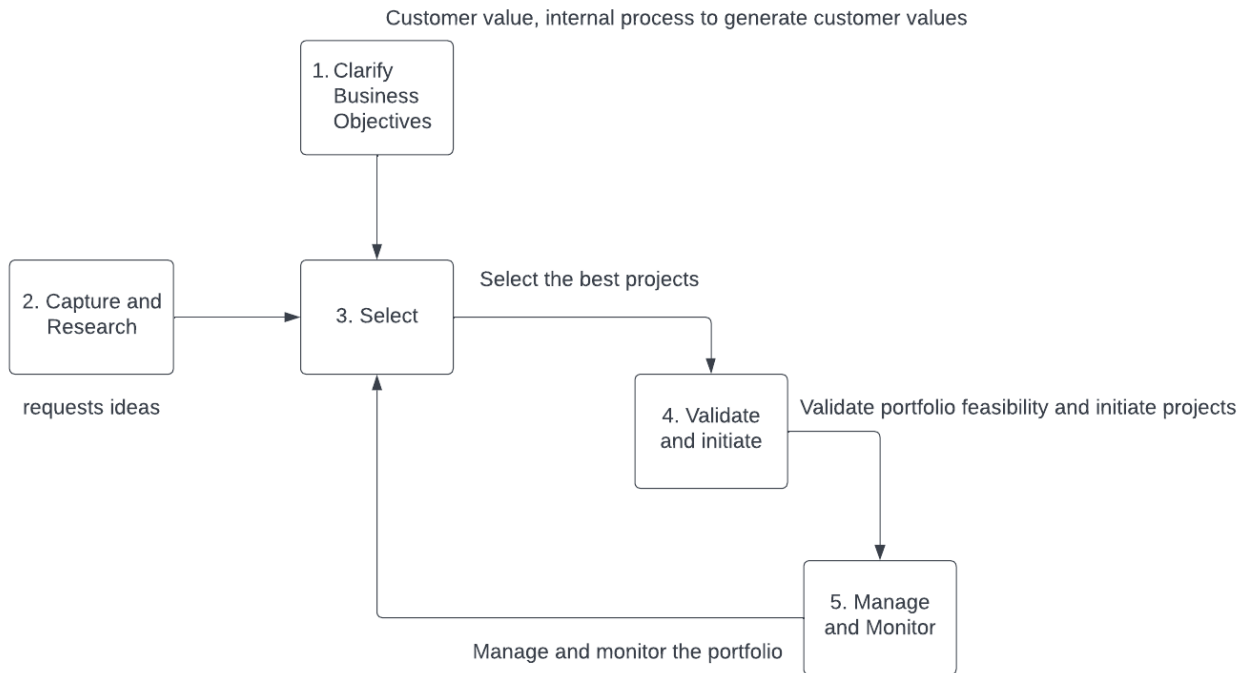
- To determine which are your best projects for your portfolio, you'll need to do a cost benefit analysis.
- There are a variety of aspects for project selection such as the payback period, net present value, or risk level.

4) Validate Project Portfolio Feasibility

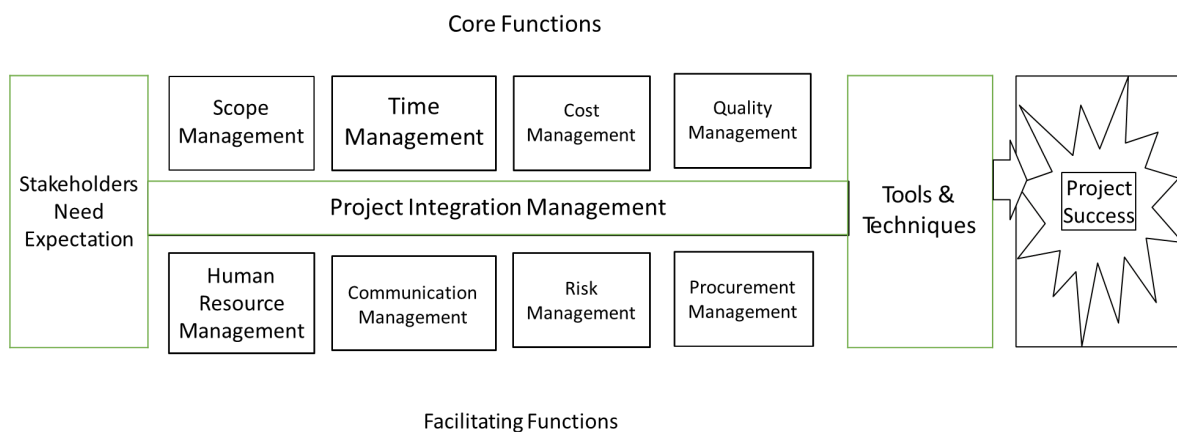
- Now that you've chosen the projects that are the best fit for your portfolio, it's time to do a feasibility study.

5) Execute and Manage your Project Portfolio

- Now coordinate the execution of the projects and programs in the portfolio simultaneously by working with project and program managers.



3.2.2 PMI framework



Knowledge Areas of PMI Framework

- PPM essentially gives you the ability to manage a portfolio of projects similar to how you may manage a portfolio of various investments, such as stocks.
- By maintaining a balanced portfolio, we can reduce the risks of individual projects and produce an overall higher rate of return.
- PPM allows managers to identify project risks and quickly address them,

The knowledge areas are classified into following categories.

1) Stakeholders need and Expectation

- A stakeholder is any individual, group, or party that has an interest in an organization and the outcomes of its actions.
- Common examples of stakeholders include employees, customers, shareholders, suppliers, communities, and governments.

2) Core Knowledge Areas : Includes scope, time, cost and quality management

3) Facilitating Knowledge areas: Includes Human resources, communication, risk, procurement management

4) Integrative Knowledge Areas: Include project integration management

5) Tools and Techniques

- Tools and techniques can help to increase understanding in any of above knowledge areas
- They are most vital in core knowledge areas
- Although project management software facilitates the use of these tools and techniques, the same techniques have long been used with paper and pencil.

3.2.3 Differences between project management and portfolio management

SN	Project Management	Portfolio Management
1	It refers to “Do projects right”.	Do the right projects
2	Application of knowledge, skills, tools and techniques to project activities to meet project goals	It is the collection of projects to meet organization strategy
3	Project management is about directing a single project successfully	Project portfolio management is about selecting and successfully executing the right projects for the organization.
4	It is a temporary process but unique.	It is an ongoing process that has to be performed on a daily basis.
5	It does not make strategic plans and prioritize projects.	It makes strategic plans and prioritizes projects.

6	Its main goal is to complete a single project and provide service.	Its main goal is to look at all projects and in turn improves return of investment as well as reduction of costs.
7	This management is basically used for small scale projects and more detailed one.	This management is basically used for not only large scale projects but also for more diffused one.

3.2.4 Why Project Portfolio Management?

- 1) It helps to predict the outcome and plan for projects that will offer the best results.
- 2) It helps to understand which risk of projects and can offer the most rewards.
- 3) The project manager gets the global view of each project which makes them able to predict the future problems.
- 4) It provides a way to improve accuracy for estimating and managing financial resources of a group of projects.
- 5) Project Managers can make business based decisions.
- 6) Reduces cost and increase productivity
- 7) PPM helps managers monitor project progress in real time and provides detailed data
- 8) Enable people to succeed

3.2.5 Types of PPM User

1) Execution-focused PPM users

- Manage the technical factors of project execution
- use reporting tools to communicate progress and expenses among business sponsors and executives

2) Project portfolio-level PPM users

- Create project-related decision frameworks
 - Select specific projects based on those frameworks
 - Plan the delivery of those projects or investments
 - Track those investments at a high level
 - Report those above activities

3.4 Project Management Office

- It is a group or department within an organization that defines and maintains standards of project management
- Provides guidance and standards in the execution of projects
- PMO is established to create and maintain procedures and standards for project management.
- A PMO can either be internal or external.



Successful PMO

- combination of good people, process and tasks

3.4.1 Benefits of PMO

- Strategic Planning and Governance
- Best practices and processes
- Common languages, culture
- Resources management
- Creating and maintaining project artifacts and tool set



3.4.2 Types of PMO

A PMO can be one of three types from an organizational exposure perspective:

- 1) Enterprise PMO
- 2) Organizational (departmental) PMO
- 3) Special-purpose PMO.

Some other types

1) Functional PMO

- This type of PMO is used in one functional area or division of an organization, such as information systems.
- Main goal: resource management
- The PMO may or may not actually manage project

2) Corporate (or Strategic) PMO

- This type of PMO services the entire company and focuses on corporate and strategic issues rather than functional issues.
- If this type of PMO does management projects, it is for cost reduction efforts.

3) Customer Group PMO

- This type of PMO is for better customer management and customer communications.
- Common customers or projects are clustered together for better management and customer relations.
- This acts like a company within a company.
- This type of PMO will have a permanent project manager assigned and managing projects.

3.4.3 Functions of PMO

- 1) Standardization in estimating, planning, scheduling, controlling and reporting
- 2) Simplification of project management roles and responsibilities
- 3) Developing project management templates and methodology
- 4) Recommending and implementing changes and improvements to the existing project management methodology
- 5) Identifying project management standards
- 6) Identifying best practices in project management
- 7) Performing strategic planning for project management
- 8) Coordinating and/or conducting project management training programs
- 9) Transferring knowledge through coaching and mentorship
- 10) Assessing risks in projects
- 11) Planning for disaster recovery in projects
- 12) Performing or participating in the portfolio management of projects
- 13) Acting as the guardian for project management intellectual property

3.5 Project Management Institute (PMI)

- Project Management Institute (PMI) is a U.S.-based not-for-profit professional organization for project management.
- PMI considers the PMO as a major catalyst for organizational excellence.

3.6 Drivers of Project Success

- 1) Top Management Support
- 2) Clear Goals and Objectives
- 3) Client Support
- 4) Realistic Plan
- 5) Appropriate Resources (4M; Man, Machine, Material, Money)
- 6) Ownership
- 7) Formal methodology
- 8) Hard working, focused staff
- 9) Standard and structured ICT support Infrastructure
- 10) Effective Communication
- 11) Experienced PM

3.7 Inhibitors of Project Success

- 1) Poor Communication
- 2) Lack of Leadership
- 3) Unclear Expectations
- 4) Poor Upfront Planning
- 5) Inbuilt Negative Attitude Towards IT
- 6) Changing Business Strategies
- 7) Poor Top Management Support
- 8) Conflicts of Objectives
- 9) Inadequate Resource
- 10) Financial Limitations
- 11) Lack of Historical Data

Chapter 4:

Project Management(7 Marks)

4.1 Project Management

- It is the use of methods, techniques, skills, knowledge, and experience to power activities of planned projects.
- Project management involves project planning and monitoring

4.1.1 Advantages of Project Management

- 1) Increased control of financial, physical and human resources
- 2) Improved customer relations
- 3) Higher quality outcome
- 4) Enhanced Reliability on solutions
- 5) Increased profit margins
- 6) Improved productivity at work
- 7) Better internal coordination
- 8) Higher Work Morale
- 9) Shorter Development Time
- 10) Lower Costs

Q. Projects cannot run in Isolation Why?

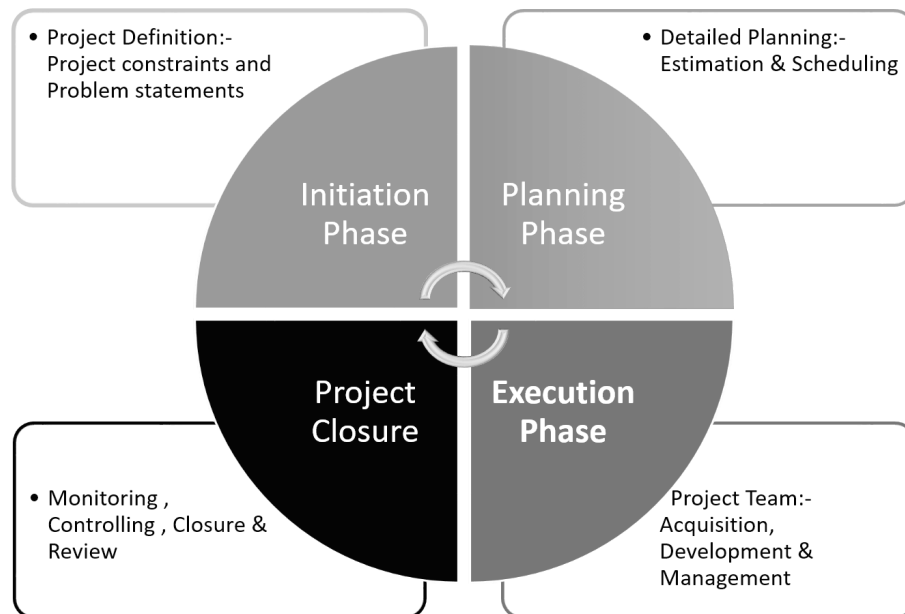
- Projects must operate in a broad organizational environment
- Project managers need to use systems thinking
- Senior managers must make sure projects continue to support current business needs

4.2 Project Management in Context of PMI

- As per PMI, project management is subdivided into different phases such as initiating, planning, monitoring and closing.
- It helps to decrease uncertainty.
- Each phase is marked by completion of one or more deliverables.

4.3 Project Life Cycle

- A **project life cycle** is a collection of project phases that defines
 - what work will be performed in each phase
 - what deliverables will be produced and when
 - who is involved in each phase, and
 - how management will control and approve work produced in each phase
- A **deliverable** is a product or service produced or provided as part of a project



1) Initiation Phase

- The initiation phase defines those processes that are required to start a new project.
- The purpose is to determine what the project should accomplish.
- Mainly include two activities:
 - Develop a Project Charter and
 - Identify Stakeholders
- All the information related to the project is entered in the Project Charter and Stakeholder Register. When the project charter is approved, the project becomes officially authorized.

2) Planning Phase

- The Project Planning phase covers about 50% of the whole process.
- Planning phase determines the scope of the project as well as the objective of the project.
- It begins with the outputs of the initiation phase and the output of the planning phase serves as the input for the execution phase.
- Important aspects:
 - The planning phase should not be executed before your initial planning is finished
 - Until the execution process does not start, you should not stop revising plans

3) Execution Phase

The executing phase consists of those activities that are defined in the project management plan. This process involves managing stakeholder expectations, coordinating with people and resources, as well as performing other activities related to project deliverables.

Action taken in the execution phase may affect the project management plan or documents.

4) Monitoring, Controlling and Closing Phase

- This stage involves tracking, reviewing and regulating the progress in order to meet the objective of the project.
- It also ensures that the built products are according to the project management plan.
- The main focus of this step is to identify any changes made from the point of project management plan to determine appropriate preventive action.



4.3.1 Project Life Cycle in different fields

Construction

- Feasibility Study
- Planning and Design
- Construction
- Turnover and Startup

Pharmaceuticals

- Discovery and Screening
- Preclinical Development
- Registration workup
- Post-submission Activity

Government Office

- Feasibility Study
- Concept and Technology Development
- System Development and Demonstration
- Production and Deployment
- Support

Defense

- Strategic Planning
- Concept and Technology Development
- System Development and Demonstration
- Production and Deployment
- Support

4.3.2 Characteristics of Project Life Cycle

- 1) Project should pass through each of the project phases
- 2) Conclusion of a project phase is done by a review on primary outputs and project performance
- 3) Cost and staffing levels are
 - a) low at start,
 - b) higher toward the end and
 - c) drop rapidly as the project close
- 4) Risk and Uncertainty are
 - a) highest at the start and
 - b) less at the end
- 5) Ability of the stakeholders to influence the “final characteristics of the projects product” and “final cost” is
 - a) highest at the start and
 - b) gets progressively lower as the project continues

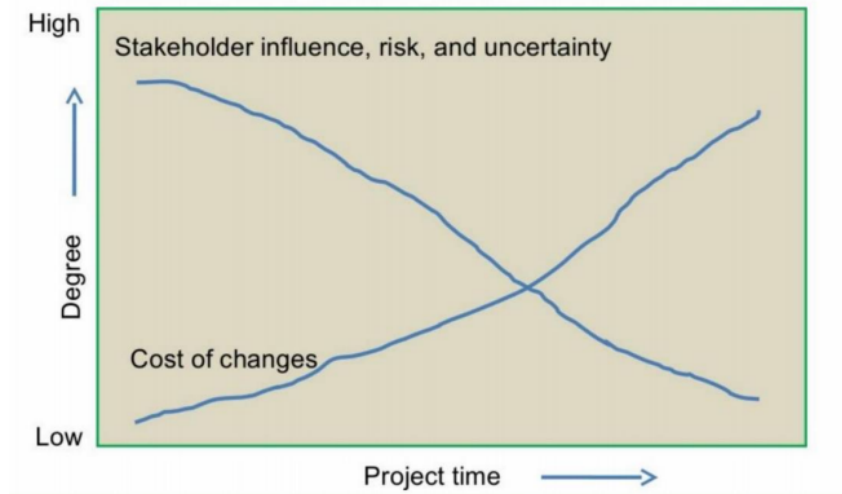


fig: Characteristics of Project Life Cycle

4.3.3 IT Product Development Life Cycle

- Basically called software development Life Cycle
- Software Development Life Cycle (SDLC) is a process used by the software industry to design, develop and test high quality softwares.
- The SDLC aims to produce high-quality software that meets or exceeds customer expectations, reaches completion within time and cost estimates.

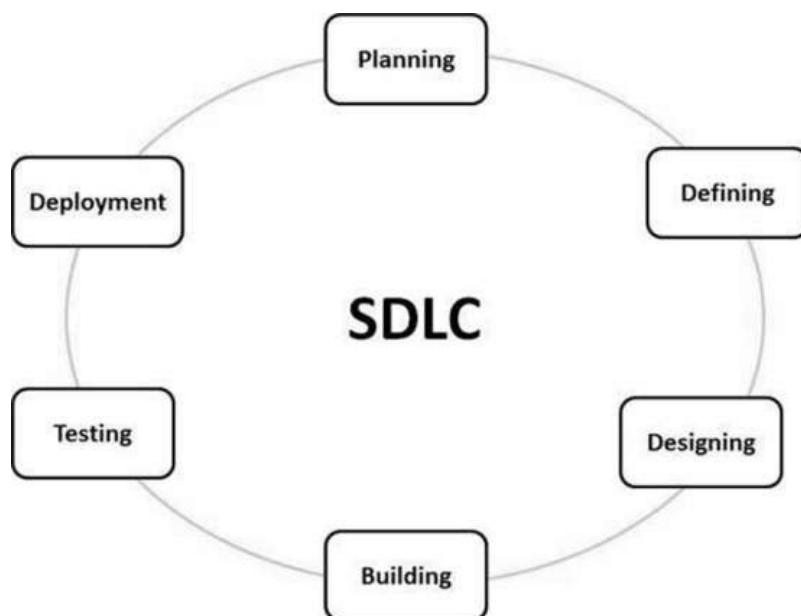


fig: General Representation of SDLC

Stage 1: Planning and Requirement Analysis

- Requirement analysis is the most important and fundamental stage in SDLC.
- Focus: plan the basic project approach and to conduct product feasibility study in the economical, operational and technical areas.
- The outcome of the technical feasibility study is to define the various technical approaches that can be followed to implement the project successfully with minimum risks.

Stage 2: Defining Requirements

- Once the requirement analysis is done the next step is to clearly define and document the product requirements and get them approved from the customer.
- This is done through an SRS (Software Requirement Specification) document which consists of all the product requirements to be designed and developed during the project life cycle.

Stage 3: Designing Product Architecture

- Make design of a product
- Design approach clearly defines the architecture models of a product

Stage 4: Developing Product

- Actual Development starts in this phase.
- Developers start writing their code for the modules.

Stage 5: Testing Product

- Once the code is ready, they are compiled together to make a build.
- This build is now tested by software testing.
- Here product defects are reported, tracked, fixed and retested, until the product reaches the quality standards defined in the SRS.

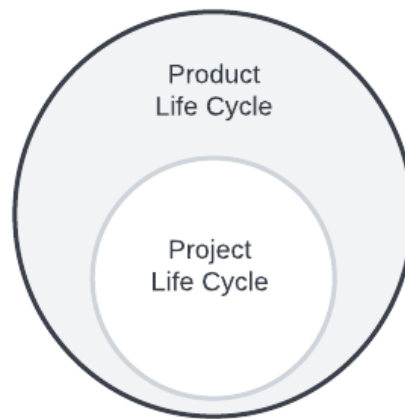
Stage 6: Deployment

- Once the product is tested, and ready to be deployed, it is finally released in the market. Feedback is taken from the external environment.
- After the product is released in the market, its maintenance is done for the existing customer base.

4.4 Product Life Cycle vs. Project Life Cycle

The product life cycle describes the stages a product passes through from conception to retirement.

The stages of the product life cycle are development, introduction, growth, maturity, and retirement. These phases are sequential and do not overlap. The project life cycle can be a part of one or more phases in the product life cycle.



SN	Product Life Cycle	Project Life Cycle
1	The product life cycle is always longer than that of a project.	shorter than product
2	Superset of Project Life Cycle	can be subset of product Life Cycle
3	The product life cycle may or may not end compared to the project life cycle	has a definite end.
4	Phases: Development, Introduction, Growth, Maturity and Decline	Phases: Initiation, Planning, Executing, Monitoring and Controlling, Closing
5	Phases in the product life cycle are always sequential and can never overlap.	Phases of the project life cycle usually overlap
6	There can be more than one project life cycle in one product life cycle.	Single project during life cycle

4.5 Representative Project Life Cycle

Software Development (Spiral)

- Proof-of-concept cycle
- First build cycle
- Second build cycle
- Final Cycle

Software Development (Waterfall)

- Concept
- Analysis
- Requirements
- Design and Development
- Programming
- Testing
- Verification
- Maintenance

4.6 System Development Methodology

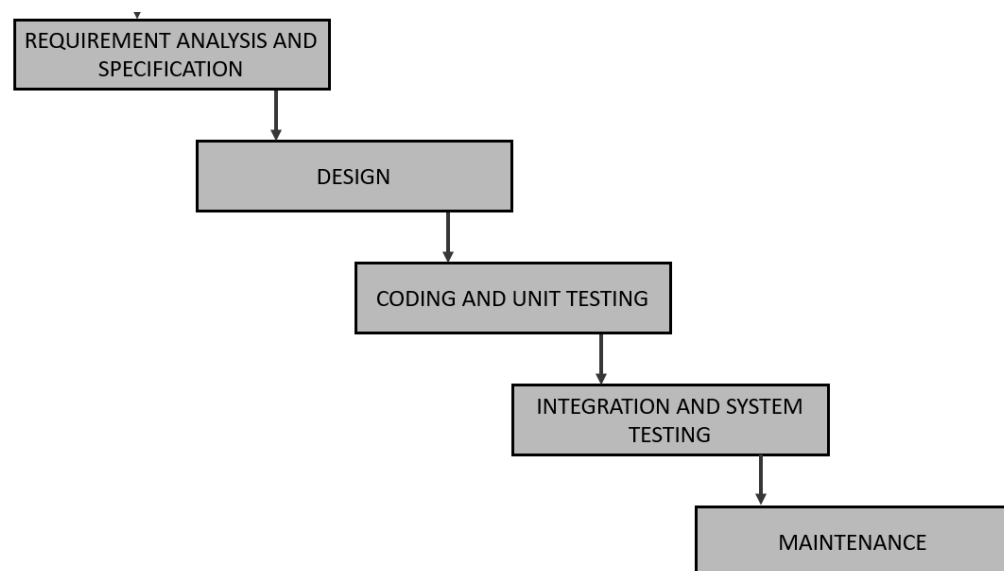
- SDLC is a framework for describing the phases involved in developing and maintaining information systems
- A formalized approach to implement SDLC is called Methodology.
- Examples: Waterfall model, Royce Model, Iterative & Incremental Model, Agile Model, Lean Model,

Choosing a model based on requirements and technology

		Requirements	
		Stable	Unstable
Technology	Novel	Agile & Plan Based Iterative Model	Agile & Lean Based Iterative Model
	Well-Known	Plan Based Iterative Model	Lean & Plan Based Iterative Model

4.6.1 Waterfall Model

- Waterfall Model is a sequential model that divides software development into pre-defined phases.
- Each phase must be completed before the next phase can begin with no overlap between the phases.
- It is also referred to as a linear-sequential life cycle model.
- It is very simple to understand and use.
- Because of the cascade from one phase to another, this model is known as the waterfall model.
- We should plan and schedule all of the process activities before starting work on them. Principle stages of the waterfall model are listed below:-



1) Requirements analysis and definition:

- All possible requirements of the system to be developed are captured in this phase and documented in a requirement specification document.

2) System and software design:

- The requirement specifications from the first phase are studied in this phase and the system design is prepared.
- This system design helps in specifying hardware and system requirements and helps in defining the overall system architecture.

3) Implementing and unit testing:

- With inputs from the system design, the system is first developed in small programs called units, which are integrated in the next phase.
- Each unit is developed and tested for its functionality, which is referred to as Unit Testing.

4) Integration and system testing:

- all the units developed in the implementation phase are integrated into a system after testing of each unit.
- Post integration the entire system is tested for any faults and failures.

5) Deployment

- Once the functional and non-functional testing is done; the product is deployed in the customer environment or released into the market.

6) Maintenance

- There are some issues which come up in the client environment.
- To fix those issues, patches are released.
- Maintenance is done to deliver these changes in the customer environment.

Advantages

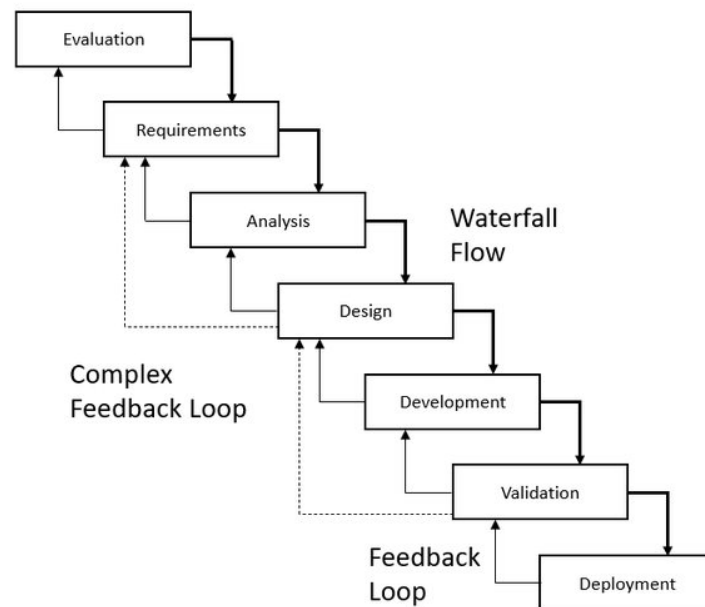
- Simple and easy to understand and use
- Easy to manage due to the rigidity of the model.
- Works well for smaller projects where requirements are very well understood.
- Process and results are well documented

Disadvantages

- Not suitable for the projects having changeable requirements
- Not suitable for large projects
- High amounts of risk and uncertainty.
- Error can be fixed only during the phase

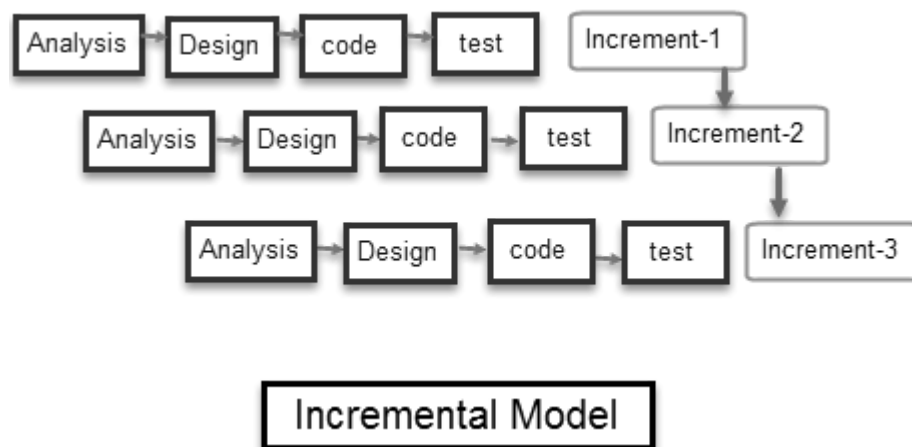
4.6.2 Conventional Waterfall Model - Royce's Model

- Waterfall model but has feedback



4.6.3 Evolutionary Methodology

- An evolutionary methodology follows an iterative and incremental approach.



Iterative and incremental development is a process that combines the iterative design method with the incremental build model. It is used by software developers to help manage projects.

Project is started with a comparatively small task or component and increments are made in each cycle of the iterations until desired product is reached.

Advantages of iterative and incremental approach

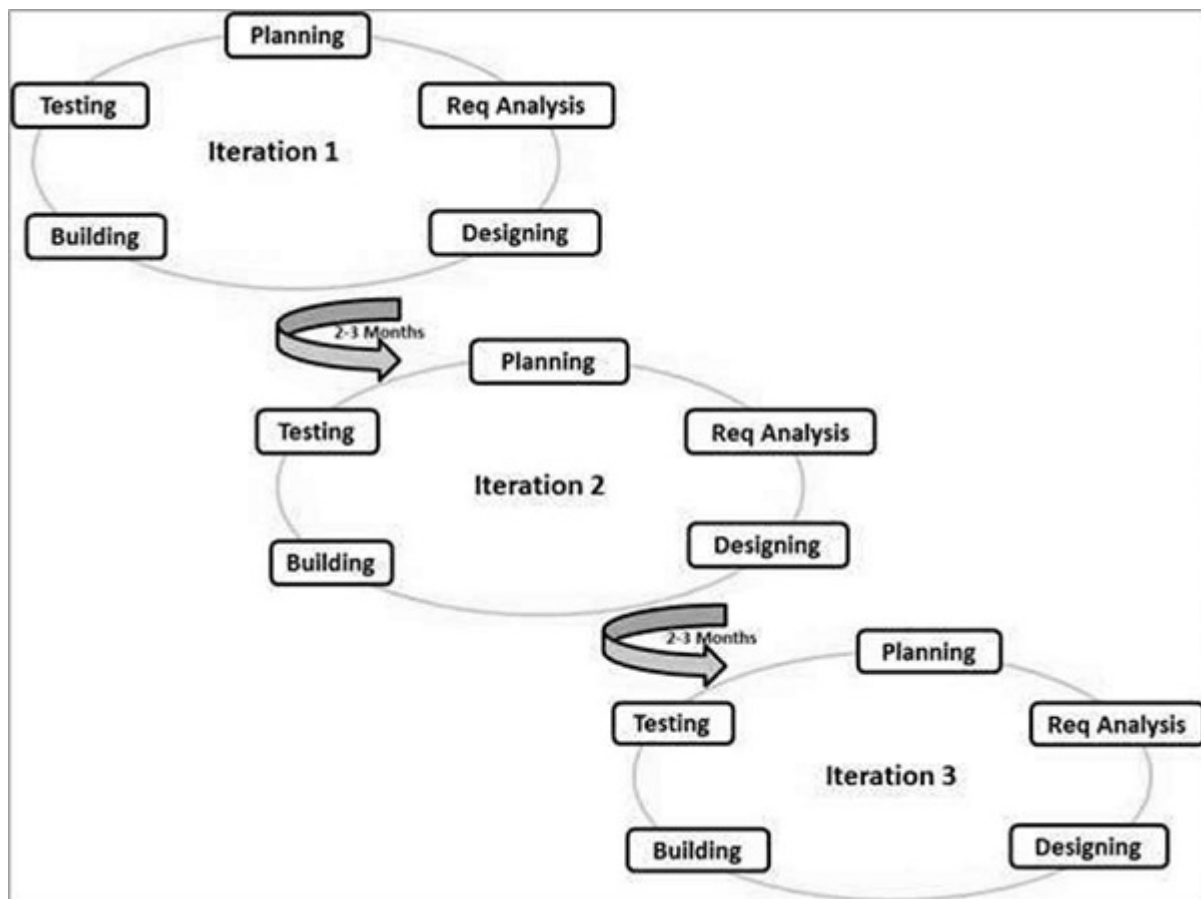
- Initial product delivery is faster
- Customers get important functionality early.
- Lowers initial delivery cost.
- Customer can provide feedback to each product increment
- Supports changing requirements
- Testing and debugging is easy.

Disadvantages of iterative and incremental approach

- More resources may be required
- Requires effective planning of iterations.
- More management attendance is required
- Not suitable for small projects
- Management Complexity is high

4.6.4 Agile Model

- Agile SDLC model is a combination of iterative and incremental process models
- Agile Methods break the product into small incremental builds. These builds are provided in iterations.
- Each iteration typically lasts from about one to three weeks.
- Every iteration involves cross functional teams working simultaneously on various areas like:
 - Planning
 - Requirements Analysis
 - Design
 - Coding
 - Unit Testing and
 - Acceptance Testing.
- At the end of the iteration, a working product is displayed to the customer and important stakeholders.



Advantages of Agile Model

- Resource requirements are minimum.
- Suitable for fixed or changing requirements
- Little or no planning required.
- Easy to manage.
- Gives flexibility to developers.

Disadvantages of Agile Model

- Due to the shortage of formal documents, it creates confusion and crucial decisions taken throughout various phases
- Due to the lack of proper documentation, maintenance of the finished project can become a difficulty.
- Not suitable for handling complex dependencies.
- Depends heavily on customer interaction, so if the customer is not clear, the team can be driven in the wrong direction.

4.6.4 Lean Model

- is a systematic approach to identify and eliminate waste through continuous improvement.
- Lean software development method is based on the principle “Just in time production”.
- It aims at increasing the speed of software development and decreasing cost.

Principles of lean Model

1) Eliminating Waste

- Only deliver what is needed to the end user
- remove waste that doesn't add value to the product such as unnecessary code, quality issues, etc

2) Build in quality

- Various tactics are used to ensure quality is built into the Lean process, such as pair programming and test-driven development.

3) Amplify learning

- Knowledge gained by a software engineer must be shared with every engineer on the software development team.
- This occurs through code review and sharing at meetings.

4) Decide as late as possible

- The goal is to experiment and learn as much as possible before committing to irreversible decisions.

5) Deliver as fast as possible

- Developers launch a product quickly, receive customer feedback fast and use that feedback to create a strategy for improvement.
- The idea behind this approach is to fail fast and learn from the results.

6) Motivate team

7) Optimize the whole

- The team examines the process from start to finish to make the Lean value stream as efficient as possible.

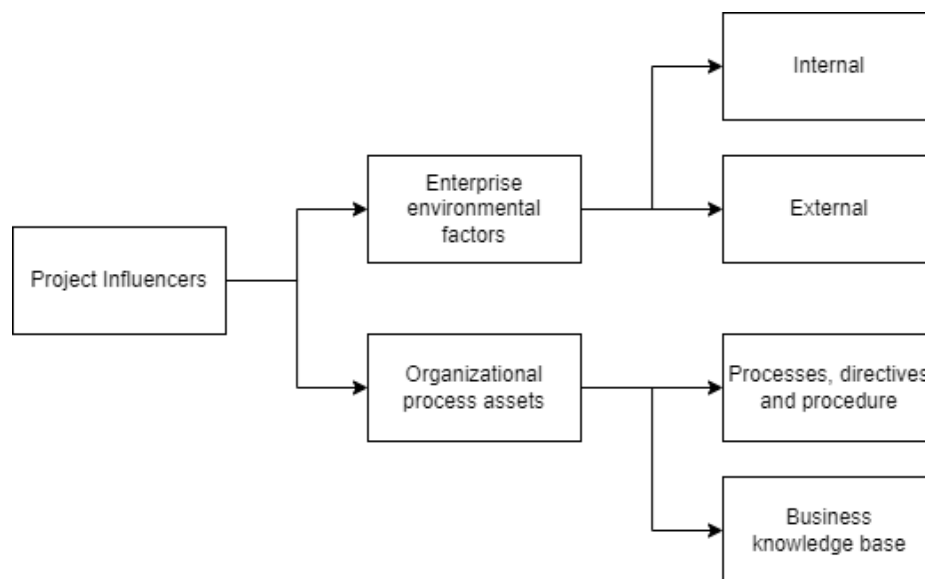
Advantages of Lean Model

- Increased customer satisfaction
- Increased productivity
- Improved utilization of work areas
- Improved ability to react to changes

Disadvantages of Lean Model

- High Implementation Cost
- Keeping high level of team cohesion

4.7 Project Influencers



1) Organizational process assets (OPA)

- OPA are valuable information, documents and knowledge tools that the organization accumulates over time.
- OPA are simply the lessons learned from the past.
- An organization gradually gains and develops more Organizational Process Assets as it gets involved in various projects.

2) Enterprise Environmental Factors (EEF)

- Forces that drive this change in the business are known as internal and external environmental factors.
- **Internal Factors**
 - Factors that occur inside the organization cause internal changes

- Examples:
 - events, people, systems, structures and conditions inside an organization
 - Corporate mission, corporate culture, and leadership style
 - Staff, budget,
- **External Factors**
 - factors that occur outside the organization but which can cause internal changes
 - Customers, competition, economy, technology, political and social conditions, and resources are common external factors that influence the organization
 - external corporate environmental factors: the economy, competition, politics, customers, suppliers

4.8 Roles and Responsibilities of Key Project Members

- Project Teams is a group of people who perform activities for accomplishing a common goal.
- Project team consisting of various skilled workers from various domain to work on a project
- Project teams must have the right combination of skills, abilities and personality types for successful projects.

1. Functional Manager

- a. Provide all necessary support services to the project including purchases
- b. Managing HR and Administration of the performing organization
- c. For IT Projects, Finance Manager performs duties of Functional Manager

2. Project Sponsor

- a. Assures availability of essential project resources
- b. Approves the budget
- c. Lead the project board
- d. Has Power and Authorities to make Decisions & Settle Disputes/Conflicts
- e. Possible Candidates: • Executive Director • Director Finance

3. Project Manager

- a. Managing project deliveries in time with the project plan
- b. Developing and Maintaining detailed project plan
- c. Detailed project planning and control
- d. Managing coordination of the partners and working groups
- e. Recruiting project staffs
- f. Recording and Managing project issues
- g. Resolving cross-functional issues
- h. Managing project scope
- i. Monitoring project progress and performance
- j. Possible Candidates: • Senior Manager • General Manager

4. System Administrator

- a. Data Migration
- b. Interfaces with other system
- c. Reporting configuration and deployment
- d. Set-up and maintenance of security
- e. Development and operation of technical testing programs
- f. Contributing to technical strategy, policy and procedures

5. Project Champion

- a. Helps focus attention on the project from technical perspective
- b. Usually someone with a great deal of technical expertise and industrial knowledge
- c. Possible Candidate: • Manager Technical • CTO • Technical Lead

Chapter 5:

Project and Organization structure (4 Marks)

5.1 Organization

- Organization is a group of people having different knowledge and skills working together to achieve an organizational goal.
- Projects can't be run in isolation. Projects must be operated in a broad organizational environment.
- Project Managers need to take the holistic/ system view of a project and understand how it is simulated within the larger organization.

5.2 System View of Project Management

- Project Management System is an organized set of tools and procedures for helping managers and teams to achieve project goals.

Three parts of projects' system view include:

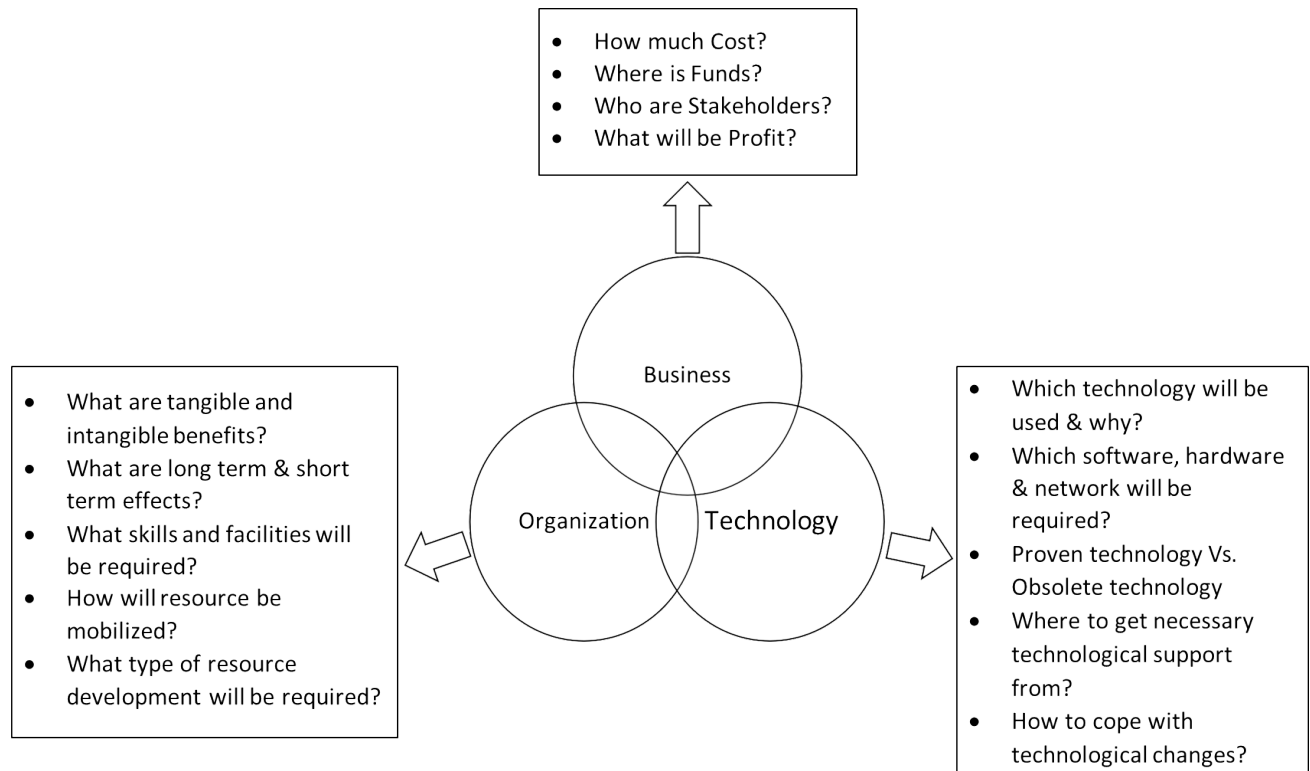
- **Systems Philosophy:**
 - View things as systems,
 - interacting components working within the environment to fulfill some purposes.
- **Systems Analysis:**
 - Includes Problem Solving Approach
- **Systems Management:**
 - Address business, technological and organizational issues before making changes to systems.

Three crucial elements

- Structured Approach: Organizational Structure
- Disciplined Approach: Coordinated Management Processes
- Defined Approach: Purposefully defined inputs and outputs for each process

Three Sphere Model for Systems Management

- Business Sphere : refers to What are the answer to the business goals
- Technological Sphere: refers to proper hardware and software issues to be resolved
- Organizational Sphere: issues involving the stakeholders



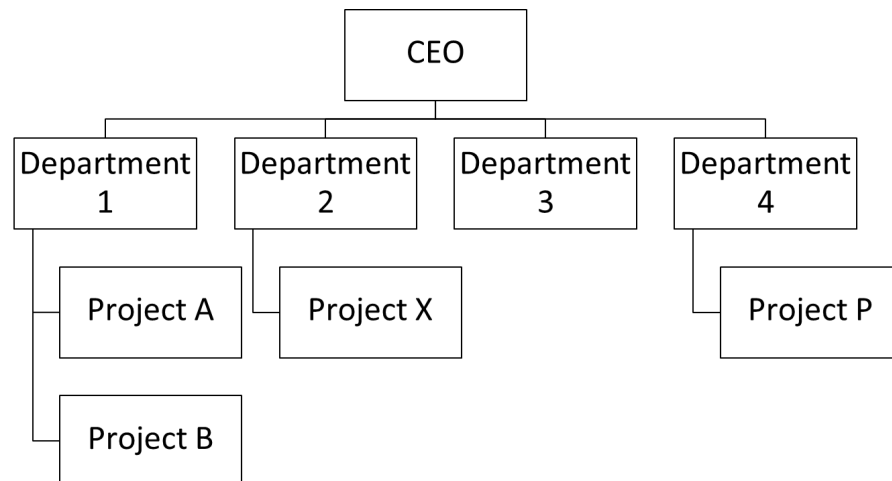
5.3 Organization Structure

- Organization structure refers to the division of labour as well as the patterns of coordination, communication, workflow and formal power that direct organizational activities.
- **Division of Labour:** refers to subdivision of work into separate jobs assigned to different people.
- **Coordinating work activities:** everyone work harmoniously

5.4 Forms of Organizational Structures

- 1) Functional Organization
- 2) Pure Project or Projectized Organization
- 3) Matrix Organization

5.4.1 Functional Organization



- It is a common type of organization structure in which organization is divided into smaller groups based on specialize
- An upgrade to line organization
- overcome the disadvantages of line organizations: principle of specialization is fully used
- Activities are divided into various functions such as production, marketing, finance, etc.
- One persons will be head of many workers according to functional authority
- Head will be specialist on job

Advantages of Functional Organization

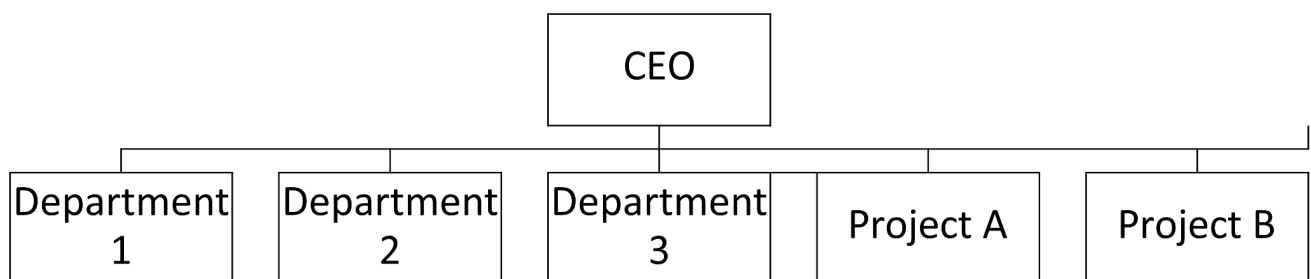
- 1) It groups employees based on functional skills for a higher degree of performance.
- 2) Helps in increasing efficiency of workers and staffs
- 3) Reduces burden of top managers
- 4) Offers a greater scope for training and development of employees
- 5) Provides better control and supervision in the organization
- 6) No duplication of work as each department is different. Also, the job description is clear.

Disadvantages of Functional Organization

- 1) Violates the principle of unity of command. Workers are not under one head, they are under many heads.
- 2) It can be difficult to get quick decision because functional manager has to report to top authority
- 3) Problems and difficulties of multiple command
- 4) Not easy to understand as line organization
- 5) Employees are treated as machines
- 6) There may be conflict that may encourage indiscipline in organization

5.4.2 Projectized Organization Structure

- also called Pure Project Organization
- Unlike Functional Organization, Project Organization is separate from functional units.
- The Project Manager has his own line organization with project authority and responsibility.
- Projects are directly reported to the head of the organization.
- There is problems while coordinating with departmental functions



Advantages of Pure Project Organization

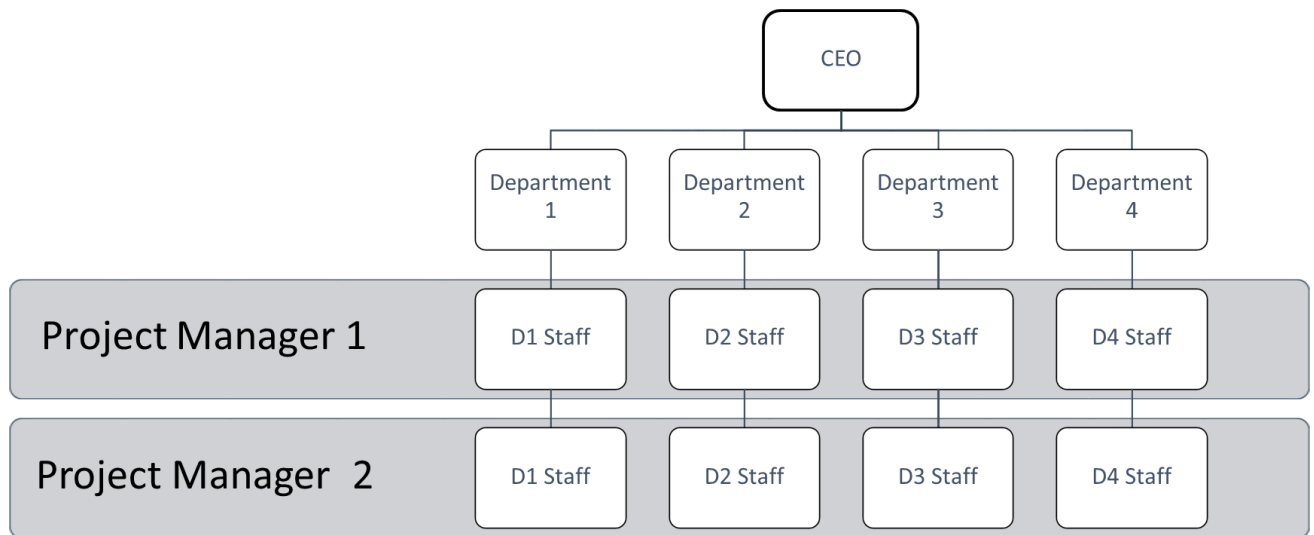
- Focus on project objectives
- Clear authority
- Effective communication
- Use of labor force as required

Disadvantages of Pure Project Organization

- Duplication of facilities and efforts
- Lack of job security
- Project specific benefit may create conflict between project and non project staff
- Useful for large projects only

5.4.3 Matrix Organization

- modern organization type
- is established to face the challenges of fast changing market
- mobilizes the resources of organization in systematic manner
- A matrix structure provides for reporting levels both horizontally as well as vertically
- You can further classify the Matrix Organization into
 - **Strong**
 - Project Managers are more Powerful in terms of Authority
 - PM rules but members are under functional managers
 - **Balanced**
 - Power is balanced between Project manager and Functional manager
 - Staff are assigned for project according to project needs by Project manager
 - Functional Manager facilitates all support to the project
 - **Weak**
 - Project managers are simply Project Coordinator having limited Authority
 - Functional manager assigns work Project manager negotiates
 - Project Manager facilitates the projects



Advantages of Matrix Organization

- 1) It helps in sharing resources efficiently
- 2) Decision making is balanced and flexible
- 3) Staff members can communicate with each other across boundaries
- 4) Flexible use of resources
- 5) Fast response to change
- 6) Strong project coordination

Disadvantages of Matrix Organization

- 1) Costly to maintain as it has many managers
- 2) Potential confusion over authority and responsibility.
- 3) The dual reporting structure add confusion and results in conflicts
- 4) You need to maintain resources throughout the project, no matter how long it takes
- 5) Create issues when there is no coordination between functional and project managers

5.5 Organizational Structure Influences on Projects

Project Characteristics	Functional Organization	Weak Matrix Organization	Balanced Matrix Organization	Strong Matrix Organization	Pure Project Organization
PM's Authority	Little or None	Limited	Low/ Moderate	Moderate/High	High to Full
Involvement of Team Member from Overall Organization	Virtually None	0-25 %	15-60 %	50-95 %	85-100 %
PM's Role	Part Time	Part Time	Full Time	Full Time	Full Time
Common Title for PM's Role	Project Coordinator/ Leader	Project Coordinator/ Leader	Project Manager/ Officer	Project /Program Manager	Project /Program Manager
PM's Administrative Staff	Part Time	Part Time	Full Time	Full Time	Full Time

- Organization Structure also represents the top level view of “Structure of Ownership” and “ Main Line of Reporting Order” for the project.
- The organizational structure influences the project manager’s authority, but project manager’s need to remember a disciplined and definite approach.

Chapter 6:

Project Management Process Groups (4 Marks)

6.1 Process

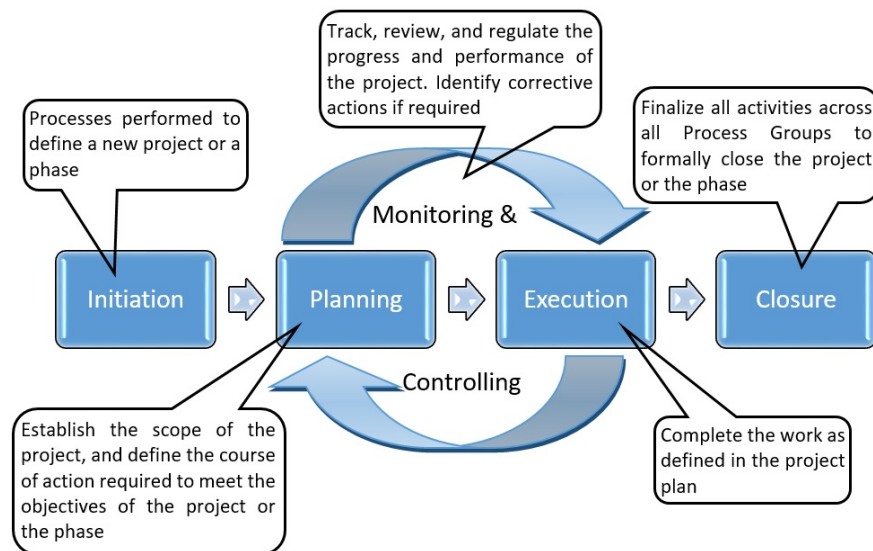
- A process is a sequence of steps taken to achieve a particular goal.
- Processes are described in terms of inputs, tools & techniques and outputs.
 - **Inputs:** Documents, Plans, Designs, Artifacts, etc.
 - **Tools and Techniques:** Mechanisms Applied to Inputs
 - **Output:** Documents, Products, Service, Knowledge, Artifacts.

For Successful Project:

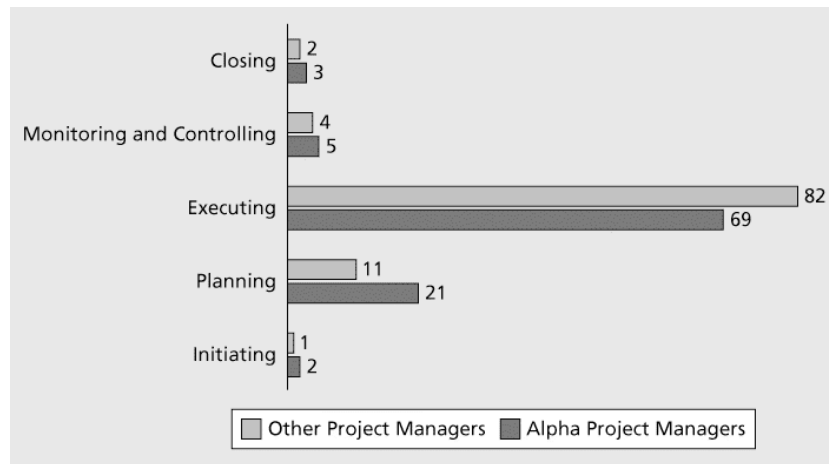
- 1) Select appropriate process required to meet the project objectives
- 2) Use a method that is well-defined and flexible to satisfy requirements.
- 3) Establish and maintain appropriate communication and engagement with stakeholders
- 4) Follow requirements to meet stakeholder needs and expectations
- 5) Balance the competing constraints of scope, schedule, budget, quality, resources, and risk to produce the specified product, service, or result

6.2 Project Management Process

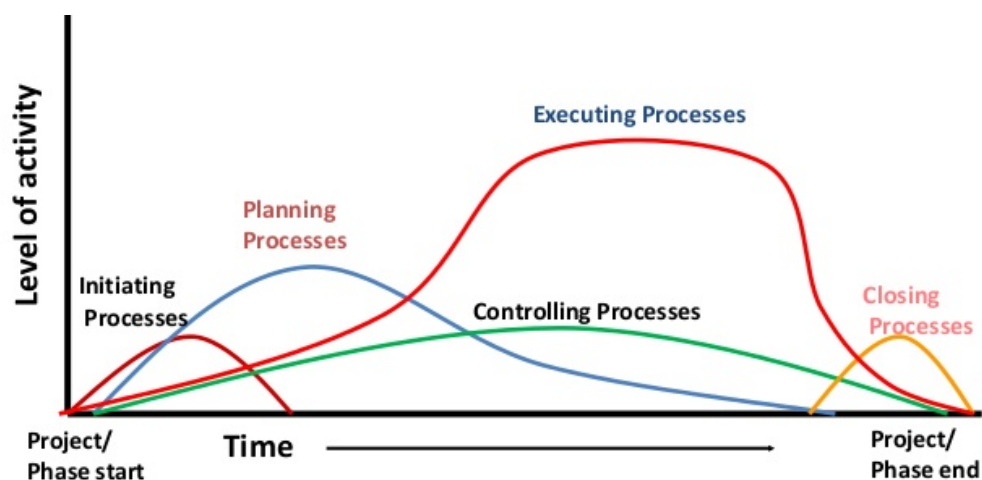
- These are concerned with describing and organizing the work of a project.
- There are mainly five processes. They are:
 1. **initiating processes:** recognize when project or phase should begin
 2. **planning processes** :- design and maintain a plan which leads to successful completion of a project
 3. **executing processes** : coordinate with people and resources to follow plan
 4. **monitoring and controlling processes:** monitor and measure progress and take right actions when necessary
 5. **closing processes** : It analyzes completion of the project or phase and its acceptability.



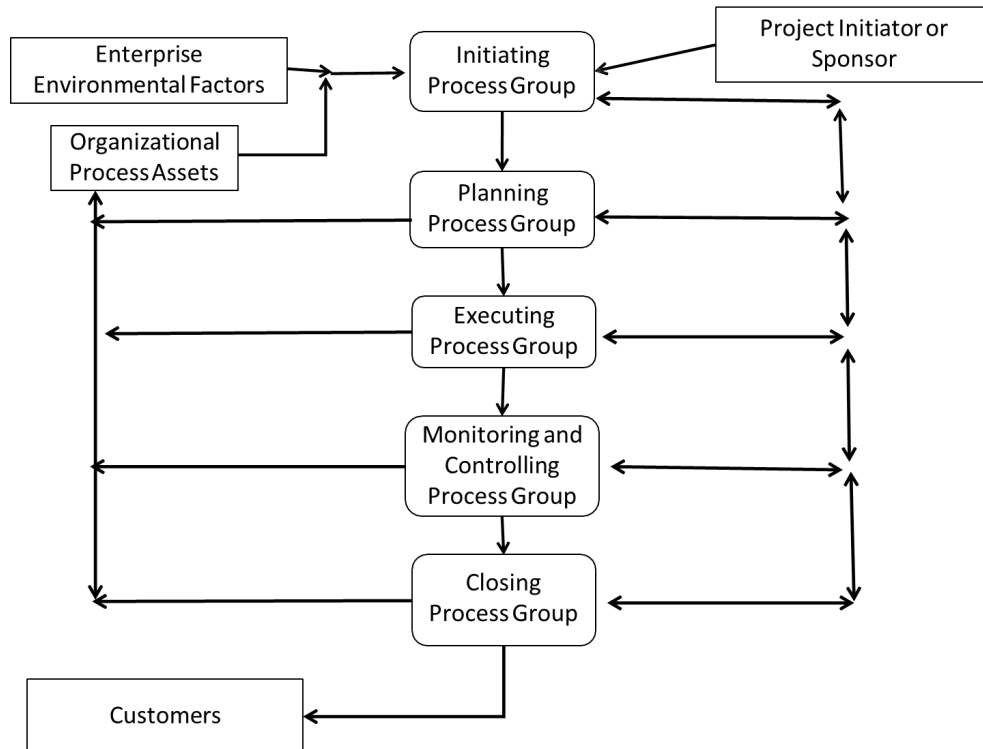
6.3 Percentage of Time Spent on Each Process Group



6.4 Overlap of Process groups in phase



6.5 High-level process group interaction –PMI view (Not imp)



6.6 Mapping of Project Management Process Group to Area of Knowledge

- Knowledge areas provide detailed description of the processes.
- The knowledge areas may cross various process groups.

Process Groups	Initiating	Planning	Executing	Controlling	Closing
Knowledge Area					
4. Project Integration Management		4.1 Project Plan Development	4.2 Project Plan Execution	4.3 Integrated Change Control	
5. Project Scope Management	5.1 Initiation	5.2 Scope Planning 5.3 Scope Definition		5.4 Scope Verification 5.5 Scope Change Control	
6. Project Time Management		6.1 Activity Definition 6.2 Activity Sequencing 6.3 Activity Duration Estimating 6.4 Schedule Development		6.5 Schedule Control	
7. Project Cost Management		7.1 Resource Planning 7.2 Cost Estimating 7.3 Cost Budgeting		7.4 Cost Control	
8. Project Quality Management		8.1 Quality Planning	8.2 Quality Assurance	8.3 Quality Control	
9. Project Human Resource Management		9.1 Organizational Planning 9.2 Staff Acquisition	9.3 Team Development		
10. Project Communications Management		10.1 Communications Planning	10.2 Information Distribution	10.3 Performance Reporting	10.4 Administrative Closure
11. Risk Project Management		11.1 Risk Management Planning 11.2 Risk Identification 11.3 Qualitative Risk Analysis 11.4 Quantitative Risk Analysis 11.5 Risk Response Planning		11.6 Risk Monitoring and Control	
12. Project Procurement Management		12.1 Procurement Planning 12.2 Solicitation Planning	12.3 Solicitation 12.4 Source Selection 12.5 Contract Administration		12.6 Contract Closeout

Chapter 7:

Project Integration Management (7 Marks)

7.1 Project Integration Management

- Integration Management is a collection of the processes and activities needed to identify, define, combine, unify and coordinate the various processes and project management activities within the Project Management Process Groups.
- These actions should be applied from the start of the project through completion.
- Project Managers must coordinate all the knowledge areas throughout a project's life cycle.
- integration Processes are used to manage Integration.
- Each process has: Inputs, Tools and techniques, and Outputs

7.1.1 Need of Integration Management

- Manage change and communication
- Reduce project time and cost
- Involve stakeholders early and often
- Make results visible
- Identify problems/solutions early
- Use relevant experience as early as possible

7.2 Integration Management Processes

Process Group	Integration Management Process	Major Output
Initiating	Developing the project charter	Project Charter
Planning	Developing the project management plan	Project Management Plan
Executing	Directing and managing project work	Deliverables, work performance info, change requests
Monitoring and Controlling	(1) Monitor/Control Project Work (2) Perform Integrated Change Control	(1) Change Requests (2) Status updates
Closing	Close Project/Phase	Product Transition

7.2.1 Developing Project Charter

- Developing Project Charter is a part of **Project** Initiation Phase
- Project Charter authorizes Project Managers to start the project and to use organizational resources as required.
- The project charter serves as an agreement contract between project sponsor and project team-documenting the projects.
- Project Charters should address following:
 - Scope
 - Objectives and deliverables
 - Project team members
 - Project risks
 - Benefits or returns on investment
 - Budget
 - Business case

Develop Project Charter Overview

INPUTS	TOOLS & TECHNIQUES	OUTPUT
1. Contract Agreement	1. Project Selection Methods	1. Project Charter
2. Statement of Work	2. Project Management Methodology	
3. Enterprise Environmental Factors	3. Project Management Information System (PMIS)	
4. Organizational Process Assets	4. Expert Judgment	

Inputs

- **Agreements/Contract:** The agreement should include much of the information needed for creating a good project charter.
- **Enterprise Environmental Factors :** Managers should review these factors when developing a project charter. Factors are:
 - Culture/Infrastructure, Tools, Human Resource, Personnel Policies, Marketplace Conditions

- **Organizational Process Assets**

- Organizational process assets include formal and informal plans, policies, procedures, guidelines, information systems, financial systems, management systems, lessons learned, and historical information that can influence a project's success.

- **Statement of Work**

- The statement of work (SOW) is a legally binding document that captures and defines all the work management aspects of your project.
- Statement of Work Includes: Business need, Product description, and Strategic Plan: Organization often perform SWOT Analysis

SWOT Analysis

Very important to have and managers from outside the IT department assist in the planning process. Let the scenario be 4 people to start a new business in the film industry.

- **S : Strengths**

- As experienced professionals, we have numerous contacts in the film industry
- Two of us have strong sales and interpersonal skills.
- Two of us have strong technical skills and are familiar with several filmmaking software tools.

- **W : Weakness**

- None of us have accounting or financial experience.
- We have no clear marketing strategy for products and services.
- We have little money to invest in new projects.
- We have no company website and limited use of technology to run the business

- **O: Opportunities**

- A potential client has mentioned a large project she would like us to bid on.
- The film industry continues to grow.
- There are two major conferences this year where we could promote our company.

- **T: Threats**

- Other individuals or companies can provide the services we can.
- Customers might prefer working with more established individuals and organizations.
- There is high risk in the film business.

Q) Prepare a statement of work (SOW) and IS project which is to be developed as Engineering College MIS. You are free to make your own assumptions regarding the system.

Reports consists of following sections:

1) Section 1: Introduction

- a) explained basic things
- b) What is the type of work that is being done? Is it a service that's being performed or a product that's being built? Who are the parties involved?

2) Section 2: Project Overview and Objectives: why this project is being done

3) Section 3: Scope of work

- a) the work that needs to be done in order to complete the project.

4) Section 4: Task list : break your tasks down into phases.

5) Section 5: Project Schedule

6) Section 6: Project Deliverables

7) Section 7: Adoption plan : An adoption plan is the process for how the deliverables will be put into place.

8) Section 8: Project Management: includes payment, administrative works

9) Section 9: Success Criteria and Sign-off

Tools and Techniques

Techniques to select Projects

1. Focus on broad organizational needs

- Three important criteria for projects:
 - Need: Do people in the organization agree that the project NEEDS to be done?
 - Funding: Is the needed FUNDING secure for the project?
 - Will: Is there a strong WILL to make the project succeed?

2. Categorize IT projects

- One categorization is done whether the project addresses
 - a problem
 - an opportunity, or
 - a directive

3. Perform net present value or other financial analyses

Three primary methods for determining the projected financial value of projects:

1) Net present value (NPV) analysis

- Net present value (NPV) is the difference between the present value of cash inflows and the present value of cash outflows over a period of time.
- All expected future cash inflows and outflows are depreciated to the present point in time.
- If $NPV > 0$ Accept If $NPV < 0$ Reject If $NPV = 0$ Indifferent

$$NPV = \frac{R_t}{(1 + i)^t}$$

NPV = net present value

R_t = net cash flow at time t

i = discount rate

t = time of cash flow

2) Return on investment (ROI)

- calculated by subtracting the project costs from the benefits and then dividing by the costs
- $ROI = \text{Net Income} / \text{Cost of Investment}$

3) Payback analysis

- Payback analysis is a mathematical methodology to determine the payback period for an investment.
- The payback period is how long it will take to pay off the investment.
- Break Even Point $\Rightarrow$ Net Income = 0, Profit = Loss
- Must be less

4. Use a weighted scoring model

Project A is selected.

Project	Weight=5	Weight=3	Weight=1	Total Weighted Score
	Profit Potential	Marketability	Ease to Produce/Serve	
Project A	5	4	4	41
Project B	5	3	3	37
Project C	3	4	2	29

5. Implement a balanced scorecard

Project Management Methodology

It is a set of Project Management Process Groups where their related control functions are merged and combined into a working cohesive system.

Project Management Information System

- Now-a-days various enterprise project management systems are available.
- They are even accessible via the Internet and tie into other systems, such as financial systems.
- Even in smaller organizations, project managers or other team members can create Gantt charts.

Expert Judgment

- This is a way of making a judgment based on expertise, skill, or specialized knowledge in a specific area.
- The expertise may have gained knowledge via training, or career experience.
- Anyone who has worked on a large, complex project appreciates the importance of expert judgment in making good decisions.
- Project managers should not hesitate to consult experts on different topics, such as
 - what methodology to follow,
 - what programming language to use, and
 - what training approach to follow

7.2.2 Develop Preliminary Project Scope Management

- It is one of the output of project initiation process group
- **Purpose:** To identify the high level project objectives
- Project objectives must be clear, actionable and measurable
- Includes:
 - Project Objectives, Constraints and Assumptions
 - Product requirements
 - Initial risk
 - Preliminary Estimated Cost
 - Approval Requirements and so on.

PRELIMINARY Project Scope Statement Overview		
INPUTS	TOOLS AND TECHNIQUES	OUTPUT
1. Project Charter	1. Project Management Methodology	1. Preliminary Project Scope Statement
2. Project Statement of Work	2. Project Management Information Systems	
3. Enterprise Environmental Factors		
4. Organizational Process Assets	3. Expert Judgment	

7.2.3 Develop Project Management Plan

- Developing Project Management Plan is a part of **Planning** process group
- Primary use of project plans are to document assumptions and decisions, facilitate communication among stakeholders, and document approved.
- Includes:
 - Project Charter
 - Project Management Approach
 - Scope Statement
 - Budget, Schedule and Resources

Project Management Plan Overview		
INPUTS	TOOLS AND TECHNIQUES	OUTPUT
1. Preliminary Project Scope	1. Project Management Methodology	1. Project management Plan
2. Project Management Processes	2. Project Management Information Systems	
3. Enterprise Environmental Factors		
4. Organizational Process Assets	3. Expert Judgment	

- **Common elements to most project management plan**
 - Introduction/overview of the project
 - Project organization
 - Management and technical processes
 - Work to be performed (scope)
 - Schedule and budget information
 - References to other project planning documents

Project Planning Matrix (PPM)

- also called logical framework
- a tool used in project planning to develop a well designed, objectively described and valuation able project
- Provides summary of project design documents

Characteristics of PPM

- Project Logic moves vertically from Inputs to Goals
- Horizontal Logic moves from Narrative Summary to Important Assumptions
- All cell items should be linked with proven non-causal relationship
- Better cause effect linkage between cell items for better project design

	Narrative Summary	Objectively Verifiable Indicators (OVI)	Means of verification (MOV)	Assumptions / Risk
Goal				
Purpose				
Outputs				
Activities				

Stakeholder Analysis

- Process of Identifying the individuals or groups that are likely to affect or to be affected by proposed system and sorting them according to their impact
- Stakeholder analysis document has information such as:
 - Stakeholders name and organization
 - Roles on the Project
 - Unique facts about stakeholders
 - Level of Influence and interest in the project
 - Suggestions for Managing relationships and Nature of Reports

Types of stakeholders

- 1) **Primary Stakeholders** : Primary stakeholders are those individuals, groups or entities that are involved with the monetary transactions of an organization.
- 2) **Secondary Stakeholders**: Secondary stakeholders are those individuals, groups or entities that are invested in the social transactions of an organization.
- 3) **Key Stakeholders**:
 - a) Key stakeholders are those who help companies make strategic decisions, minimize risks and grow their business.
 - b) have significant influence upon or importance within an organization

Baseline Management

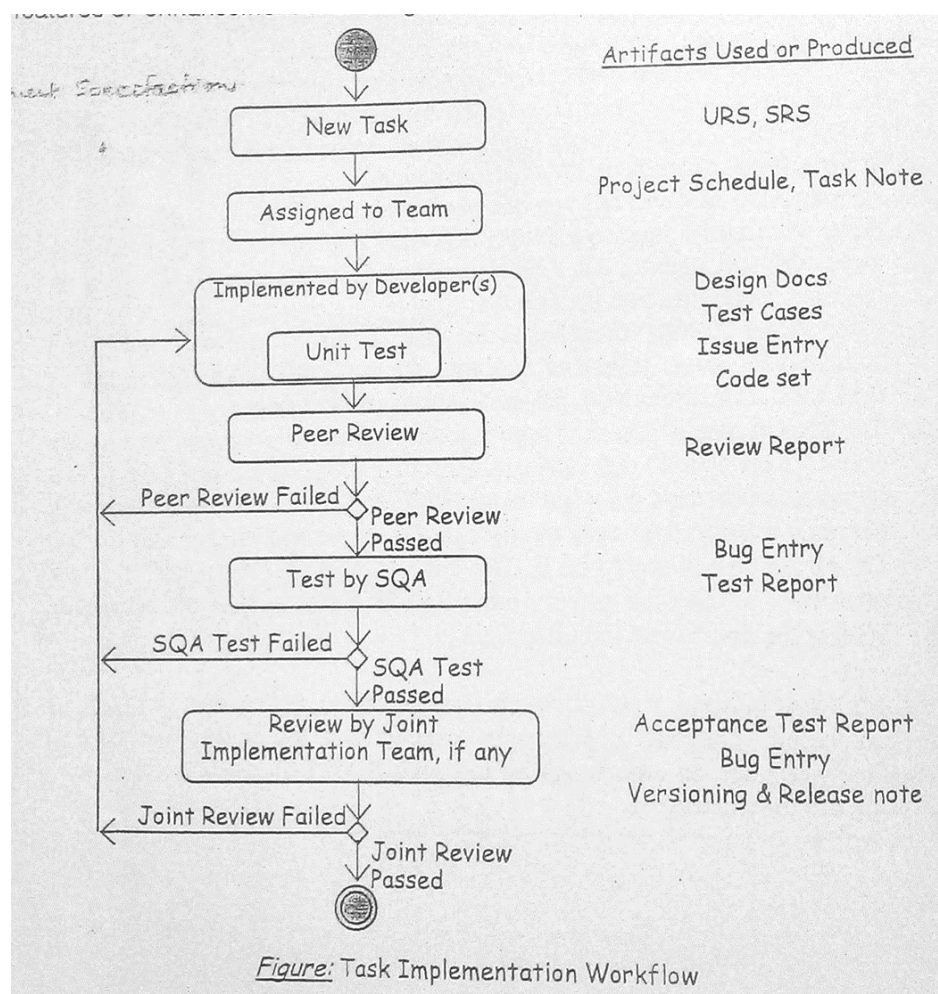
- It is a fixed reference point to measure and compare your project's progress against.
- This allows you to assess the performance of your project over time.
- Project Baseline includes
 - Schedule Baseline
 - Cost Baseline
 - Scope Baseline
 - Quality Baseline
- Develop Baseline Project Plan following the steps:
 - Set measures
 - Record Progress
 - Calculate the Variance
 - Take Remedial Action

7.2.4 Direct & Manage Project Execution

- is a part of **Execution** process group
- Manage and perform the tasks described in the project management plan
- Project planning and execution are related and inseparable activities

INPUTS	TOOLS AND TECHNIQUES	OUTPUT
1. Project Management Plan	1. Project Management	1. Deliverables in Project Plan
2. Approved Corrective Actions	Methodology	2. Requested Changes
3. Approved Preventive Actions	2. Project Management Information	3. Implemented Change Request
4. Approved Change Request	Systems	4. Implemented Corrective Action
5. Approved Defect Repair		5. Implemented Preventive Actions
6. Validate Defect Repair		6. Implemented Defect Repair
7. Administrative Closure Procedure		7. Work Performance Information

Development Task Implementation Workflow in IT Projects



Work Performance Info

It Includes:

- Schedule Progress
- Completed, Pending and Incomplete Deliverables
- Extent to which quality standards are being met

- Resource Utilization Details
- Lessons Learnt
- Conflicting Issues that need to be addressed

7.2.5 Monitor & Control Project Work

- is a part of **Monitoring and Controlling** process group
- Mostly, changes are unavoidable, so better approach is to develop and follow monitor and control strategy
- Monitoring project work includes collecting, measuring, and circulating performance information
- Goal: Watch all the activities to meet the performance objectives of the project

Monitor and Control Project Work Overview				
INPUTS		TOOLS AND TECHNIQUES		OUTPUT
1.	Project Management Plan	1.	Project Management Methodology	1. Recommended Corrective Action
2.	Requested Changes			2. Recommended Preventive Actions
3.	Work Performance Information	2.	Project Management Information Systems	3. Forecasts
		3.	Earned Value Management	4. Recommended Defect Repair
		4.	Expert Judgment	5. Requested Changes

7.2.6 Perform Integrated Change Control

- is a part of **Monitoring and Controlling** process group
- Objective:
 - Influence the factors that create changes to ensure that changes are beneficial
 - Determining that a change has occurred
 - Managing actual changes as they occur

Integrated Change Management		
INPUTS	TOOLS AND TECHNIQUES	OUTPUT
1. Project Management Plan	1. Project Management	1. Approved Change Request
2. Requested Changes	Methodology	2. Rejected Change Request
3. Work Performance Information	2. Project Management Information	3. Updated Project Management Plan
4. Recommended Corrective Actions	Systems	4. Updated Project Scope Statement
5. Recommended Preventive Actions	3. Expert Judgment	5. Approved Corrective Actions
6. Recommended Defect Repair		6. Approved Defect Repair
7. Deliverables		7. Validated Defect Repair
		8. Deliverables

Types of control

1) Change control system

- formal, documented process that describes when and how official project documents and work may be changed
- Describes who is authorized to make changes and how to make them

2) Change Control Board (CCB)

- formal group of people
- responsible for approving/rejecting changes on a project
- provide guidelines for:
 - preparing change requests,
 - evaluate change requests,
 - manage the implementation of approved changes

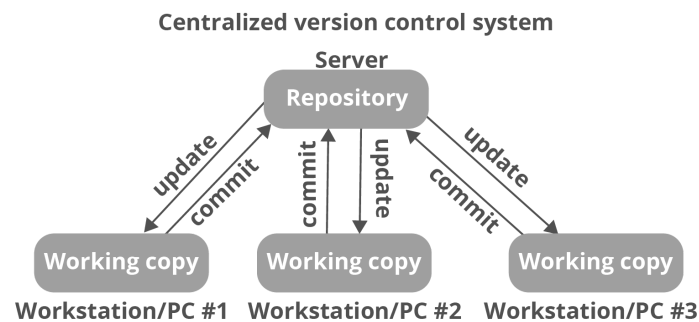
3) Configuration Management

- Ensures the project's product description are correct, complete and consistent
- Involves identifying and controlling the functional and physical design characteristics of products (and its documentation)

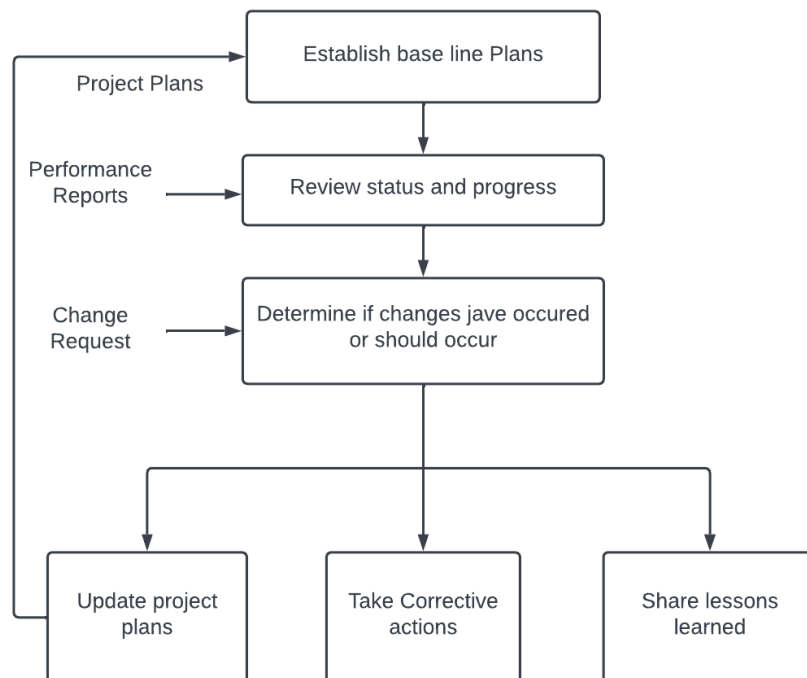
4) Version Control and Build Management

- Version control, also known as source control, is the practice of tracking and managing changes to software code.
- Version control systems are software tools that help software teams manage changes to source code over time.
- Maintains history and supports Rollback

- Build Management: Regardless of where software is generated, project managers need to formalize a check-in/check-out policy to guarantee code quality.



Block Diagram of Integrated Change Control



7.2.7 Close Project

- is a part of **Closing** process group
- To close a project or phase:
 - Must finalize all activities
 - Transfer the completed or canceled work to the appropriate people
- Main Outputs:
 - Final product, service, or result transition
 - Organizational process asset updates

Main Steps in closing Project

1) Phase end activities

- Facilitate project team while changing from one phase to other
- Include activities like Peer review, Release Management, Review Meeting, Update Project Plan/or for Next Phase, Status Reporting, and End User Education

2) Administrative closure procedure

- During closing of project Ensure project plan has proper budget/time for closure
- Populate Project Archives like
 - Formal Acceptance by Client
 - Recommendation Support from Sponsors/Support Organization
 - Maintenance/Upgrade
 - Trainings
 - Lesson Learnt

3) Contract closure procedure steps

- Verification of Contract requirement
- Formal Acceptance by Client
- Performance Evaluation by buyer of contract
- Performance evaluation by seller of contract
- Procurement Audit
- Update Project Files and historical Databases

4) Updated Organizational process assets

- Formal Acceptance documentation
- Project Files
- Project Closure Documentation
- Archiving is the last step
- Historical Information and lesson learnt

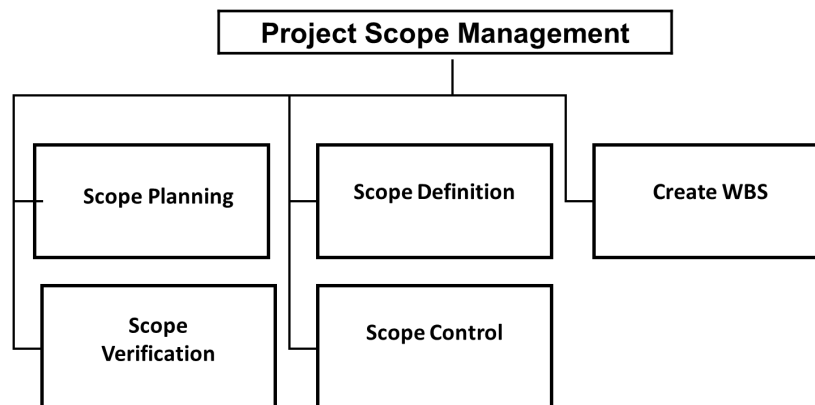
7.3 Project Scope Management

- Scope refers to all the work involved in creating the products of the project and the processes used to create them
- Project scope management includes the processes involved in defining and controlling what is or is not included in the project
- The project team and stakeholders must have the same understanding of what products will be produced as a result of a project and what processes will be used in producing them

7.3.1 Project Scope and Product Scope

SN	Project Scope	Product Scope
1	The project scope is the work that is required to deliver the product.	Product scope is the sum of all features, functions, and characteristics of the product.
2	The project scope focuses on the work needed to deliver the product.	The product scope focuses on the features of the end product
3	subset of Product Scope	superset of project scope
4	project scope is oriented towards the “how” (work-related).	Product scope is oriented towards the “what” (functional requirements)
5	Eg:- constructing a bridge	Eg: technical specifications of bridges such as length, width, etc.

7.3.2 Various Processes in PSM



- 1) **Initiation:** beginning a project or continuing to the next phase
- 2) **Scope planning:** developing documents to provide the basis for future project decisions
- 3) **Defining scope :** Reviewing the project charter requirement documents, and organization process assets to create a scope statements
- 4) **Creating the WBS:** subdividing the major project deliverables into smaller, more manageable components
- 5) **Scope verification:** formalizing acceptance of the project scope
- 6) **Scope change control:** controlling changes to project scope

7.3.3 Work Breakdown Structure (WBS)

WBS is a process of systematically breaking down the major project work into smaller, more manageable components and subcomponents in hierarchical order.

A work package is a task at the lowest level of the WBS. Eg Gantt Chart

7.3.3.1 Advantage of WBS:

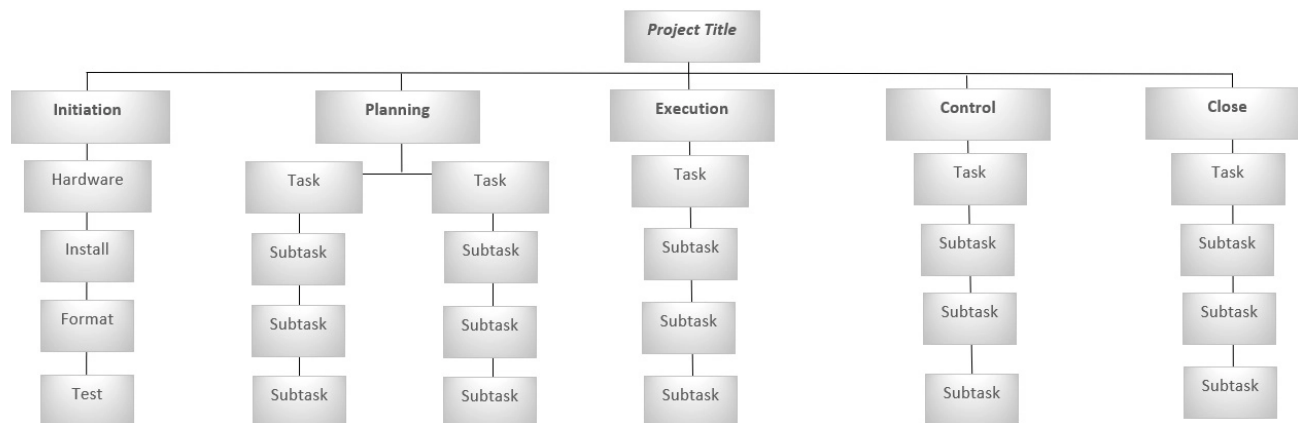
- Planning can be performed easily.
- Cost and budgets can be established
- Schedules can be established
- Responsibility can be assigned to each activity
- Can be used to analyze risk of each activity

7.3.3.2 Create WBS

- 1) Defines work (what)
- 2) Identifies time to complete a work package (how long)
- 3) Identifies a time-phased budget to complete a work package (cost)
- 4) Identifies resources needed to complete a work package (how much)
- 5) Identifies a person responsible for units of work (who)
- 6) Identifies monitoring points (milestones) for measuring success

Levels of WBS

- Total Program
 - Project
 - Task (activity)
 - Sub-tasks
 - Work Packages
 - Level of Efforts



7.3.3.3 Approaches to Developing WBSs

- 1) **Using guidelines:** some organizations , provide guidelines for preparing WBSs
- 2) **The analogy approach:** review WBSs of similar projects and customizes it to your project
- 3) **The top-down approach:** start with the largest items of the project and break them down
- 4) **The bottom-up approach:** start with the specific tasks and roll them up
- 5) **Mind-mapping approach:** mind mapping is a technique that uses branches radiating out from a core idea to structure thoughts and idea

7.3.3.4 Organizational Breakdown Structure (OBS):

- is a type of organizational chart
- OBS links cost, activity & responsibility
- It allows : work definition, owner assignment of work packages, and budget assignment to departments

7.3.3.5 Advice/Principles for Creating a WBS and WBS Dictionary

1. The unit of work should appear only once in WBS.
2. The work content of WBS items is the sum of the WBS items below.
3. A WBS item is the responsibility of only one individual, even though many people are working on it.
4. WBS must be consistent with the way in which work is actually going to be performed.
5. Project team members should be involved in developing WBS to ensure consistency.
6. Document WBS items in WBS Dictionary

100% rule

- The 100% rule is a core characteristic of the WBS
- This rule states that the WBS includes 100% of the work defined by the project scope and captures ALL deliverables – internal, external and interim – in terms of work to be completed, including project management.
- The 100% rule is one of the most important principles guiding the development, decomposition and evaluation of the WBS

8/80 rule : This rule states that the lowest level of work should be no less than 8 hours and no more than 80 hours.

7.3.4 Responsibility Assignment Matrix (RAM)

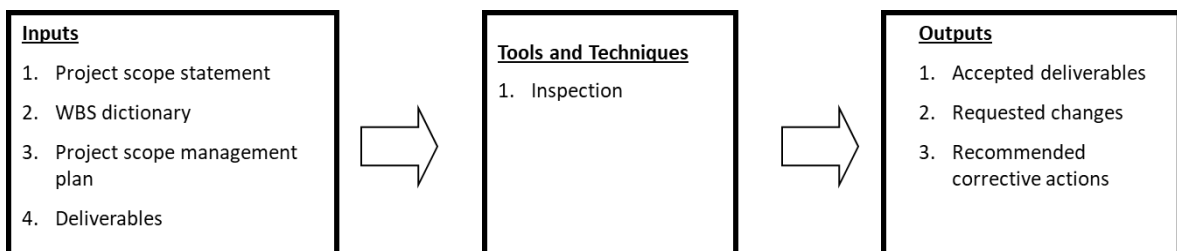
- also known as RACI matrix where
 - **R** = Responsibility: Who does the task?
 - **A** = Accountability: Who signs off on the task or has authority for it?
 - **C** = Consultation: Who has information necessary to complete the task?
 - **I** = Informed: Who needs to be notified of task status and results?
- RAM maps the work of the project, to the people responsible for performing the work
- Works are described in the WBS
- For smaller projects, it is best to assign individual people to WBS activities.
- For very large projects, it is more effective to assign the work to organizational units or teams

Example of RACI

RACI Chart		Person			
Activity	Person 1	Person 2	Person 3	Person 4	Person 5
Define	A	R	I	I,C	I
Design	I	A	R	C	C
Develop	I	A	R	C	C
Test	A	I,R	I	R	I
Implementation	I	R	A	C	I

7.3.5 Scope Verification

- The process of formally accepting the finished project outcomes is called scope verification.
- Scope Verification includes:
 - Obtaining formal acceptance of the project scope by the stakeholders (sponsor, client, customer, etc.).
 - Reviewing outcomes and work results to ensure that all were completed correctly and satisfactorily.
 - Determining completion, especially if the project is terminated early.
- The scope verification process should establish and document the level and extent of completion.



1) Inputs

- a) Project Scope Statement: describe the project's outcomes and work required to create those outcomes in detail manner
- b) WBS Dictionary: A document that describes each component of the WBS
- c) Project Scope Management Plan: A document that provides guidelines on how project scope will be defined, documented, verified, managed, and controlled by the project management team.
- d) Outcomes : Final results of a project

2) Tools and Techniques

a) Inspection

- Inspection includes activities such as measuring, examining and verifying that work and confirming that results meet the requirements and product acceptance criteria.
- Also called product review, reviews, audits, and walkthrough.

3) Outputs

- a) Accepted results
- b) Requested changes :
- c) Recommended corrective actions :

7.3.6 Scope Control

- Scope control is a key process of project management that involves controlling changes to the project scope
- Goals of scope control
 - influence the factors that cause scope changes
 - guarantees that changes are processed according to procedures developed
 - manage changes when they occur
- Variance is the difference between planned and actual performance

7.3.7 Best Practices for avoiding Scope Problems

1. Keep the scope realistic. Don't make projects so large that they can't be completed.
2. Involve users in project scope management.
3. Use off-the-shelf hardware and software whenever possible.
4. Follow good project management processes.

7.3.7.1 Scope Creep

1. Poor change control
2. Lack of proper initial identification and/or documentation of the features that are required for the achievement of project objectives
3. Weak project manager or executive sponsor

7.3.7.2 Reduce Scope Creep

1. Define requirements as “must haves” and “nice to haves”
2. Set project expectations
3. Agree on deliverables, and document
4. Document and review business requirements

7.3.7.3 Key reason for IT Project Failure

1. Poor project scope management is the key reason for IT Project failure.
2. Good project scope management should have:
 - a. strong user involvement,
 - b. executive support,
 - c. a clear statement of requirements,
 - d. a process for managing scope changes.

7.3.7.4 Estimating a Project Scope

1. First Define project scope :
 - a. Easy to estimate Time and Cost
2. Aspects of Estimation
 - a. Smaller the task - > easier to estimate
 - b. Estimation error increases project risk
 - c. Estimate by team and modification by experts
 - d. Estimate not in task duration:
 - i. Estimate in ‘work effort required by individual’

Types of Estimation

- Scope / Size Estimation
- Tie Estimation
- Cost Estimation

Chapter 8:

Project Time Management(7 Marks)

8.1 Importance of Project Schedules

- Manager's biggest challenge is to deliver projects on time
- Schedule issues are the main reason for conflicts on projects, mainly during the second half of projects
- When people compare planned completion date vs. actual project completion date, they are not always as expected.
- During the project, there may be MANY change request which may account for delay

That's why Project Schedule Management is required. It focuses on timely completion of a project

What are the policies helpful for maintaining a good schedule?

Plan Schedule Management		
Inputs	Tools and techniques	Outputs
1. Project charter	1. Expert judgment	1. Schedule
2. Project management plan	2. Data analysis	management
3. Enterprise environmental factors	3. Meetings	plan
4. Organizational process assets		

8.2 Project Time Management Processes

Process Group	Integration Management Process	Major Output
Planning	P1: Defining Activities	Activity List
	P2: Sequence Activities	Project Schedule Network Diagram
	P3: Estimate Activity Resources	Activity Resource Reqs.
	P4: Estimate Activity Duration	Activity Duration Estimates
	P5: Develop Schedule	Project Schedule
Monitoring and Controlling	MC1: Control Schedule	Work Performance Measurements

8.2.1 Activities Definition

- is a part of **Planning** process groups
- Define all the tasks necessary to produce the project's final deliverable.
- Activity definition involves developing a more detailed WBS, so we can develop realistic cost and duration estimates.
- Defining project activities is significant, because if the project team is aware of what needs to be done during the project life cycle, products can be released timely.

8.2.1.1 Decomposition of Activities

- 1) Sub divide work packages into Schedule Activities.
- 2) Define Activities to meet project objectives.
- 3) Activities should be relatively smaller and manageable components.
- 4) Activities may be work packages or lower level items.
- 5) WBS and Activity list may be developed sequentially or concurrently

Terms

- 1) An **activity list** is a table containing tasks to be listed on a project schedule, along with the activity attributes.
 - Activity attributes provide more information about each activity, such as predecessors, successors, resource requirements, constraints, and assumptions related to the activity.

8.2.2 Sequence Activities

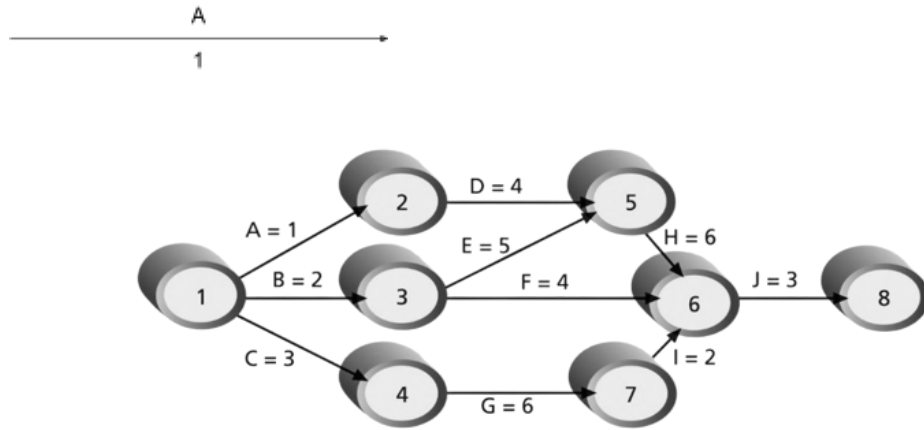
- is a part of **Planning** process groups
- Sequence Activities involves reviewing activities and determining dependencies
 - A dependency is the sequencing of project activities
 - We must determine dependencies in order to use critical path analysis

8.2.2.1 Network Diagram

- Network diagrams are the preferred technique for showing activity sequencing
- A network diagram is a schematic display of the logical relationships among project activities.
- Two main formats are the arrow and precedence diagramming methods.

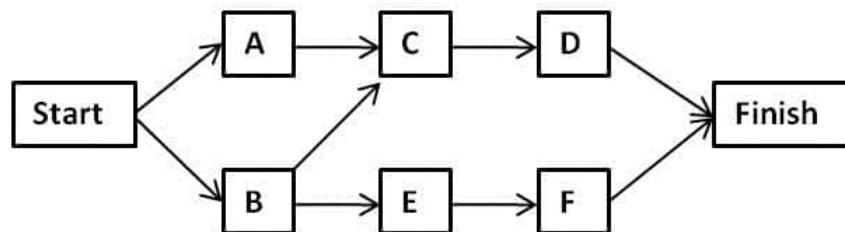
Arrow Diagramming Methods (ADM)

- Also called activity-on-arrow (AOA) network diagram.
- Activities are represented by arrows.
- Nodes or circles are the starting and ending points of activities.
- Can only show finish-to-start dependencies



Precedence Diagramming Method (PDM)

- Activities are represented by boxes.
- Arrows show relationships between activities.
- More popular than ADM method and used by project management software.
- Better at showing different types of dependencies.



8.2.3 Estimate Activity Resources

- is a part of **Planning** process groups
- **Goals:** Identify what resources are needed (as well as the quantity) to complete tasks and produce deliverables.
- People doing the work should help create estimates, and an expert should review them
- Instead of providing activity estimates as a discrete number, such as four weeks, it's often helpful to create a three-point estimate

Three point Estimates

- It is an estimate that includes an optimistic, most likely, and pessimistic estimate, such as three weeks for the optimistic, four weeks for the most likely, and five weeks for the pessimistic estimate.

8.2.4 Estimate Activity Duration

- is a part of **Planning** process groups
- **Goals:** Figure out how long each task will take to complete and estimate the duration of the entire project timeline.
- is a challenging tasks that involves
 - Stable Activities : straightforward estimating
 - Dependent Activities : amount needed for testing is dependent on a successful test or unsuccessful test, and 3-point estimates or analogous work as well.
 - Uncertain Activities

Reserve Time

- A reserve time is a percentage of the project duration or a preset number of work periods that is usually added to the end of the project schedule.
- As the project moves forward, the reserve time can be reduced or eliminated as the project manager sees fit.
- Reserve time decisions should be documented.

8.2.5 Develop Schedule

- is a part of **Planning** process groups
- Results of the other time management processes to determine the start and end dates of the project.
- **Goals:** Using the above information, create a project schedule.
- Important tools and techniques include Gantt charts, critical path analysis, critical chain scheduling, and PERT analysis.
- To develop a schedule, one needs to
 - Define the activities (WBS),
 - Put them in order of how the work will be done (activity sequencing), and then
 - Estimate the duration of each activity (activity duration estimating).

8.2.5.1 Principles for Scheduling

- 1) Don't try too tight or flexible scheduling and don't commit to something that the team can't deliver.
- 2) Eliminate uncertainty whenever it arises. Perform risk analysis, update risk management plan and use it during scheduling as required.
- 3) Create a lot of backup plans to deal with changes.
- 4) Pick the right level of granularity for "activity duration".
- 5) Schedule for the unexpected. Events or conditions that Project Managers could not foresee will interrupt the normal flows of the project.

8.2.5.1 Milestones

- A milestone is a significant event that normally has no duration.
- It often takes several activities and a lot of work to complete a milestone.
- Milestones are useful tools for setting schedule goals and monitoring progress.
- Milestones should follow SMART (• Specific • Measurable • Assignable • Realistic • Time-framed) Criteria.

8.2.5.2 Critical Path Method (CPM)

- CPM is a network diagramming technique used to predict total project duration.
 - A critical path for a project is the series of activities that determines the earliest time by which the project can be completed.
 - Techniques for shortening schedules Crashing Activities and Fast tracking Activities
 - The critical path is the longest path through the network diagram and has the least amount of slack or float.
 - Slack or float is the amount of time an activity can be delayed without delaying the project finish date.
- 1) Total float (TF) :** Total float is the amount of time an activity can be delayed without delaying the project completion date.

Total Float or Float = Late Start – Early Start or Late Finish – Early Finish

- 2) Free float (FF) :** Free Float is the amount of time that an activity can be delayed without delaying the early start of any successor activity.

Free Float = Early Start (of successor) – Early Finish (of current) or

$FF = EF(\text{current}) - ES(\text{previous}) - \text{duration}$

Total Float is how much an activity can be delayed without affecting the project Finish date, **Free Float** is about how much an activity can be delayed without affecting its successor activity.

3) Independent Float (IdF) : Interfering Float is the maximum amount of time an activity can be delayed without delaying the early start of the succeeding activities and without being affected by the allowable delay of any predecessor activity.

Independent Float (INDF) = Earliest Successors' ES – Earliest Predecessors' LF – Activity's duration

4) Interfering float (IfF) : $IfF = \text{Total float} - \text{Free Float}$

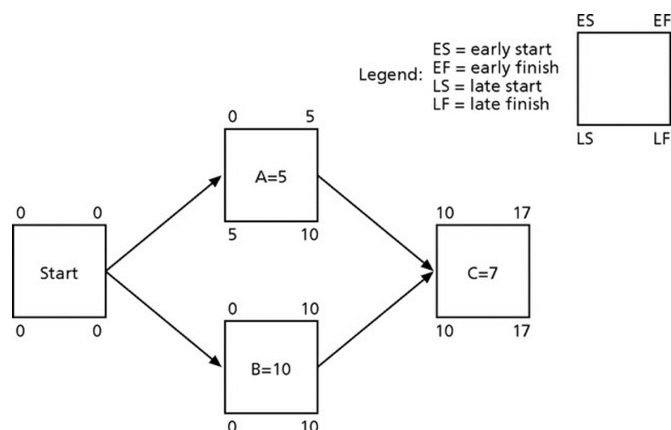
While making network diagram

1) Forward Pass :

- determines the early start and early finish dates for each activity
- Early finish = Early Start + Duration
- EF of predecessor = ES of successor
- Largest preceding EF at a merge point becomes ES for successor

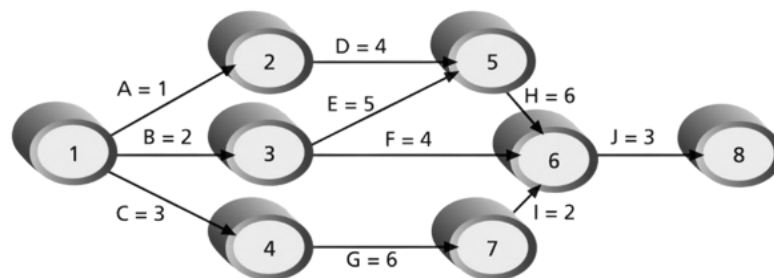
2) Backward pass :

- determines the late start and late finish dates
- Late start = late finish - Duration
- LS of successor = LF of predecessor
- Smallest succeeding LS at a burst point becomes LF for predecessor



Calculating CPM

- 1) Develop a good network diagram.
- 2) Add the duration estimates for all activities on each path through the network diagram.
- 3) The longest path is the critical path.
- 4) If one or more of the activities on the critical path takes longer than planned, the whole project schedule will slip unless the project manager takes corrective action.



Note: Assume all durations are in days.

Path 1:	A-D-H-J	Length = 1+4+6+3 = 14 days
Path 2:	B-E-H-J	Length = 2+5+6+3 = 16 days
Path 3:	B-F-J	Length = 2+4+3 = 9 days
Path 4:	C-G-I-J	Length = 3+6+2+3 = 14 days

Since the critical path is the longest path through the network diagram, Path 2, B-E-H-J, is the critical path for Project X.

Critical Chain Scheduling

- Scheduling method that considers limited resources
- Includes buffers to protect the project completion date
 - buffer is an additional time to complete a task
- Critical chain scheduling removes buffers from individual tasks and instead creates a project buffer & Feeding Buffer
 - Project Buffer: time added before the project's due date
 - Feeding Buffer: time added before tasks on critical chain

Monte Carlo Simulation

- simulates a model's outcome many times to provide a statistical distribution of the calculated results
- This can inform:
 - The probability of completing the project on any specific day
 - The probability of completing the project for any specific amount of cost

- The probability of any task actually being on the critical path
- The overall project risk

E.g Monte Carlo analysis can determine that a project will finish by a certain date only 10% of the time, and determine another date for which the project will finish 50% of the time.

8.2.5.3 PERT

- Stands for Program Evaluation and Review Technique
- It is a network analysis technique to estimate project duration when there is a high degree of uncertainty about the individual activity duration estimates
- It sets three time estimates for each activity with a view to overcome uncertainty in time estimates.
 - Optimistic time estimate
 - Pessimistic time estimate
 - Most likely time estimate

$$PERT \text{ weighted average} = \frac{\text{optimistic time} + 4 * \text{most likely time} + \text{pessimistic time}}{6}$$

1) Optimistic time estimate :

- minimum time required to complete the activity in ideal situation
- it is assumed that everything is in favorable conditions

2) Pessimistic time estimate :

- maximum time required to complete the activity in worst situation
- it is assumed that everything is in worst conditions

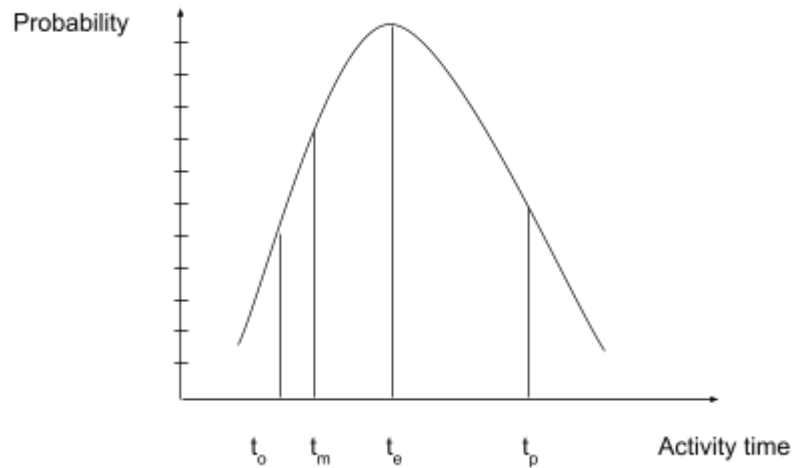
3) Most likely time estimate :

- time required to complete the activity in normal situations.
- it is assumed that conditions are neither favorable nor unfavorable but normal

Expected time estimate (t_e)

Expected time estimate can be determined by using formula :

$$t_e = \frac{t_o + 4 * t_m + t_p}{6}$$



The reliability of expected time(t_e) depends upon the variability of two time estimators (t_p) and (t_o).

The range for the time estimate is $t_p - t_o$.

There are two measures of variation of possible times. They are:

- 1) Variance
- 2) Standard Deviation

They can be calculated as:

$$SD (\sigma) = \frac{t_p - t_o}{6}$$

$$\text{Variance } (\sigma^2) = \left(\frac{t_p - t_o}{6} \right)^2$$

Differences between PERT and CPM

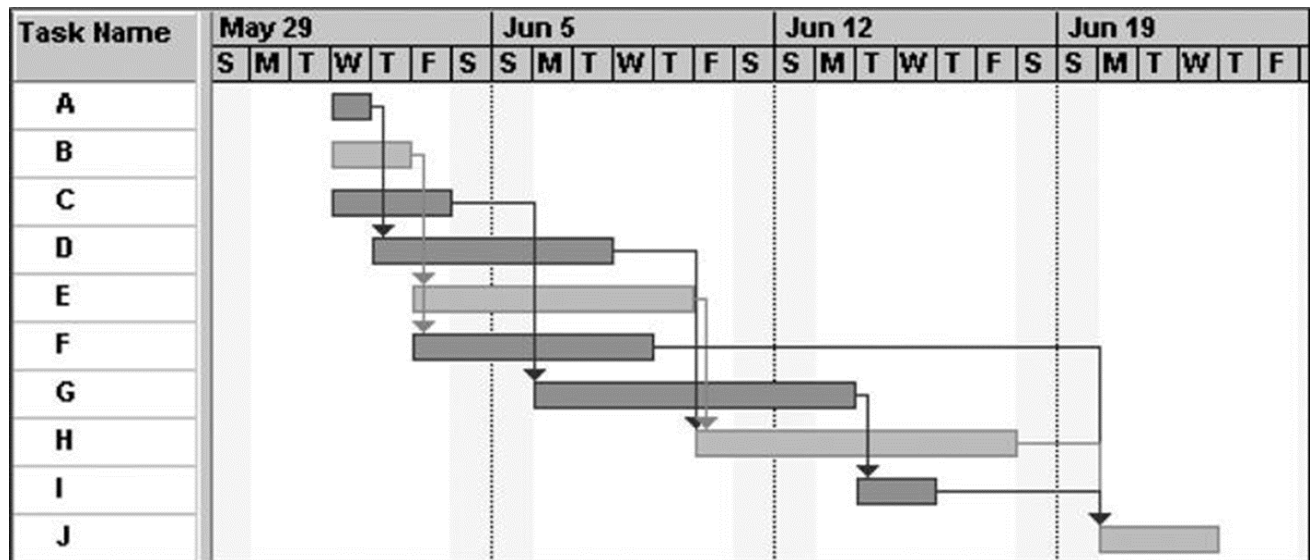
SN	PERT	CPM
1	PERT is that technique of project management which is used to manage uncertain activities of any project.	CPM is that technique of project management which is used to manage only certain activities of any project.
2	It is an event oriented technique i.e the network is constructed based on events.	It is an activity oriented technique.
3	It is a probability model.	It is a deterministic model.
4	It is appropriate for high precision time estimation.	It is appropriate for reasonable time estimation.
5	It has Non-repetitive nature.	It has a repetitive nature.
6	There is no chance of crashing as there is no certainty of time.	There may be crashes because of certain time boundation.
7	it doesn't use any dummy activities.	It uses dummy activities for representing a sequence of activities.
8	It is suitable for projects which require research and development.	It is suitable for construction projects.
9	Costly to maintain	Easy to maintain

8.2.5.4 Gantt charts

- Gantt charts
 - display project schedule information
 - shows project activities and Projects' start and finish dates in a calendar format
- Symbols include:
 - A black diamond: a milestones
 - Thick black bars: summary tasks
 - Lighter horizontal bars: durations of tasks
 - Arrows: dependencies between tasks

Adding Milestones to Gantt Charts

- Many people like to focus on meeting milestones, especially for large projects.
- Milestones emphasize important events or accomplishments in projects.
- You typically create milestones by entering tasks that have a zero duration, or you can mark any task as a milestone.



8.2.6 Control Schedule

- is a part of **Monitoring and Controlling** process groups
- **Goals:** Develop a strategy to mitigate risk and address changes to the project schedule when it's executed.
- Perform reality checks on schedules.

Reality Checks on Scheduling

- Review the draft schedule or estimated completion date in the project charter.
- Prepare a more detailed schedule with the project team.
- Make sure the schedule is realistic and followed.
- Alert top management well in advance if there are schedule problems.

Examples

Example 8.1

A project consist of 6 activities A, B, C, D, E, F with their time of completion as follows:

Activity	Duration (weeks)	❖ Activity A and B can be performed in parallel ❖ Activity C and D can not start until B is complete ❖ Activity E can not start until half the work of C is completed. ❖ Activity F can start after D is complete.
A	2	
B	4	
C	2	
D	4	
E	6	
F	4	

Solutions:

Activity / Week	1	2	3	4	5	6	7	8	9	10	11	12
A												
B												
C												
D												
E												
F												

Example 8.2

A project consists of 7 activities with their time required and precedence relationships as follows. Draw a bar chart and find out project completion time.

Activity	A	B	C	D	E	F	G
Preceding activity	-	-	A, B	A	C	A, B	F
Time (days)	40	20	15	25	15	25	10

Solutions:

Activity / Week	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85
A																	
B																	
C																	
D																	
E																	
F																	
G																	

NOTES: First see example 8.8

There are two methods for drawing networks.

- Arrow Network Diagram
- Precedence Network Diagram (Recommended because easy, see example : 8.8)

All the mentioned problems are copy pasted, so try to solve using PND using eg 8.8.

Example 8.3

SN	Activity	Predecessors	Successor
1	A	-	B, C, D
2	B	A	E
3	C	A	F
4	D	A	G
5	E	B	-
6	F	D	-
7	G	C	-

Draw an arrow network.

Solutions :

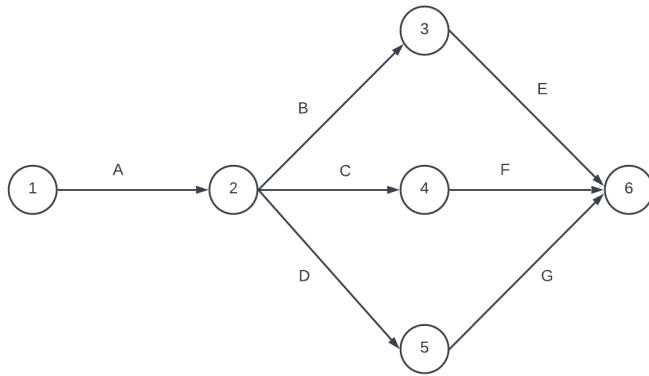
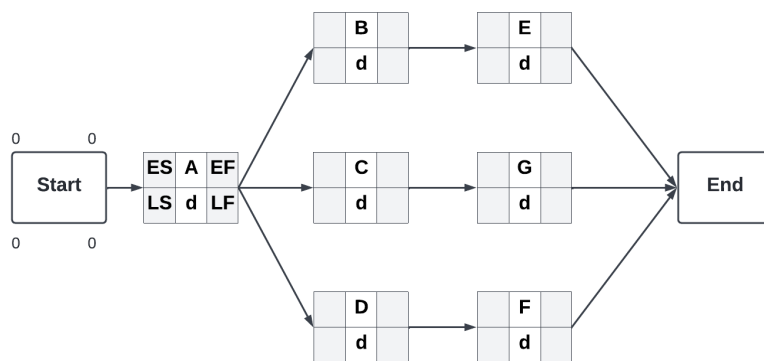


fig: arrow network diagram

Drawing precedence network diagram,



Example 8.4

Draw a network with the following details. Number the events using Fulkerson's rule.

SN	Activity	Predecessors
1	A	-
2	B	A
3	C	A
4	D	B
5	E	C
6	F	C, D
7	G	E, F

-----> represents dummy activities

Solutions :

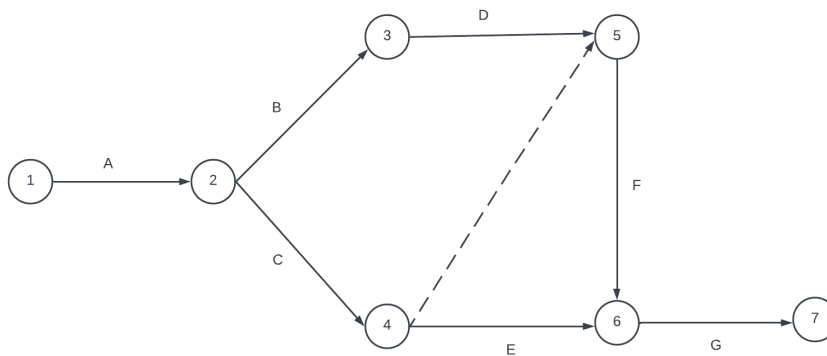


fig: arrow method

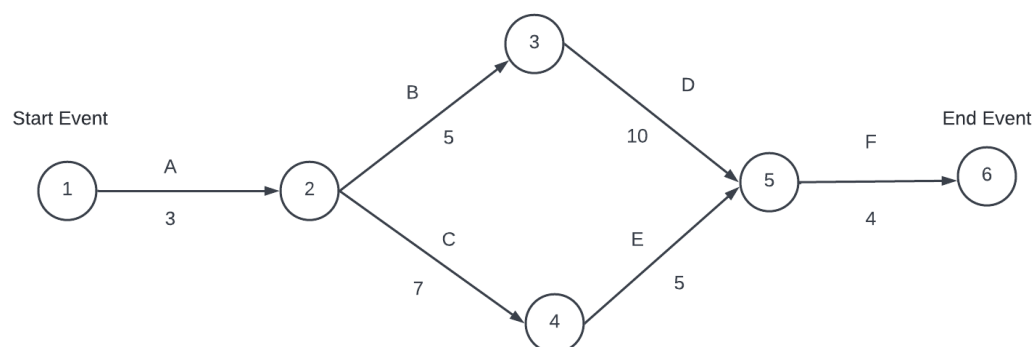
Example 8.5

Draw a network with the following details. Determine the critical path and project completion time.

Activity	Predecessor Activity	Duration (Weeks)
A	-	3
B	A	5
C	A	7
D	B	10
E	C	5
F	D,E	4

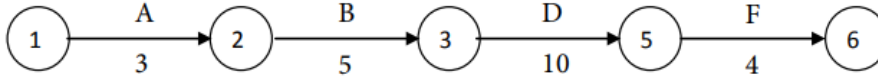
Solutions :

First let us construct the network diagram for the given project. We mark the time estimates along the arrows representing the activities. We obtain the following diagram:



Consider the paths, beginning with the start node and stopping with the end node.
There are two such paths for the given project. They are as follows:

Path I



with a time of $3 + 5 + 10 + 4 = 22$ weeks.

Path II



with a time of $3 + 7 + 5 + 4 = 19$ weeks.

Compare the times for the two paths. Maximum of $\{22, 19\} = 22$.

We see that path I has the maximum time of 22 weeks. Therefore, path I is the critical path. The critical activities are A, B, D and F. The project completion time is 22 weeks.

We notice that C and E are non-critical activities.

Time for path I - Time for path II = $22 - 19 = 3$ weeks.

Therefore, together the non-critical activities can be delayed upto a maximum of 3 weeks, without delaying the completion of the whole project.

Example 8.6

Draw a network with the following details. Determine the critical path and project completion time.

Activity	immediate Predecessors	Duration in days		
		Optimistic	Most Likely	Pessimistic
A	-	2	4	6
B	A	3	6	9
C	A	8	10	12
D	B	9	12	15
E	C	8	9	10
F	D, E	16	21	26
G	D, E	19	22	25
H	F	2	5	8
I	G	1	3	5

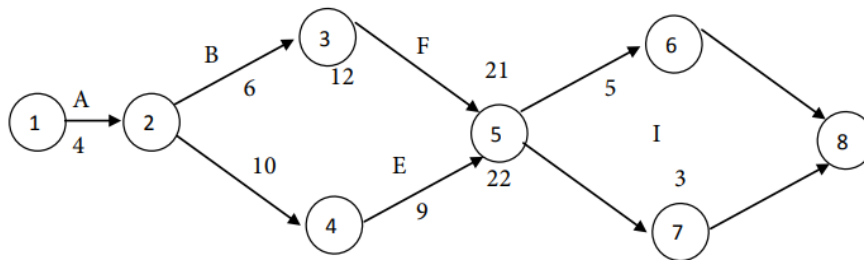
Solutions :

From the three time estimates t_p , t_m and t_o , calculate t_e for each activity. We obtain the following table:

Activity	immediate Predecessor	Duration in days			t_e
		Optimistic	Most Likely	Pessimistic	$\frac{t_o + 4 * t_m + t_p}{6}$
A	-	2	4	6	4
B	A	3	6	9	6
C	A	8	10	12	10
D	B	9	12	15	12
E	C	8	9	10	9
F	D, E	16	21	26	21

G	D, E	19	22	25	22
H	F	2	5	8	5
I	G	1	3	5	3

Using the single time estimates of the activities, we get the following network diagram for the project.



For path 1 - 2 - 3 - 5 - 6 - 8, Time for the path: $4+6+12+21+5 = 48$ days

For path 1 - 2 - 3 - 5 - 7 - 8, Time for the path: $4+6+12+6+3 = 31$ days

For path 1 - 2 - 4 - 5 - 6 - 8, Time for the path: $4+10+9+21+5 = 49$ days

For path 1 - 2 - 4 - 5 - 7 - 8, Time for the path: $4+10+9+ 6+3 = 32$ days.

- Maximum of {48, 31, 49, 32} = 49.

We see that Path III has the maximum time.

- Therefore the critical path is Path III. i.e., 1 2 4 5 6 8.
- The critical activities are A, C, E, F and H.
- The non-critical activities are B, D, G and I.
- Project time (Also called project length) = 49 days

For variance and standard deviation, calculate for each nodes, then

total variance = sum of all variance

standard deviation = $\sqrt{\text{total variance}}$

Example 8.7

A project consists of seven activities with the following time estimates. Find the probability that the project will be completed in 30 weeks or less.

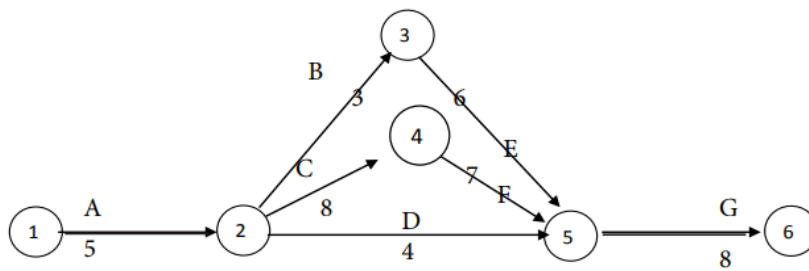
Activity	immediate Predecessors	Duration in days		
		Optimistic	Most Likely	Pessimistic
A	-	2	5	8
B	A	2	3	4
C	A	6	8	10
D	A	2	4	6
E	B	2	6	10
F	C	6	7	8
G	D, E, F	6	8	10

Solutions :

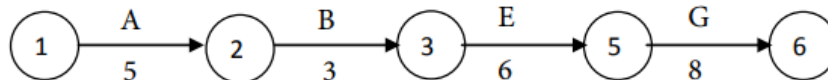
From the three time estimates , and , calculate for each activity. The results are furnished in the following table:

Activity	Optimistic time estimate (to)	4 x Most likely time estimate	Pessimistic time estimate (tp)	to+ 4tm + tp	Time estimate $t_e = \frac{t_o + 4t_m + t_p}{6}$
A	2	20	8	30	5
B	2	12	4	18	3
C	6	32	10	48	8
D	2	16	6	24	4
E	2	24	10	36	6
F	6	28	8	42	7
G	6	32	10	48	8

With the single time estimates of the activities, the following network diagram is constructed for the project.

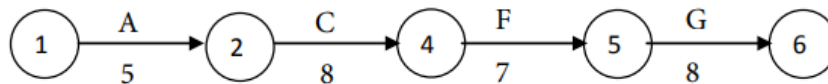


Path I



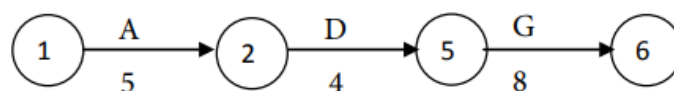
Time for the path: $5+3+6+8 = 22$ weeks.

Path II



Time for the path: $5+8+7+8 = 28$ weeks.

Path III



Time for the path: $5+4+8 = 17$ weeks.

Compare the times for the three paths. Maximum of {22, 28, 17} = 28.

It is noticed that Path II has the maximum time. Therefore the critical path is Path II. i.e., 1 2 4 5 6.

The critical activities are A, C, F and G. The non-critical activities are B, D and E. Project time = 28 weeks.

Calculation of Standard Deviation and Variance for the Critical Activities:

Critical Activity	Optimistic time estimate (to)	Most likely time estimate (tm)	Pessimistic time estimate (tp)	Range (tp - to)	Standard deviation = $\sigma = \frac{t_p - t_o}{6}$	Variance $\sigma^2 = \left(\frac{t_p - t_o}{6}\right)^2$
A: 1 2	2	5	8	6	1	1
C: 2 4	6	8	10	4	$\frac{2}{3}$	$\frac{4}{9}$
F: 4 5	6	7	8	2	$\frac{1}{3}$	$\frac{1}{9}$
G: 5 6	6	8	10	4	$\frac{2}{3}$	$\frac{4}{9}$

$$\text{Variance} = 1 + 4/9 + 1/9 + 4/9 = 2$$

$$\text{Standard deviation of the critical path} = \sqrt{\text{variance}} = \sqrt{2} = 1.414$$

Use Z Score formula i.e

$$Z = \frac{x - \mu}{\sigma}$$

For less than or equal to 30 weeks,

$$x = 30, \mu = \text{total time completion} = 28 \text{ and } \sigma = 1.414$$

Therefore, $Z = 1.41$, Corresponding to Z , probability is 0.4207 (selected Q)

Calculator trick for Casio fx-991EX, Classwiz

- 1) Press Menu
- 2) Goto Statistics (option 6)
- 3) Don't need any equation, so press AC.
- 4) Press OPTN.
- 5) Press the down button.
- 6) Press 4 and Select Q) or P) for -ve z and write P value. If you select Q, add 0.5 later.
- 7) You will get probability.

So, total probability = $0.5 + 0.4207 = 0.9207 = 92.07\%$. (always add 0.5 in probability)

i.e. There is a 92% chance that the project will be completed before 30 weeks. In other words, the chance that it will be delayed beyond 30 weeks is 8%

If $Z > \text{value}$, then

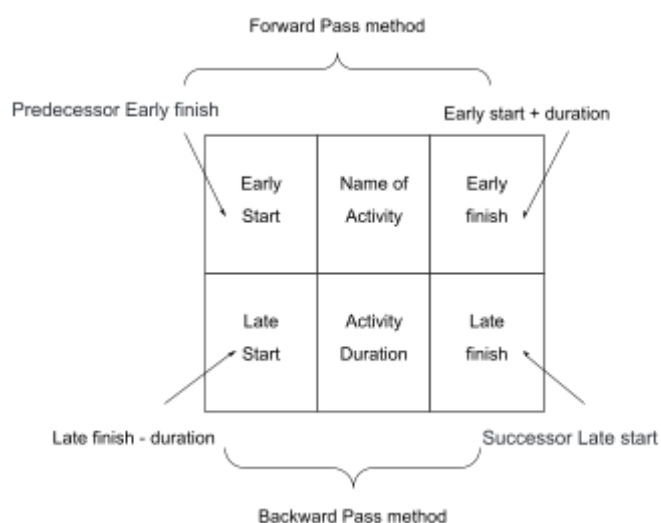
Example 8.8 (figure method)

Determine the earliest and latest times, the total float for each activity, the critical activities and the project completion time

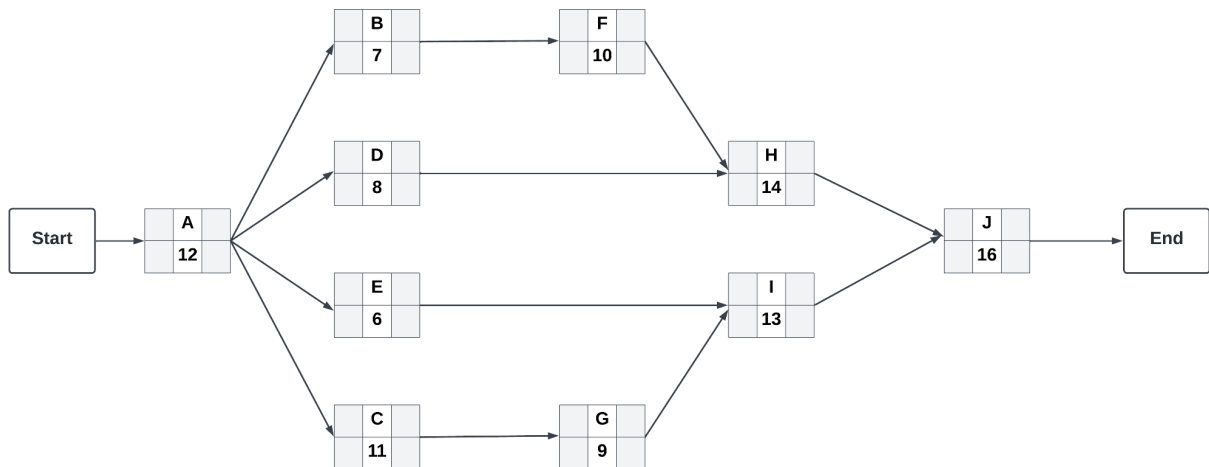
Activity	Predecessor Activity	Duration (Weeks)
A	-	12
B	A	7
C	A	11
D	A	8
E	A	6
F	B	10
G	C	9
H	D, F	14
I	E, G	13
J	H, I	16

Solutions:

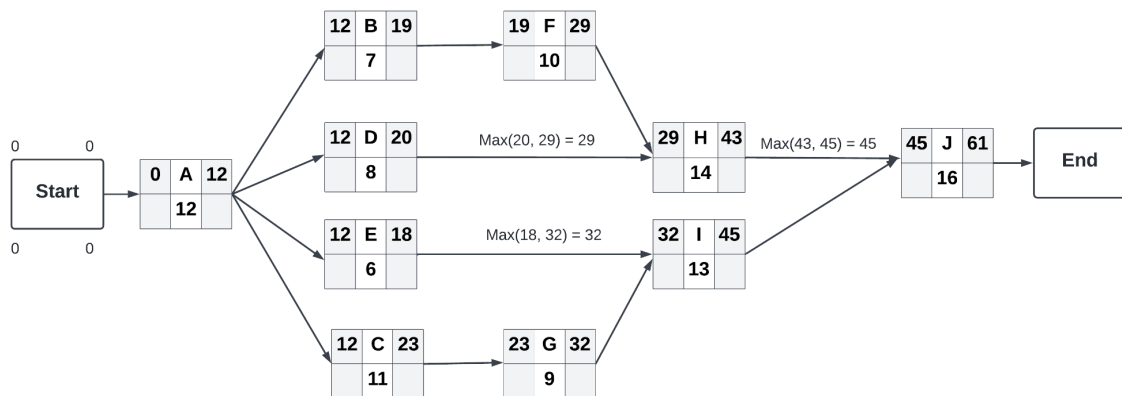
Note :



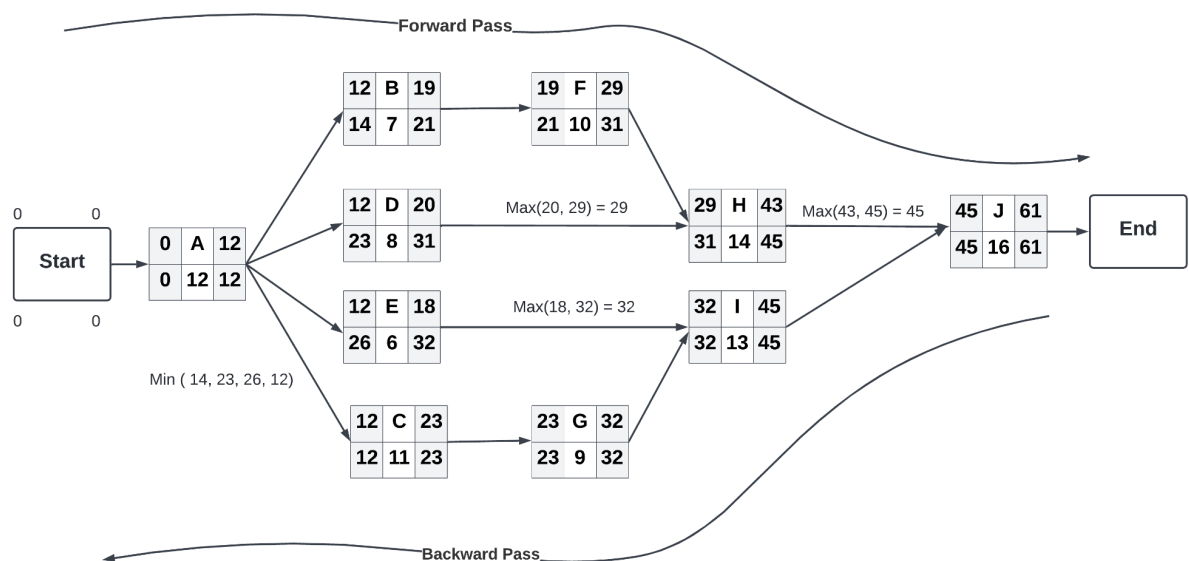
Drawing precedence network diagram, we get



Using forward pass, we get



Using backward pass, we get



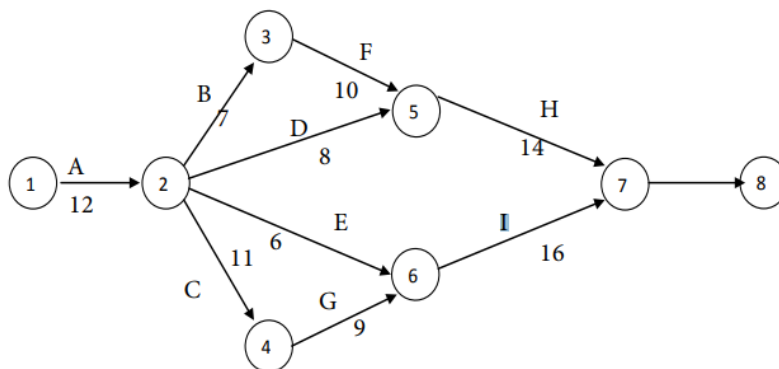
Task	A	B	C	D	E	F	G	H	I	J
Float	0	2	0	11	14	2	0	2	0	0

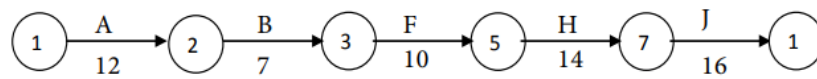
So path is A - C - G - I - J

Determine the earliest and latest times, the total float for each activity, the critical activities and the project completion time

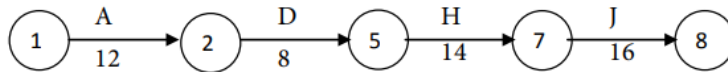
Activity	Predecessor Activity	Duration (Weeks)
A	-	12
B	A	7
C	A	11
D	A	8
E	A	6
F	B	10
G	C	9
H	D, F	14
I	E, G	13
J	H, I	16

With the given data, we construct the following network diagram for the project.

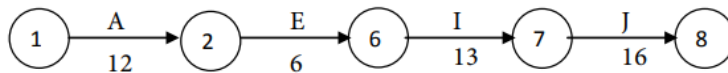


Path I

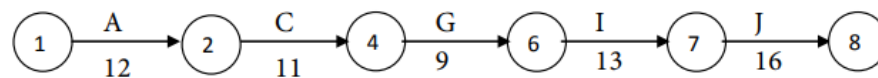
Time of the path = $12 + 7 + 10 + 14 + 16 = 59$ weeks.

Path II

Time of the path = $12 + 8 + 14 + 16 = 50$ weeks.

Path III

Time of the path = $12 + 6 + 13 + 16 = 47$ weeks.

Path IV

Time of the path = $12 + 11 + 9 + 13 + 16 = 61$ weeks.

Compare the times for the four paths. Maximum of $\{51, 50, 47, 61\} = 61$.

We see that the maximum time of a path is 61 weeks.

For forward pass

Node	Earliest Time of Occurrence of Node
1	0
2	Time for Node 1 + Time for Activity A = $0 + 12 = 12$
3	Time for Node 2 + Time for Activity B = $12 + 7 = 19$
4	Time for Node 2 + Time for Activity C = $12 + 11 = 23$
5	Max {Time for Node 2 + Time for Activity D, Time for Node 3 + Time for Activity F} = Max { $12 + 8, 19 + 10$ } = Max {20, 29} = 29
6	Max {Time for Node 2 + Time for Activity E, Time for Node 4 + Time for Activity G} = Max { $12 + 6, 23 + 9$ } = Max {18, 32} = 32
7	Max {Time for Node 5 + Time for Activity H, Time for Node 6 + Time for Activity I} = Max { $29 + 14, 32 + 13$ } = Max {43, 45} = 45
8	Time for Node 7 + Time for Activity J = $45 + 16 = 61$

Using the above values, we obtain the Earliest Start Times of the activities as follows:

Activity	Earliest Start Time (Weeks)
A	0
B	12
C	12
D	12
E	12
F	19
G	23
H	29
I	32
J	45

For backward pass

Node	Latest Allowable Time of Occurrence of Node
8	Maximum time of a path in the network = 61
7	Time for Node 8 - Time for Activity J = $61 - 16 = 45$
6	Time for Node 7 - Time for Activity I = $45 - 13 = 32$
5	Time for Node 7 - Time for Activity H = $45 - 14 = 31$
4	Time for Node 6 - Time for Activity G = $32 - 9 = 23$
3	Time for Node 5 - Time for Activity F = $31 - 10 = 21$

2	Min {Time for Node 3 - Time for Activity B, Time for Node 4 - Time for Activity C, Time for Node 5 - Time for Activity D, Time for Node 6 - Time for Activity E} = Min {21 - 7, 23 - 11, 31 - 8, 32 - 6} = Min {14, 12, 23, 26} = 12
1	Time for Node 2 - Time for Activity A = $12 - 12 = 0$

Using the above values, we obtain the Latest Finish Times of the activities as follows:

Activity	Latest Finish Time (Weeks)
J	61
I	45
H	45
G	32
F	31
E	32
D	31
C	23
B	21
A	12

Calculation of Total Float for each activity:

Activity	Duration (Weeks)	Earliest Start Time	Earliest Finish Time	Latest Start Time	Latest Finish Time	Total Float = Latest Finish Time - Earliest Finish Time
A	12	0	12	0	12	0
B	7	12	19	14	21	2
C	11	12	23	12	23	0
D	8	12	20	23	31	11
E	6	12	18	26	32	14
F	10	19	29	21	31	2
G	9	23	32	23	32	0
H	14	29	43	31	45	2
I	13	32	45	32	45	0
J	16	45	61	45	61	0

The activities with total float = 0 are A, C, G, I and J. They are the critical activities.

Project completion time = 61 weeks.

Chapter 9:

Project Cost Management(7 Marks)

9.1 Need of Project Cost Management

- IT projects have a poor track record for meeting budget goals
- A 2011 Harvard Business Review study says that:
 - an average cost overrun of 27%.

9.2 Project Cost Management Basics

- Cost management is the process of planning and managing the budget of a business or project.
- In the case of a project, it helps the project manager to estimate what the project will cost and set controls to reduce the chances of the project going over budget.

9.2.1 Cost

- Costs refers to a resource spent in order to accomplish a particular goal.
- Costs are usually measured in monetary units like Rupees, Dollar.

Types of cost

- 1) Direct Cost : Cost of labor, Resource, Machinery
- 2) Overhead Cost : Cost of building, rent, Maintenance, Insurance, Vacation
- 3) Administrative Cost : Cost due to Management Expenses, HRM & HRD activities

9.2.2 Project Cost

Project Cost is the cost of running a project. It primarily involves labor cost, resource cost and special machinery cost.

$$\text{Project Cost} = \text{Project's Labor Cost} + \text{Resource Cost} + \text{Special Machinery Cost}$$

If a company wants to launch a new project then it must allocate human resources, machineries and other resources to it. The overall cost of these materials is project cost.

9.2.3 Product Cost

Product cost is the cost involved in manufacturing a product. It involves various costs such as Material Cost, Manufacturing Cost. and Overhead Cost. It is also known as sales price.

Material Cost + Manufacturing Cost + Overhead Cost = Sales Price, where

Overhead Cost = Project other overhead costs + Project Cost + Profit

9.3 Basic principles of Project Cost Management

- 1) Profits = revenues - expenses
- 2) Profit margins are the ratio of profits to revenue.
- 3) Life Cycle costing is the total development cost plus support costs for a project.
- 4) Cash flow analysis is determining the estimated annual costs and benefits for a project
- 5) Benefits and costs can be tangible or intangible
- 6) Sunk cost should not be a criteria in project costing.

A cost management plan includes:

- Level of accuracy
- Units of measure
- Organizational procedure links
- Control thresholds
- Rules of performance measurement
- Reporting formats
- Process descriptions

9.4 How to Manage Project Costs/ Cost Management Processes

- 1) Resource Planning
- 2) Cost Estimating
- 3) Cost Budgeting
- 4) Cost Control

9.4.1 Resource Planning

- Resource planning is the process of predicting future resource requirements for a business, project, or scope of work.
- To create a resource plan, we need to start by defining the project scope, a document that details the project activities that will be done.
- Once the project activities have been defined, project managers usually rely on historical data, expert opinions, and resource planning tools to estimate the resources that will be needed.

9.4.2 Cost Estimating

- Cost estimating is the process of creating an estimate for the price of the resources required to finish the project's operations.
- Cost estimating also includes identifying and considering cost alternatives.
- Cost estimating process is a part of "Project Planning Phase"

Types of Estimates	When Done	Why Done	How Accurate
Rough order of Magnitude (ROM)	Very early in the project cycle, often 3-5 years before project completion	Provides rough ballpark of cost for selecting decisions	-25% to +75%
Budgetary	Early, 1-2 years cost	Puts dollars in the budget plans	-10% to +25%
Definitive	Later in the project < 1 year out	Provides details for purchases, estimate actual costs	-5% to +10%

Cost Estimation Tools & Techniques

Basic tools and techniques for cost estimates are:

- 1) Top-down estimates (Analogous estimates)
 - is based on top down approach
 - makes comparison with other projects
 - is less accurate than other tools
- 2) Bottom-up estimates
 - It is activity-based costing
 - also based on WBS
 - more accurate than other tools
 - Creates a detailed estimate of each work components
 - Would be better for controlling for cost

3) Three-point estimates

- involve estimating the most likely, optimistic, and pessimistic costs for items.
- use formula like PERT weighted average (discussed in previous chapter)

4) Parametric estimating

- Mathematical modeling tool for cost estimation
- Similar to analogous top-down
- Better to be used with historical info

5) Vector bid analysis

- This method estimates the cost based on the amount of bids the vendor makes.

6) Reserve Analysis

- used to determine if the amount of remaining reserve left is adequate
- Compares the amount of risk remaining at any time in project

Typical Problems with IT Cost Estimates

- Estimates are done too quickly
- People lack estimating experience
- Human beings are biased toward underestimation
- Management desires accuracy

Contingency reserve is often used to reduce the risk from these problems

9.4.3 Cost Budgeting

- Cost budgeting is the process of allocating the project cost estimate to individual work items over time
- The WBS is a required input to the cost budgeting process since it defines the work items
- Based on your cost estimates, you can now create a project budget, which is simply the sum of all your project costs.
- Once the project starts, the project budget is a baseline that's used to compare actual costs vs. estimated costs.
- Therefore, project budgets allow project managers to quickly understand if their costs are too high and if there's a risk of cost overrun.

Principles of cost budgeting

- 1) Plan for low cost for relatively less important items
- 2) Estimate separately for different categories of events and functions
- 3) Estimate separately for aggregated task or activities
- 4) Estimate separately for emerging cost to deal with contingency budget or reserver budget
- 5) Maintain “Promise Low, Deliver High” phrases.

Cost Aggregation

- The act of combining all of our individual cost estimates into a single, total number that represents your project's spending is known as cost aggregation.
- Benefits of cost aggregation
 - View costs across different partners, countries, and platforms
 - Defeat fragmented cost data
 - Give a clear and correct picture of ROI
 - Give a clear and correct picture of ROI

Cost baseline

- approved budget (broken into a list of salaries, materials, etc.)
- used to measure and monitor cost performance.
- Inputs for cost baseline
 - WBS
 - Task Schedule
 - Cost Estimates
 - Risk Management Plan

9.4.4 Cost Control

- Project cost control includes
 - Monitoring cost performance
 - Ensuring that only appropriate project changes are included in a revised cost baseline
 - Informing project stakeholders of authorized changes to the project that will affect costs
- Many organizations around the globe have problems with cost control

9.4.4.1 Cost Control Measures

- Influencing the factor that causes changes to the cost baseline
- Ensuring all requested changes are agreed upon
- Managing the actual changes when and as they occur
- Assuring that potential cost overruns do not exceed the authorized funding periodically or in total for the project
- Monitoring cost performance to detect and understand variances from the cost baseline
- Recording all appropriate changes accurately against the cost baseline.
- Acting to bring expected cost overruns within acceptable limits

9.4.4.2 Cost Control Methods

1) Cost Change Control System

- It defines the procedures by which the cost baseline can be changed, usually documented in the cost management plan.
- Integrated with the Integrated Change Control

2) Performance Evaluation Analysis : Earned Value Management (EVM)

9.4.4.3 Earned Value Management (EVM)

Earned value management (EVM) is a project management methodology that integrates schedule, costs, and scope to measure project performance. We must provide actual information periodically to use EVM.

Earned Value Analysis is an approach for measuring how much work has been completed in a project at a given point of time and performance.

EVM is based on:

1) Planned value (PV) : Budgeted cost of work schedule (BCWS)

- authorized budget assigned to scheduled work.

$PV = \text{Total project cost} * \% \text{ of planned work}$

PV for the complete project for 5 months = \$25,000

PV at 2 months = $\$25,000 * 40\% = \$10,000$

2) Actual cost (AC): actual cost of work performed (ACWP)

- real cost associated with the work done on an activity

let's assume, AC at the end of 2 months = \$15,000

3) **Earned value (EV): Budgeted cost of work performed (BCWP)**

- measure of work performed expressed in terms of the budget authorized for that work. EV can't be greater than the authorized PV budget
let's say you managed to just finish 30% of work.

$$EV = \text{Total project cost} * \% \text{ of actual work} = \$25,000 * 30\% = \$7,500$$

Benefits of EVM

1. EVM is an Excellent Measure of Progress
2. EVM Enables Accurate Forecasting
3. EVM Supports Management by Exception
4. EVM Promotes Good Project Management Disciplines
5. EVM is Cost Effective

Terms to understand

In order to identify variances, first of all we should understand the following basic terms:

1. Budgeted cost of work scheduled (BCWS)
2. Budgeted cost of work performed (BCWP), which is also called Earned Value
3. Actual cost of work performed (ACWP)

Cost Variance

Cost variance (CV) is the difference of Budgeted Cost of Work Performed (earned value) minus Actual Cost of Work Performed. It can be expressed as:

$$CV = BCWP - ACWP \text{ where negative variance indicates cost overrun}$$

Schedule Variance

Schedule variance (SV) is the difference of Budgeted Cost of Work Performed (earned value) minus Budgeted Cost of Work Schedule. It can be expressed as:

$$SV = BCWP - BCWS \text{ where negative variance indicates cost overrun}$$

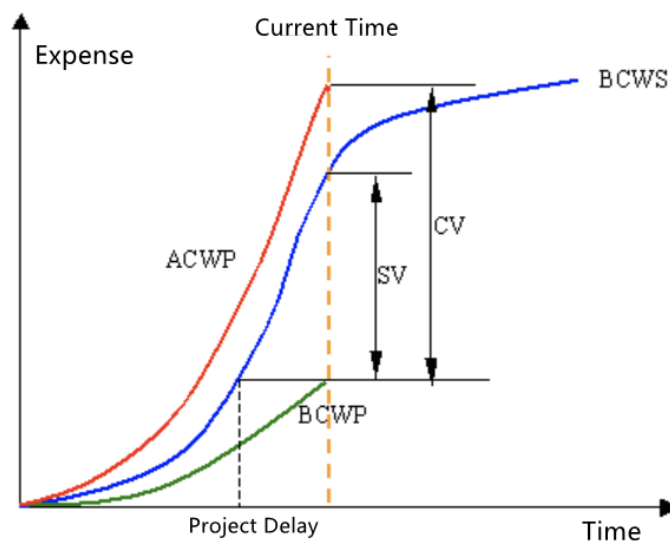
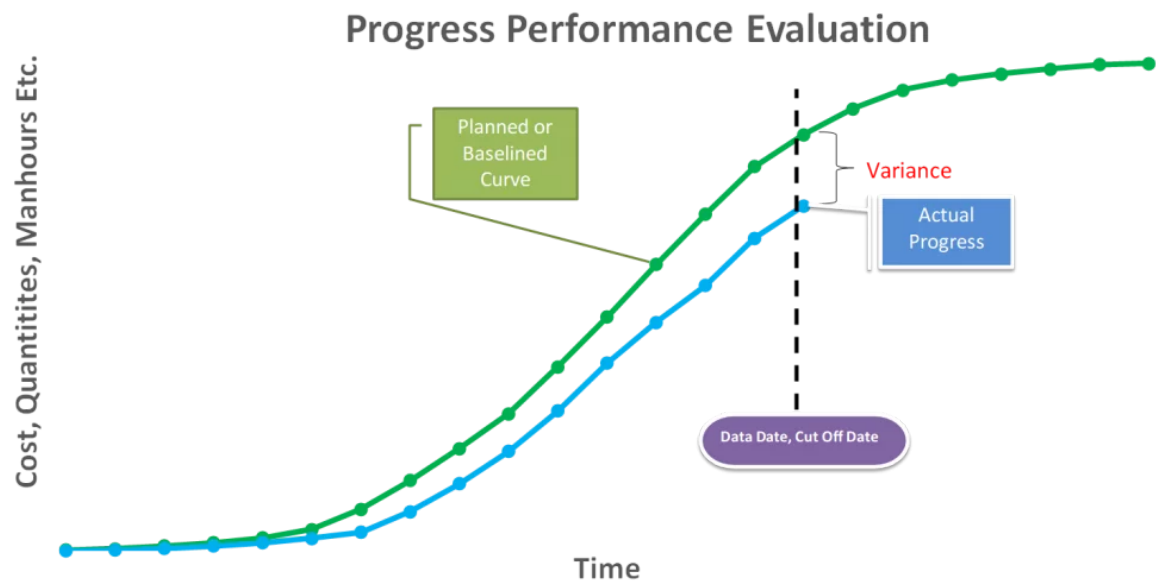
Rate of performance (RP)

Rate of performance is the ratio of actual work completed to the percentage of work planned to have been completed.

$$\text{Rate of Performance} = \frac{\text{Actual Work Completed}}{\% \text{ of work planned}}$$

S-Curve

- mathematical graph of Project cost vs. time
- effective tool in project control
- The start represents a slow, purposeful but growing acceleration, and the finish represents a slowing as the work runs out.



Cost & Schedule Performance

1) Cost Performance

Cost performance can be obtained by dividing earned value (BCWP) by Actual Cost of Work Performed (ACWP), which can be expressed as:

$$\text{Cost performance} = \frac{BCWP}{ACWP}$$

2) Schedule Performance

Schedule performance can be obtained by dividing earned value (BCWP) by Budgeted Cost of Work Schedule, which can be expressed as:

$$\text{Schedule performance} = \frac{BCWP}{BCWS}$$

9.5 Examples

Example 9.5.1

A project was scheduled for the time of 20 days . The estimated cost of the project is Rs 500,000. At the end of 5th day , evaluation is done and it is identified that 25% of work was accomplished but Rs 130,000 cost has been incurred (used). Now calculate the cost and duration required for completing the remaining works?

Solution:

a) Budget Cost of work Schedule (BCWS)

Planned expenditure in 20 days = Rs 500,000, then

Planned expenditure in 1 days = Rs 25,000

Planned expenditure in 5 days = Rs 125,000

b) Budget Cost of Work Performed (BCWP)

Work Completed = 25% = 0.25 work

Let 1 works = Rs 500,000, then

0.25 works = 0.25 * Rs 500,000 = Rs 125,000

c) Actual Cost of Work Performed (ACWP)

Total Expenses at the end of 5th day (ACWP) = Rs 130,000

- i) Schedule Variance (SV) = BCWP - BCWS = 0, so the project is not over or under the budget.

$$\text{Schedule Performance Index (SPI)} = \frac{BCWP}{BCWS} = 1$$

$$\text{Revised Schedule (Duration)} = \frac{\text{Original Estimate}}{SPI} = \frac{20}{1} = 20 \text{ days}$$

- ii) Cost Variance (CV) = BCWP - ACWP = 125,000 - 130,000 = -5000, it means budget overrun.

$$\text{Cost Performance Index (CPI)} = \frac{BCWP}{ACWP} = \frac{125,000}{130,000} = 0.96 < 1$$

< 1 means budget overrun and vice versa.

$$\text{Revised Cost} = \frac{\text{Original Cost}}{CPI} = \frac{500,000}{0.96} = \text{Rs } 520,000$$

Example 9.5.2

When 125 m³ of concrete was to be done in 10 days at the cost of Rs 1,250,000 at the end of 3rd day, it managed to complete 40m³ of concrete with expenses of Rs 375,000. Find EVM and comment on the performance.

Solutions:

- a) Budget Cost of work Schedule (BCWS)

Planned expenditure in 10 days = Rs 1,250,000, then

Planned expenditure in 1 days = Rs 125,000

Planned expenditure in 3 days = Rs 375,000

- b) Budget Cost of Work Performed (BCWP)

$$\text{Work Completed} = \frac{40}{125} = 0.32$$

Let 1 works = Rs 500,000, then

$$0.32 \text{ works} = 0.32 * \text{Rs } 1,250,000 = \text{Rs } 400,000$$

c) Actual Cost of Work Performed (ACWP)

Total Expenses at the end of 3rd day (ACWP) = Rs 375,000

- iii) Schedule Variance (SV) = BCWP - BCWS = 400,000 - 375,000 = 25000, under budget

Schedule Performance Index (SPI) = $\frac{BCWP}{BCWS} = \frac{400,000}{375,000} = 1.067 > 1$, ahead the schedule

$$\text{Revised Schedule (Duration)} = \frac{\text{Original Estimate}}{SPI} = \frac{10}{1.067} = 9.36 \text{ days}$$

- iv) Cost Variance (CV) = BCWP - ACWP = 400,000 - 375,000 = 25,000, it means under budget.

$$\text{Cost Performance Index (CPI)} = \frac{BCWP}{ACWP} = \frac{400,000}{375,000} = 1.067 > 1$$

> 1 means under budget and vice versa.

$$\text{Revised Cost} = \frac{\text{Original Cost}}{CPI} = \frac{1,250,000}{1.067} = \text{Rs } 1,171,509$$

Example 9.5.3

50 units of plantation have to be done in a two weeks period. Per unit cost of plantation is estimated as Rs. 200 of which progress monitoring was done one week after the work was started. Only 40 % work was found completed and the account record showed that the actual expenditure (cost) for plantation per unit was Rs. 250.

Solution:

Here,

$$BCWS = 25 * 200 = \text{Rs. } 5000$$

$$BCWP = 20 * 200 = \text{Rs. } 4000$$

$$ACWP = 20 * 250 = \text{Rs. } 5000$$

Now,

$$\text{Cost variance (CV)} = BCWP - ACWP = 4000 - 5000 = - 1000 \text{ (Indicating cost overrun)}$$

Schedule variance (SV) = BCWP – BCWS = 4000 – 5000 = - 1000 (Indicating time overrun)

Number of the key parameters used in variance analysis is the same as BCWP.

Earned value is a forecasting variable used to predict whether the project will finish over or under the budget.

Cost performance = BCWP/ACWP

= 4000/5000 = 0.80

Schedule performance = BCWP/BCWS

= 4000/5000 = 0.80

Since the cost performance is 0.80, the final cost would be (50 units * Rs 200)/0.80 = Rs. 12500 (instead of estimated Rs. 10,000)

Since the schedule performance is 0.80, the time it requires for completion would be:
2 weeks/0.80 = 2.5 weeks (instead of scheduled time 2 weeks)

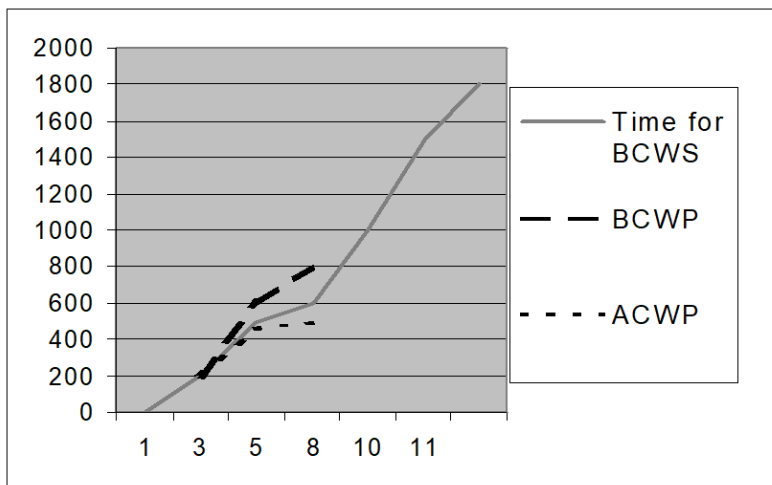
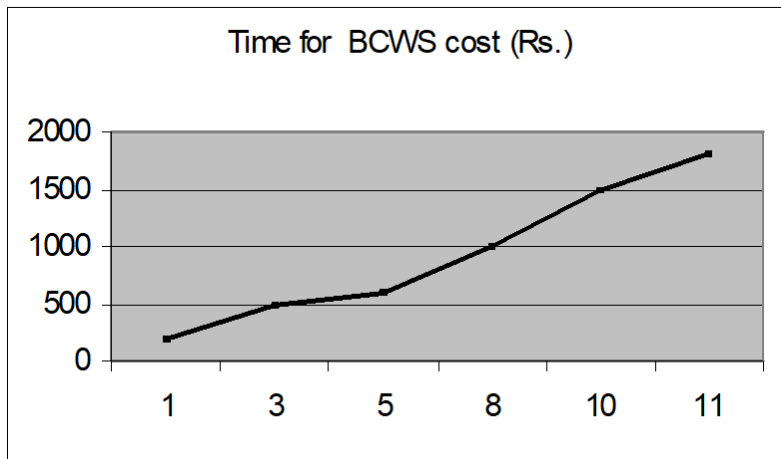
We can graphically present both the cost and schedule variances by using S – Curve, which is also used as one of the effective tools in project control.

Example 9.5.4

Sketch the S Curve of following.

Activity	Cost (Rs.)	Duration (Week)	BCWS (Rs.)	BCWP (Rs.)	ACWP (Rs.)
A	200	1	200	200	200
B	300	3	500	600	450
C	100	5	600	800	500
D	400	8	1000		
E	500	10	1500		
F	300	11	1800		

Solution:



Chapter 10:

Project Quality Management (5 Marks)

10.1 Quality

Quality refers to how good something is compared to other similar things. In other words, its degree of excellence.

$$\text{Quality} = \frac{\text{Performance or Result}}{\text{Expectations or Requirements}}$$

Common misconceptions:

- Quality is difficult to define, but you can recognize it when you see it
- Quality is expensive
- Quality is craftsmanship
- Quality is luxury
- Quality is in short supply

10.1.1 3 C's of Quality

- 1) Commitment
- 2) Competence
- 3) Communication

Project quality includes all activities of the overall management function that determine the

- quality policy, objectives, and responsibilities and implement them by means such as
- planning, quality control, quality assurance and quality improvement, within the quality system.

10.1.2 Lean Production Theory

- Lean production is an approach to management that focuses on cutting out waste, whilst ensuring quality.
- Waste brings cost.
- This approach can be applied to all aspects of a business – from design, through production to distribution

Principles of Lean Production

- 1) **Value** : Before you start identifying and eliminating waste, you need to define what's valuable for the customer. Then remove all the unnecessary work and components associated with it.
- 2) **Process improvement** as removing unnecessary work leads to improved production lead time
- 3) **Map the value stream** : This principle involves recording and analyzing the flow of information or materials required to produce a specific product or service with the intent of identifying waste and methods of improvement.
- 4) **Create flow** : Eliminate functional barriers and identify ways to improve lead time
- 5) Pursue perfection with continual process improvement, or Kaizen.

10.1.3 KAIZEN Theory

- Kaizen is a Japanese word that literally means improvement.
- Kaizen means improvement, but more than just that, it means **continual improvement**.
- In terms of business, kaizen means that all activities must be continuously improving. That means everything in the organization, from employees to the CEO.
- The idea of always making small improvements is similar to lean manufacturing, especially with the emphasis of eliminating waste.
- Kaizen, like a lean or agile approach, works with iterations. It keeps companies flexible.

Principles of Kaizen

- 1) **Know Your Custom**: Companies should identify their customer's interests to enhance their experience.
- 2) **Let it flow** : This applies to the target of achieving zero waste.
- 3) Follow **Leadership** and knowing what is happening at every level of the organization
- 4) **Motivate People**
- 5) **Be Transparent**: Performance and improvements must be tracked with real data

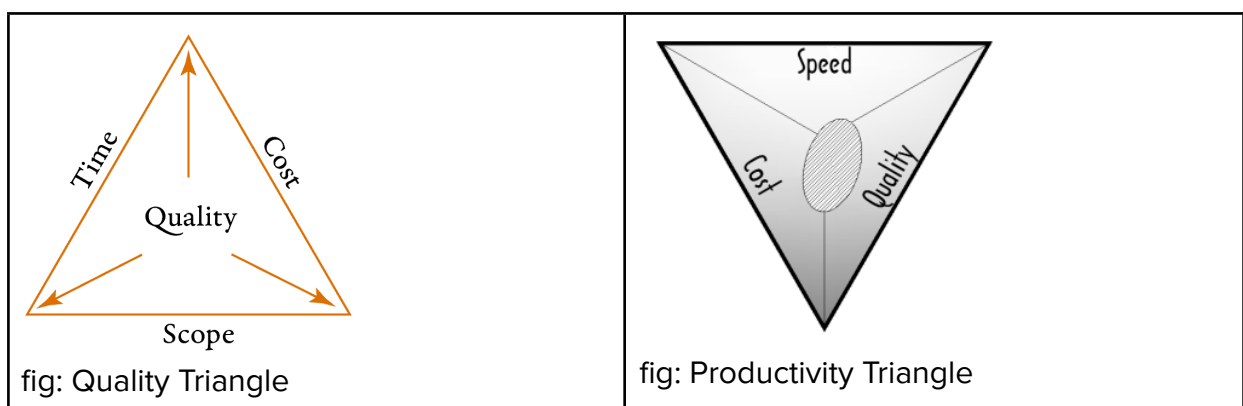
10.1.4 Just in Time (JIT) Production

- The main principle of JIT is to focus on eliminating all waste.
- By waste, it means all those tasks which do not contribute any value to the activity you are doing.
- Then being efficient means eliminating those non-value-adding activities.
- There are various types of waste:
 - Waste from overproduction
 - Wasting time
 - Transportation waste
 - Processing waste
 - Inventory level waste
 - Waste in motion
 - Waste from product defects
- JIT is about continuous improvement and addressing anything that doesn't add value to the project.

Steps for JIT Production

- 1) Build Trust With Suppliers
- 2) Have Strong Communication. Manufacturers need to strengthen their internal communications.
- 3) Work in Small Lots for better quality
- 4) Efficient Workflows to handle our production flow
- 5) Backward Scheduling

10.1.5 Quality vs Productivity

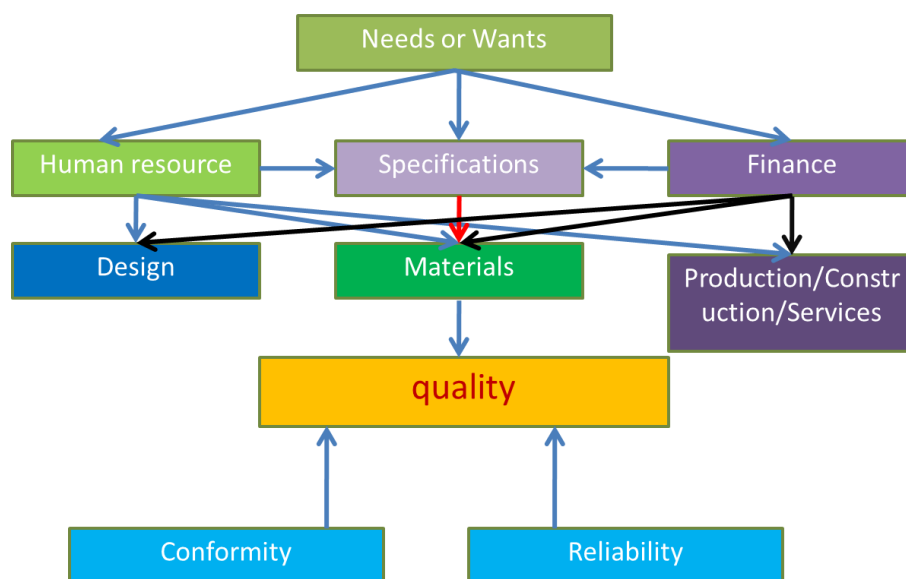


Quality and Productivity are different aspects of a manufacturing cycle. Productivity means the amount of product versus the cost of making it. While Quality means that the product is fit for the purpose.

The faster you make a product to increase risk, the higher the risk of poor quality. While, quality can be determined by the Call Center regarding customer services.

Higher Quality goods take more time and labour and higher quality raw material to produce. So a drive to greater productivity almost always means a cut of quality.

10.2 Quality Determinants



10.2.1 Quality Standards

Quality management standards are details of requirements, specifications, guidelines and characteristics that products, services and processes should consistently meet in order to ensure:

- their quality matches expectations
- they are fit for purpose
- they meet the needs of their users

Standards are an essential element of quality management systems.

Examples of quality management standards

ISO international standards are by far the most widely accepted set of quality standards in the world. ISO 9001:2015 specifies the requirements for a quality management system that businesses can use to develop their own quality agenda.

Other standards related to quality include:

- the remaining ISO 9000 family, including ISO 9000 and ISO 9004
- the ISO 14000 family for environmental management systems
- ISO 13485 for medical devices
- ISO 19011 for auditing management systems
- ISO/TS 16949 for automotive-related products

10.3 Project Quality Management

Project quality management ensures that the project will satisfy the needs for which it was undertaken. The activities related to quality management are as follows:

1. Quality assurance
2. Quality planning
3. Quality control

10.3.1 Quality Planning

- Quality planning is the process of identifying which quality standards are applicable to the project and determining how to satisfy them.
- Expected Level of Quality can only be achieved through necessary quality planning during project initiation
- It involves planning how to fulfill product quality requirements.
- It has following principles:
 - a. Quality is determined by the requirements of customers.
 - b. It is better to avoid mistakes than to repair the defects.
 - c. Cost of quality must be approved by management.

10.3.1.1 Project Quality Requirements

- Quality requirements should be a part of project scope definition.
- Quality requirements should be optimized for cost and schedule estimates.
- It defines the
 - customer's expectations for quality,
 - internal attributes that indicate quality factors being satisfied or not and
 - measures to be used to give visibility to the levels of quality being achieved.
- - It is defined in terms of quality factors, quality criteria and quality metrics.

Attributes of Quality Requirements

- 1) Completeness Criteria
- 2) Correctness Criteria
- 3) Usefulness Criteria

Project Quality requirements in IT Projects

- 1) **Functionality** : degree to which system can performs its intended functions
- 2) **Features** : special characteristics of a system that appeal to users
- 3) **System Outputs** : Important to define what the screens look like
- 4) **Performance** : Addresses how well a product or services performs
- 5) **Reliability** : Ability of products to perform as expected under normal conditions
- 6) **Maintainability** : addresses the easiness of maintaining product

10.3.1.2 Cost of Quality

Cost of quality or COQ is defined as estimation for prevention, detection and elimination of defects to improve product quality. Cost of quality includes the cost of conformance and non-conformance.

- **Conformance** : cost for providing products that meet requirements and fitness for use
- **Non-conformance** : cost for taking responsibility for failures or not meeting quality expectations

The cost of quality is generally classified into four categories.

- 1) **External failure cost** : A cost that relates to all errors not detected and corrected before delivery to the customer
- 2) **Internal failure cost**: This cost is the cost incurred to correct an identified defect before the customer receives the product.
- 3) **Inspection cost** : This is the cost of evaluating processes and their outputs to ensure that a project is error-free
- 4) **Prevention cost** : This is the cost of planning and executing a project so that it is error-free

10.3.1.3 Quality Management Plan

A Quality Management plan should describe how the project management team will implement its quality policy. The quality management plan should address following attributes:

- 1) Responsibility of management
- 2) Document management
- 3) Requirement scope
- 4) Design control
- 5) Testing and quality assurance
- 6) Quality audits
- 7) Training requirements

10.3.2 Quality Assurance

- It is a process that is used to ensure the quality of products or services.
- A constant effort is made to enhance quality practices in the organization, so that continuous improvements are expected.
- A dedicated quality assurance team is formed.
- The QA team has following responsibilities:
 - 1) Define a process and procedure to achieve and improve quality.
 - 2) The team should go through formal industrial training.
 - 3) Ensure traceability of test cases to requirements.
- QA is performed through benchmarking and quality audits.

10.3.2.1 Quality Audit Process

- Quality Audit is a structured review of process management and product development activities.
- Goals: To identify lessons learned that can improve performance of project within performing organization
- Quality audits are an important tool to help organizations grow and prosper.
- They provide mechanisms to evaluate the efficiency of the business.
- Quality Audits may be post project review, internal project process review, external regulator, internal group audits.

Approach to quality audit

- 1) **Planning** : Planning begins with the preparation of a plan, which will guide the execution of the audit.
- 2) **Preparation**: This is the moment when the auditors can learn more about the company's QMS, taking a closer look at the system documentation.
- 3) **Execution** : The execution of audits takes place with the collection of information, which determines quality standards.
- 4) **Conclusion and Follow up**: Once the audit is completed, the "real" work begins. The audit team meets to review problem areas and to determine recommendations for correcting quality problems.

10.3.3 Quality Control

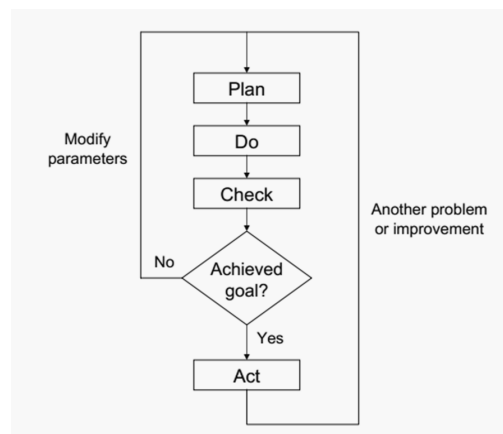
- It is the process of monitoring specific project results to determine whether it complies with relevant quality standards and identifying ways to eliminate causes of unsatisfactory performance.
- The main outputs of quality control are:
 - a. Acceptance decisions
 - b. Rework
 - c. Process adjustments

- **Seven Basic Tools of Quality Control**

- a. Flowchart
- b. Run Chart
- c. Scatter Diagram
- d. Pareto Diagram
- e. Control Chart
- f. Histogram
- g. Cause and Effect Diagram :

10.3.3.1 Flowchart

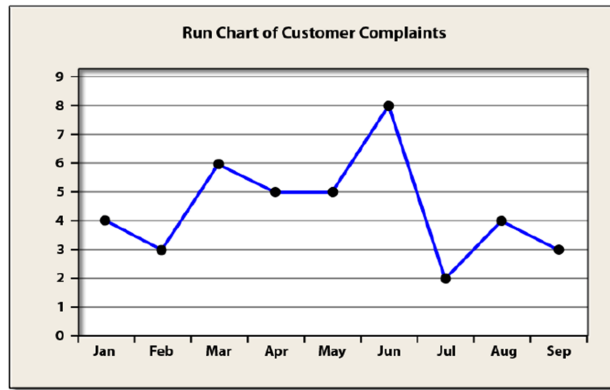
A flowchart is a picture of the separate steps of a process in sequential order. It represents a workflow or process. Flowcharts show the activities, decision points, branching loops, parallel paths, and the overall order of processing



10.3.3.2 Runchart

A run chart is a line graph of data plotted over time. It displays the history and pattern of variation of process over time. By collecting and charting data over time, you can find trends or patterns in the process.

Because they do not use control limits, run charts cannot tell you if a process is stable. However, they can show you how the process is running. The run chart can be a valuable tool at the beginning of a project, as it reveals important information about a process before you have collected enough data to create reliable control limits.

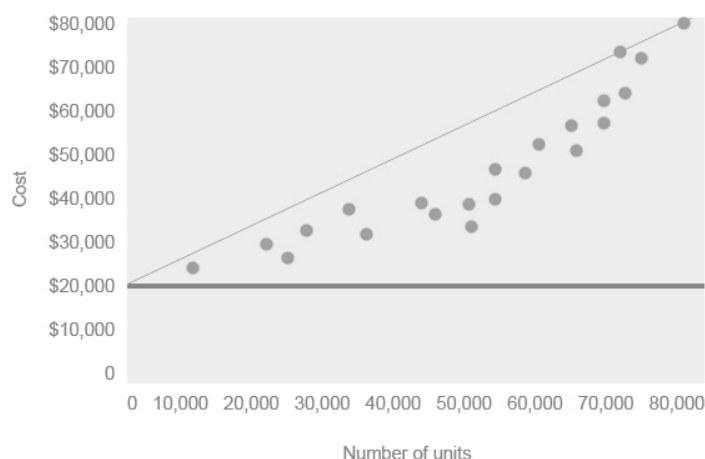


10.3.3.3 Scatter Diagram

It helps to show if there is a relationship between two variables. A scatter diagram (Also known as scatter plot, scatter graph, and correlation chart) is a tool for analyzing relationships between two variables for determining how closely the two variables are related.

One variable is plotted on the horizontal axis and the other is plotted on the vertical axis. The pattern of their intersecting points can graphically show relationship patterns.

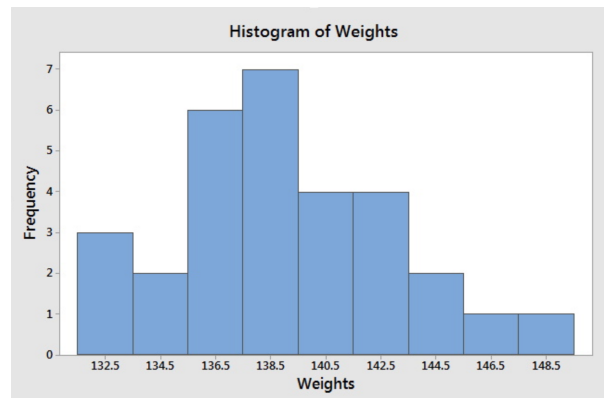
Most often a scatter diagram is used to prove or disprove cause-and-effect relationships. While the diagram shows relationships, it does not by itself prove that one variable causes the other. Thus, we can use a scatter diagram to examine theories about cause-and-effect relationships and to search for root causes of an identified problem.



10.3.3.4 Histogram

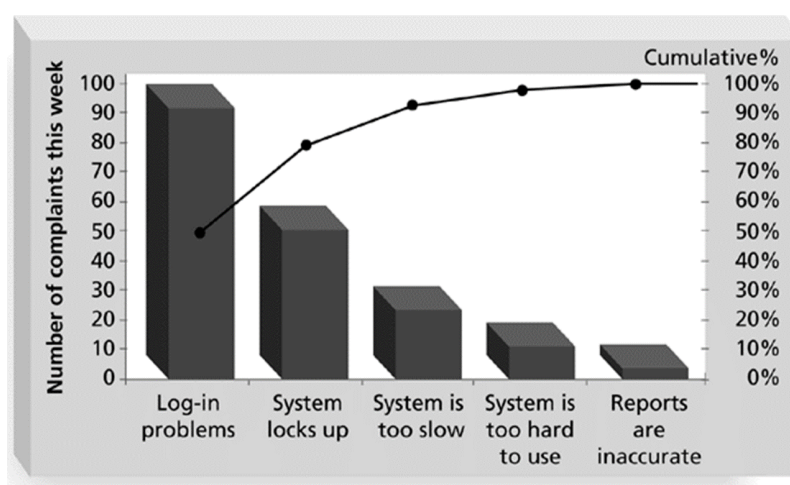
a histogram is a graphical representation of the distribution of data. The histogram is represented by a set of rectangles, adjacent to each other, where each bar represent a kind of data

It is bar graph of a distribution of graph.



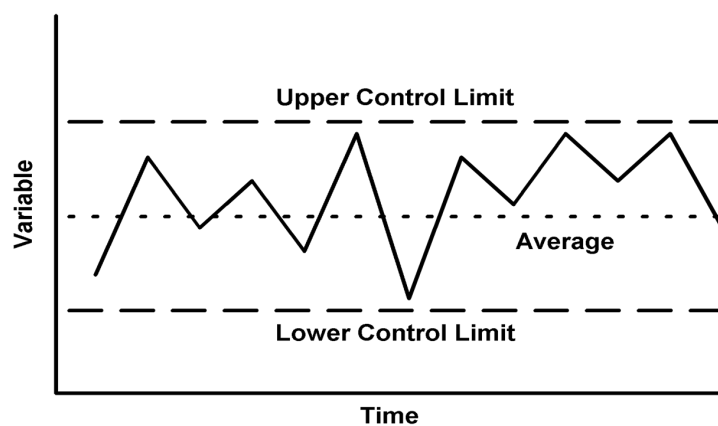
10.3.3.5 Pareto Diagram

- It is a chart containing both bar graph and line graph.
- Individual values are represented in descending order by bars.
- Cumulative total is represented by a line.
- It is used to highlight the common sources of defects, highest occurring type of defect and so on.
- It's a problem solving tool to help you decide where to focus your efforts.



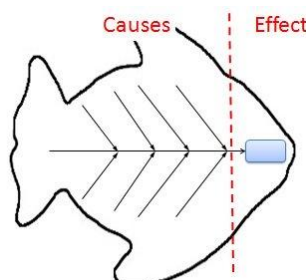
10.3.3.6 Control Chart

- A control chart is a graphic display of data that illustrates the results of a process over time.
- The main use of control charts is to prevent defects, rather than to detect or reject them.
- Quality control charts allow you to determine whether a process is in control or out of control.
 - 1) When a process is in control, any variations in the results of the process are created by random events
 - processes that are in control do not need to be adjusted.
 - 2) When a process is out of control, variations in the results of the process are caused by non-random events;
 - need to identify the causes of those non-random events and adjust the process to correct or eliminate them.



10.3.3.7 Cause and Effect Diagram

A fishbone diagram is a tool that can help you perform a cause and effect analysis for a problem you are trying to solve. This type of analysis enables you to discover the root cause of a problem. This tool is also called a cause and effect diagram or Ishikawa



The left side of the diagram is where the **causes** are listed. The causes are broken out into major cause categories.

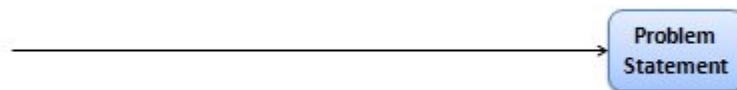
The right side of the diagram lists the **effect**. The effect is written as the **problem statement** for which you are trying to identify the causes.

How to Create a Cause and Effect Diagram

A cause and effect diagram can be created in six steps...

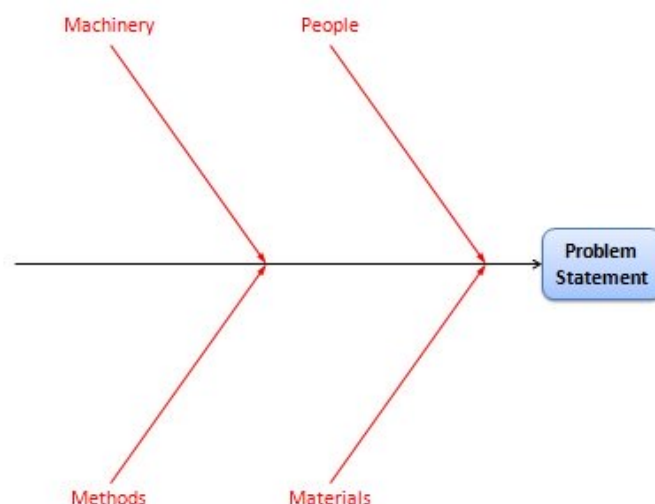
1) **Draw Problem Statement :**

- The first step of any problem solving activity is to define the problem.
- Once your problem statement is ready, write it in the box on the right hand side of the diagram.



2) **Draw Major Cause Categories :**

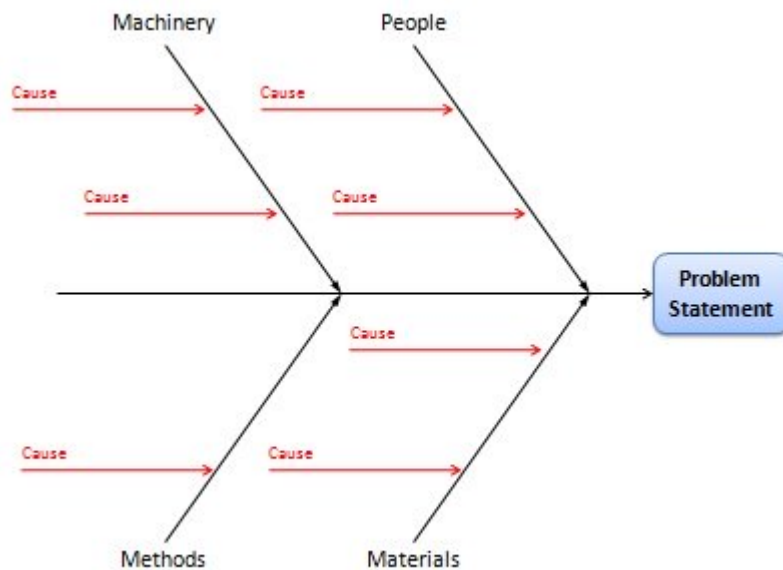
- After the problem statement has been placed on the diagram, draw the major cause categories on the left hand side and connect them to the "backbone" of the fishbone chart.



3) **Brainstorm Causes :**

4) **Categorize Causes :**

- Once your list of causes has been generated, you can start to place them in the appropriate category on the diagram.



5) **Determine Deeper Causes :**

6) **Identify Root Causes :**

- The final step for creating a fishbone diagram is to identify the root causes of the problem. This can be done in several ways...
 - Look for causes that appear repeatedly
 - Select using group consensus methods
 - Select based on frequency of occurrence

10.4 Questions and Answers

10.4.1 The Seven Run Rule

- Seven run rule states that if seven data points in a row are all below the mean, above the mean, or are all increasing or decreasing, then the process needs to be examined for non-random problems.
- It is useful for checking data patterns.
- Checking data patterns can be done by using quality control charts and the seven run rule.

10.4.2 CheckSheets

- used to collect and analyze data
- This is a generic data collection and analysis tool that can be adapted for a wide variety of purposes and is considered one of the seven basic quality tools.

Telephone Interruptions						
Reason	Day					
	Mon	Tues	Wed	Thurs	Fri	Total
Wrong number	+++			++	+++	20
Info request						10
Boss	+++		+++			19
Total	12	6	10	8	13	49

10.4.3 Total Quality Management (TQM)

- It is an approach for project quality management that takes everything related to quality under consideration.
- In TQM, the processes that produce services are thoroughly managed minimizing process variation and providing predictable quality level.
- In a TQM effort, all members of an organization participate in improving processes, products, services, and the culture in which they work

The principles used in TQM are as follows:

1. Top management
2. Training needs
3. Customer orientation
4. Involvement of employees
5. Corporate culture
6. Tools and techniques
7. Continuous improvements

The tools and techniques used for TQM are as follows:

- 1) Pareto chart
- 2) Scatter plot
- 3) Control chart

- 4) Fishbone and Ishikawa diagram (cause and effect)
- 5) Histogram
- 6) Check list
- 7) Check sheets
- 8) Assessment of customer need
- 9) Market analysis
- 10) Competition analysis
- 11) Workflow analysis
- 12) Logistic analysis

10.4.4 Possible steps to improve the quality of IT projects.

- 1) Define quality to match user needs.
- 2) Broadcast simple quality metrics.
- 3) Fine tune goals to include quality.
- 4) Get the requirements right.
- 5) Test smarter to test less.
- 6) Design applications to reduce bug risks.
- 7) Optimize the use of testing tools
- 8) Use Maturity Model such as PMMM, CMM

10.4.5 Testing Of IT Systems

- 1) Unit Test – To test each individual component to ensure they are defect free
- 2) Web Test – Series of HTTP request for testing websites
- 3) Integration Testing – Occurs between unit and system testing to test functionally grouped components
- 4) System Testing – Tests entire system as one entity
- 5) Load Testing – Used for stress testing for various load setting, network type and client configurations
- 6) User Acceptance Testing – Performed by the end user prior to accepting the delivered system

10.4.4 Software Testing Life Cycle

1. Requirement Analysis
2. Test Planning
3. Test Case Development
4. Environment Setup
5. Test Execution
6. Test Cycle Closure

10.4.7 Maturity Model

- Maturity models are frameworks for helping organization improve their processes and systems

1) Capability Maturity Model

- benchmark to measure the maturity of organization's software process
- process improvement approach that is based on a process model.



2) Project Management Maturity Model

- (PMMM) is a formal tool developed by PM Solutions and used to measure an organization's project management maturity
- Once the initial level of maturity and areas for improvement are identified, the PMMM provides
 - a roadmap, outlining the necessary steps for maturity advancement and performance improvement
- Maturity model is used to map logical ways to improve an organization's service

Five Maturity Level

1) Initial

- At maturity level 1, processes are usually ad hoc and chaotic.
- The organization usually does not provide a stable environment.
- Maturity level 1 organizations often produce products and services that work; however, they frequently exceed the budget and schedule of their projects.

2) Managed

- The projects of the organization have ensured that requirements are managed and that processes are planned, performed, measured, and controlled.
- At maturity level 2, requirements, processes, work products, and services are managed

3) Defined

- At maturity level 3, an organization has achieved all the specific and generic goals of the process areas assigned to maturity levels 2 and 3.
- At maturity level 3, processes are well characterized and understood, and are described in standards, procedures, tools, and methods.

4) Quantitatively Managed

- At maturity level 4, an organization has achieved all the specific goals of the process areas assigned to maturity levels 2, 3, and 4 and the generic goals assigned to maturity levels 2 and 3.
- At maturity level 4 Subprocesses are selected that significantly contribute to overall process performance.
- These selected subprocesses are controlled using statistical and other quantitative techniques.

5) Optimizing

- At maturity level 5, an organization has achieved all the specific goals of the process areas assigned to maturity levels 2, 3, 4, and 5 and the generic goals assigned to maturity levels 2 and 3.
- Processes are continually improved based on a quantitative understanding of the common causes of variation inherent in processes.

Chapter 11:

Project Communication Management (5 Marks)

11.1 Project Communication Management

- Project communication management is a collection of processes that help make sure the right messages are sent, received, and understood by the right people.
- Project communication management is one of the 10 key knowledge areas in the PMBOK (Project Management Book of Knowledge).
- Time to communicate is to be invested in the project.
- Everyone must understand how communications affect the project as a whole.

11.1.1 Importance of Good Communications

- 1) The greatest threat to many projects is a failure to communicate properly.
- 2) Culture of IT professionals could not make them good communicators.
- 3) Research shows that IT Professionals must be able to effectively communicate to succeed in their positions.
- 4) Strong verbal skills being a key factor in career advancement for IT Professionals.
- 5) Project managers must understand the importance of good communications as a means of achieving project success.

11.1.2 Keys to Good Communications

- 80-90% of a Project Manager's time is spent communicating on:
 - Focus on needs - both group and individual
 - Mix methods - formal and informal
 - Set the stage for communicating bad news
- Distribute important information in an effective and timely manner
- Determine the number of communication channels
 - As the number of people involved (n) increases, the complexity of communications increases because there are more communications channels through which people can communicate.

11.1.3 Communication Technology Determination

Factors contributing to determining the communication technology to be used:

- Availability
- Project environment
- Project Length
- Urgency
- Preparation Level

11.1.4 Communication Methods

- Various methods such as
 - Face to Face
 - Telephone Calls
 - Electronic Mails and other electronic tools
 - Radio Broadcasting
 - Social Media Networking
 - Paper Advertisement
- Based on the need and experience any method can be used

11.1.5 Importance of Face-to-Face Communication

Face-to-face communication is the distinction of being able to see the other party or parties in a conversation.

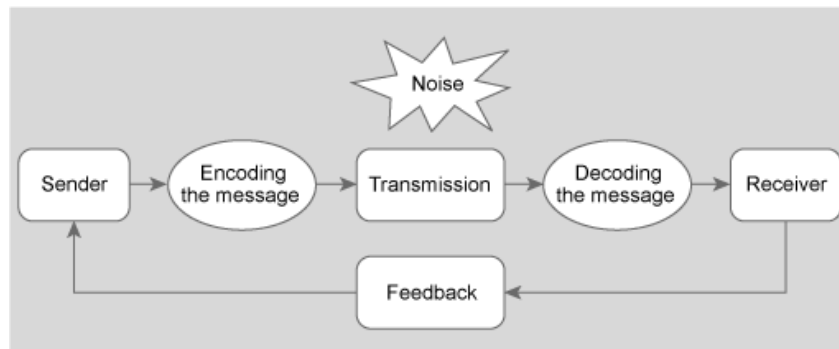
In face to face communication, Communication needs to be adjusted depending on the channel. Short, frequent meetings are often very effective in IT projects.

Benefits are:

- Keep remote workers in the loop
- Boost overall productivity
- Easier to convince people.
- Improve communication reliability
- Improve value and reduce redundancy
- Stronger Connections.
- Quicker and more effective.
- Better non-verbal understanding.

11.1.6 Communication Model

- 1) Encode - Translate thought to language
- 2) Message - Output of encoding
- 3) Medium - Method to convey message
- 4) Noise - Interference with transmission/understanding
- 5) Decode - Translation back to thought/idea



11.2 Project Communications Management Processes

- 1) **Communications planning:** determines the information and communications needs of the stakeholders
- 2) **Information distribution:** making needed information available in a timely manner
- 3) **Performance reporting:** collecting and disseminating performance information
- 4) **Administrative closure:** generating, gathering, and circulating information to formalize phase or project completion

11.2.1 Communications Planning

- Communication Planning determines the information and communications needs for the stakeholders
- Every project should include some type of communications management plan, a document that guides project communications
- Creating a stakeholder analysis for project communications also aids in communications planning
- It includes to identify the following:
 - Who needs what information,
 - When will they need it, and
 - How will it be given to them

Communication Planning Overview			
INPUTS		TOOLS & TECHNIQUES	OUTPUT
1.	Enterprise Environmental Factors		
2.	Organizational Process Assets	1. Communication Requirement Analysis	1. Communication Management Plan
3.	Project Scope Statement	2. Communications Technologies	
4.	Project Management Plan		
5.	Constraints		
6.	Assumptions		

Communication Requirement Analysis

- The total of the project stakeholders' requirements is the outcome of this study.
- Project managers should consider the number of potential communication channels as an indicator of the complexity of a projects' communication.
- Total number of channels = $n(n-1)/2$
 - Eg: 2 people need 1 communication channel, 3 need 3, and 4 need 6

Information required to determine communication requirements:

- Organizational Charts
- Project organization and stakeholders responsibility relation
- Disciplines, departments, specialties
- Logistics and personals
- Internal/Externals Information needs
- Stakeholders information
- Project Scope and Project Management Plans

11.2.2 Information Distribution

- Needed information are made available to project stakeholders in a timely manner
- It includes implementation of a communication management plan, as well as responding to unexpected requests of information.
- "Getting the right information to the right people at the right time and in useful format" is very important.
- Important considerations include
 - using technology to enhance information distribution
 - formal and informal methods for distributing information

- Formal means project report and informal means face to face communication

INPUTS	TOOLS & TECHNIQUES	OUTPUT
1. Communication Management Plan	1. Communication Skills 2. Information Gathering and Retrieval System 3. Information Distribution Methods 4. Lesson Learned Process	1. Updated Organizational Process Assets 2. Requested Changes

Distribution Techniques: written and verbal

11.2.3 Performance Reporting

- Collecting and Circulating Performance Information
- Performance reporting keeps stakeholders informed about how resources are being used to achieve project objectives
- Includes:
 - Latest status reports, progress reports and forecasting
- Performance report common types: Gantt Charts, S-Curve, Histograms, tables, etc.
- The work performance measures communicated can have impacts throughout the knowledge areas including scope, time, cost and quality

Performance Reporting Overview		
INPUTS	TOOLS & TECHNIQUES	OUTPUT
1. Work Performance Info 2. Performance Measurement 3. Forecasted Competition 4. Quality Control Measurement 5. PM Plan 6. Performance Measurement Baseline 7. Approved Change Request 8. Deliverables	1. Information Presentation Tools 2. Performance Information Gathering and Compilation 3. Status Review Meeting 4. Time Reporting Systems 5. Cost Reporting Systems	1. Performance Reports 2. Forecasts 3. Recommended Corrective Actions 4. Updated Organizational Process Assets 5. Requested Changes

11.2.4 Administrative closure

- generating, gathering, and circulating information to formalize phase or project completion

11.3 Running Effective Meeting

Before organizing a meeting, think about it.

- Do you really need a meeting?
- Who needs to attend?
- How much time do you need?

Steps to conduct Effective Meeting

- 1) Determine if a meeting can be avoided
- 2) Define the purpose and intended outcome of the meeting
- 3) Determine who should attend the meeting
- 4) Provide an agenda to participants before the meeting
- 5) Prepare handouts, and slides ahead of time
- 6) Start the meeting on time
- 7) Run the meeting professionally
- 8) Build relationships with attendees

Using Email Effectively

- 1) Be sure to send the email to the right people
- 2) Use meaningful subjects
- 3) Limit the content to one main subject, and be as clear and concise as possible
- 4) Limit the number and size of attachments
- 5) Delete email you don't need, and don't open it if you question the source
- 6) Make sure your virus software is up to date
- 7) Respond to and file emails quickly
- 8) Learn how to use important features

11.4 Communication Management Plan

It is an outcome plan of the project management plan. It provides

- Stakeholders communication requirements
- Information to be communicated
- Person responsible for communicating the information
- Person or groups who will receive information
- Frequency of the communication such as weekly
- A project schedule for producing information
- Access methods for obtaining information

11.5 Integrated Reporting System

- Huge time is invested in gathering, evaluating, formatting and disseminating status information.
- Most of the work is of Repetitive and Recurring type of work. That's why an integrated reporting system was developed.
- Integrated Reporting System will collect data during operation, store them in a relational database and provide them in a ready to analysis manner whenever required
- An integrated reporting system tool has a lot of importance in this context.

11.6 Manage Stakeholders Process

- Refers to the management of communications to satisfy the needs of, and resolve issues with, project stakeholders.
- Project Manager is responsible for managing stakeholders.\

Manage Stakeholders Overview				
INPUTS		TOOLS & TECHNIQUES		OUTPUT
1.	Communication Management Plan	1.	Communication Methods	1. Resolved Issues
		2.	Issue Logs	2. Approved Change Requests
2.	Organizational Process Assets			3. Approved Corrective Actions
				4. Updated Organizational Process Assets
				5. Updated PM Plan

11.7 Lessons Learned Process

- Lessons learned must be used in future
- Project status review meeting and Project Post Mortem meeting helps in discussing lessons learned
- Will be More fruitful if project could not fulfill the expectations

Results from lesson learned:

- Updated lesson learned knowledge base
- Input to KMS
- Updated policies, procedures and processes
- Improved business skills
- Overall Product and service improvements
- Updated risk management plan

11.8 Traceability Matrix

- It ties together requirements, functional specifications, tasks, test cases, source codes and other relevant artifacts.
- Since communications among the development groups is more constrained, creating such system is a standard practice

Table: Traceability Matrix									
Proposal	Scope			Development		Quality Assurance		Implementation	Latest Status
Goal	URS	SRS	Functions	Code Set	Unit Test	Peer Review	Tests	End User	

- A traceability matrix helps the team to visualize the relationships among various elements of the system and the requirements.
- It enables the study of the impact of proposed changes on the system when required.

Chapter 12:

Project Risk Management (7 Marks)

11.1 Understanding Risk

- A risk is a possible future event that may affect our project - positively (i.e. Opportunity) or negatively (threat).
- Risk is the possibility of loss or damage. It makes it harder to achieve the project's goals. Characteristics of Risks: • Loss • Uncertainty
- A Risk is an event that may occur or not, and if occurs generally causes loss.
- Risk is also joint function of **Risk = f(likelihood, impact)**

11.1.1 Types of Risk

- 1) **Project Risk** : occurs due to technical aspect of work or work product
- 2) **Process Risk** : occurs due to formation, processes of the project or methodologies used in the project
- 3) **Product Risk or Technical Risks**: affect the quality or performance or even implementation possibilities.
- 4) **Business Risks**: affect the organization developing or procuring the software.
- 5) **Known Risks**: easily uncover able risks.
- 6) **Predictable Risks**: extrapolated from past project experience.
- 7) **Unpredictable Risks**: May or may not occur and are extremely difficult to identify

11.1.2 Project Risk

- Project Risk is an uncertainty that can have a negative or positive effect on meeting project objectives
- Risks are possible in any project. It depends on the nature of risk and environment.
- As the project progresses, risk and its potentiality do change
- Risk evaluation should be done for events and necessary preparation have to be made
- Customer's tolerance to risk must be considered
- Risks do have consequences
- Risk Planning must be done for all kinds of risks - good or bad

Common Risks in IT Projects

- Market risk
- Financial risk
- Technology risk
- People risk
- Structure/process risk

11.1.3 Risk Triangle

- Risk Triangle represents impact of project's risk and issues on project schedule



11.2 Risk Management

Risk management is the process involved with identifying, analyzing, and responding to risk. It includes maximizing the results of positive risks and minimizing the consequences of negative events

11.2.1 Terms

- 1) **Risk Tolerance** - The amount of acceptable risk
- 2) **Risk Averse** - Someone that does not want to take risks
- 3) **Risk Factors**
 - a) Probability of occurrence
 - b) Range of possible outcomes
 - c) Expected timing of event
 - d) Anticipated frequency of risk events from that source

11.2.2 Need of Risk management

- Risk involves changes in mind, opinion, actions or place.
- Risk is the possibility of loss or damage. It hinders in meeting the project objectives
- Risk concerns future happenings.
- Risk contains choice and the uncertainty and that the choice itself entails.
- Hence, risk is certain (like death and taxes). So, we need to Manage Risks.

Risk management is often overlooked in projects. Many project problems can be reduced by using risk analysis. Risk Management procedures help in finding and avoiding problems, but also helps software project managers prevent surprises, improve negotiations, meet customer requirements, and reduce schedule slips and cost overruns.

11.2.3 Risk Audit:

- Risk Audit is the process of evaluating and documenting how well risk responses address "recognized hazards and their core causes."
- It also examines and documents the effectiveness of the overall risk management process.

11.3 Risk Management Planning Process

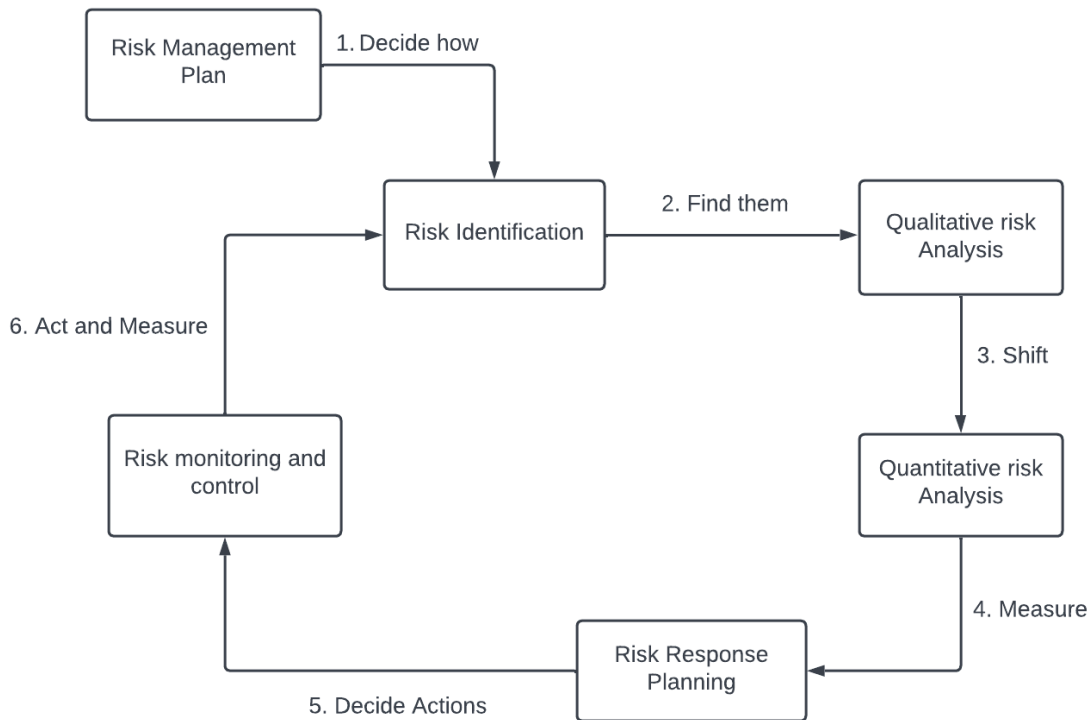
It is the process of deciding how to plan the risk management activities for a project. It describes "How do I plan for risks in my project?". It is a part of "Project Planning" phase.

It is all about

1) Planning

- Planning risk management
- Identifying risks
- Performing qualitative and quantitative risk analysis
- Planning & Implementing risk responses

2) Controlling and Monitoring : Controlling risk



11.3.1 Risk Management Plan

- is the description of the procedure that will be used to manage risk throughout the project
- It should cover who is responsible for managing various areas of risk.
- It is the role and responsibilities of Different Stakeholders
- Work done
 - Budgeting for Risk Management
 - Timing/Scheduling for Risk Management Activities •
 - Risk Categories
 - Determining Risk Probabilities and Impact
 - Probability and Impact Matrix
 - Revised stakeholders' tolerances
 - Reporting formats
 - Tracking of related tasks and issues

11.3.2 Risk Identification

- It is part of the Project Planning phase.
- Identifying risks is the process of understanding what potential events might hurt or enhance a particular project
- The main output of the risk identification process is a list of identified risks and other information needed to begin creating a risk register.

Risk Register

- A risk register is a tool in risk management and project management. It is used to identify potential risks in a project or an organization,
- Contains the outcomes of the risk management processes as they are conducted.
- Risk register contains:
 - List of identified risks
 - List of potential responses
 - Root causes of risk
 - Updated risk categories
- Risk database is the electronic form of risk register.
- Checklist could be used for the identification process.

Risk Register includes:

- a) **Risk identification ID:** Risk name or ID
- b) **Risk description:** A brief explanation of the risk.
- c) **Risk categories:** There are many risk categories that can impact a project such as schedule, budget, technical and external risks.
- d) **Risk analysis:** The purpose of risk analysis is to determine the probability and impact of a risk. You can either do a qualitative risk analysis or a quantitative risk analysis.
- e) **Risk probability:** You'll need to estimate the likelihood of each risk and assign a qualitative or quantitative value.
- f) **Risk priority:** The risk priority is determined by assigning a risk score to each risk, which is obtained by multiplying the risk impact and probability values. If you're using qualitative measurements, you'll need to prioritize risks with the highest impact and highest probability.

- g) **Risk response:** Each risk needs a risk response to mitigate its effect on your project. Those risk responses are also documented in a risk response plan.
- h) **Risk Ownership:** Each risk needs to be assigned to a team member who becomes a risk owner. The risk owner is responsible for deploying the appropriate response and supervising it.

Project Name	Created By	Date Created
Risk Register for an IS Project	Prabin Gautam	4/24/2079

Example of risk register : -

Risk ID	Risk Name	Risk Description	Impact Level	Probability	Priority Label	Status	Mitigation	Owner
Risk 1	Product Delivery	Failure of Supply Fail to deliver product in time	High	60%	1	Ongoing	Consider all plans and do work effectively	Prabin Gautam
Risk 2								

Risk identification tools and techniques

1) Brainstorming

- Brainstorming is a technique by which a group attempts to generate ideas or find a solution for a specific problem by collecting ideas without thinking and judgment
- don't overuse - can lead to groupthink and fewer ideas

2) The Delphi Technique

- lot similar to brainstorming
- People participated in this technique don't have to be located in the same place and can participate anonymously
- We can use email to facilitate this technique easily.

- Delphi Technique is used to make decisions about complex issues based on individual opinions
- In this method, a group of experts writes down and shares their thoughts about a problem with a facilitator.
- Each expert's view is compiled into a summary report by the facilitator.

3) Interviewing

- Interviewing is a fact-finding technique for collecting information in face-to-face, phone, e-mail, or instant messaging discussions
- Interviewing people with similar project experience is an important tool for identifying potential risks
- Very useful techniques to collect risk

4) SWOT analysis:

- SWOT analysis is the process of examining the project from each of the characteristic's point of view.
- SWOT helps to identify the broad negative and positive risks that apply to a project.
- SWOT analysis can also be used during risk identification.
- Strengths, Weaknesses, Opportunities and Threats

Example of SWOT Analysis

For example, a technology project may identify SWOT as:

- 1) **Strengths:** The technology to be installed in the project has been installed by other large companies in our industry.
- 2) **Weaknesses :** We have never installed this technology before.
- 3) **Opportunities :** The new technology will allow us to reduce our cycle time for time-to-market on new products. Opportunities are things, conditions, or events that allow an organization to differentiate itself from competitors and improve its standing in the marketplace.
- 4) **Threats:** The time to complete the training and simulation may overlap with product updates, new versions, and external changes to our technology portfolio.

5) Nominal Group Technique

- Another technique similar to Delphi technique
- Participants are grouped in the same room and given pen and paper.
- They are asked to write down what risks they think the project faces.
- The papers are given to facilitators and stick to the wall or white board.
- The risks are then reviewed and prioritized.
- At the end, we have a complete list of risks.

11.3.3 Risk Analysis

- Risk Analysis is the process of analyzing the identified risks, their probability of occurrence and their impacts.
- Results are stored in an updated risk register containing the priority of risk to be dealt with.
- Two methods of Analysis: Qualitative and Quantitative Risk Analysis

11.3.3.1 Qualitative Risk Analysis Process

- A part of the planning process group
- Qualitative Risk Analysis is the process of evaluating the impact and possibility of identified risks.
- It prioritizes risks according to their potential effect on the project.
- Risks are analyzed on qualitative factors only.
- Risk quantification tools and techniques include:
 - Probability/impact matrixes
 - The Top Ten Risk Item Tracking
 - Expert judgment

A **watch list** is a list of risks that are low priority, but are still identified as potential risks

Probability Impact Matrix

- It is a tabular representation of risk level based on possibility of occurrence and potential harm to project normal states

Probability\Impact	Very Low	Low	Moderate	High	Very High
Very High	Medium	Medium	High	High	High
High	Low	Medium	Medium	High	High
Moderate	Low	Medium	Medium	Medium	High
Low	Low	Low	Low	Medium	High
Very Low	Low	Low	Low	Medium	High

11.3.3.2 Quantitative Risk Analysis Process

- A part of the planning process group
- Quantitative Risk Analysis is the process of determining numerically the probability of each risk and its consequences of project objectives.
- Identifying risk requires the most attention.
- This process quantifies risk exposure and determines the size of cost and schedule contingency reserves that may be needed

Calculating Risk Exposure

- Determine probability of occurrence for each risk component
- Determine the impact for each component based on the criteria shown from the table
- Prepare a risk table and analyze the result i.e. overall risk exposure

$RE = P * C$: where, P is probability and C is cost to the project should the risk occur

Differences between qualitative and Quantitative Risk Analysis

SN	Qualitative Risk Analysis	Quantitative Risk Analysis
1	Qualitative Risk Analysis is the process of evaluating the impact and possibility of identified risks.	Quantitative Risk Analysis is the process of determining numerically the probability of each risk and its consequences of project objectives.
2	It prioritizes risks according to their potential effect on the project.	Deals only with the risks marked for further analysis by qualitative risk analysis
3	Complex as it does not involve straightforward math	Direct calculating methods and tools are available making the process simple
4	Time-consuming to identify each risk, record and rank them	Less Time Consuming as tools are used
5	Can be used easily as there is no need for any tool	Reliability on tool sometimes makes it difficult for the team to use it
6	Used in all projects irrespective of the complexity of the project	Used only in complex projects
7	Risk is ranked between 0 to 1	Risk closer to 1 ranking are taken first and calculation is done to predict the project outcome based on the effects of risk

Example: Find RE, if Risk Chance = 70%, Risk probability = 80% (assume), Risk Impact = 60 components were there i.e. 70% of 60 = 42, so, 18 components to be built.

Solutions:

If components = 100 LOC and LOC cost = Rs 100, then

Overall cost = $18 \times 100 \times 100 = \text{Rs } 180,000$

Now, Risk Exposure (RE) = 80% of 180000 = Rs 144,000

Modeling Techniques

1) Sensitivity Analysis

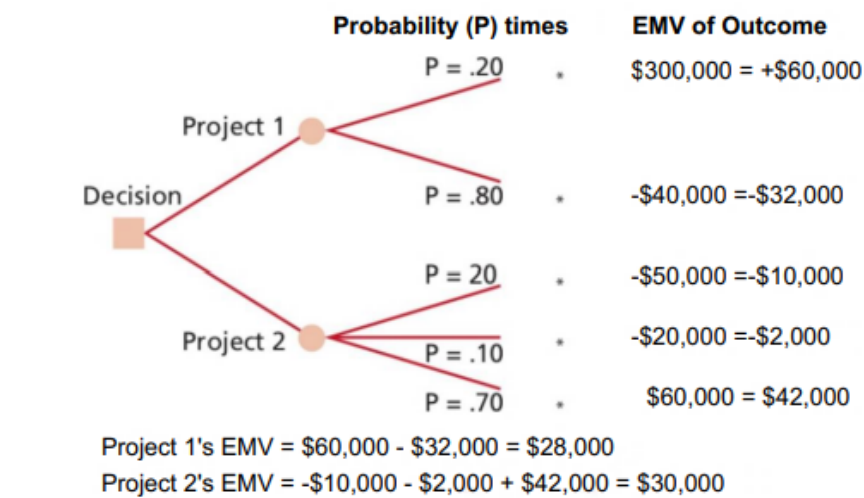
- Sensitivity analysis is a technique used to show the effects of changing one or more variables on an outcome

2) Expected Monetary Value

- Expected Monetary Value (EMV) is a project management metric used in risk analysis for determining the overall resource plan required for a project plan.
- When you make a plan, it can go better or worse than you expected.
- Eg: Decision Tree and Risk Exposure methods

Decision Tree

- A decision tree, also known as a decision tree analysis, is a diagram that will help to identify outcomes due to a collection of related choices.
- They are used for EVM study.



3) Monte Carlo Approach

- It is a mathematical technique, which is used to estimate the possible outcomes of an uncertain event.
- It includes Simulation and Modeling.
 - Simulation uses a representation or model of a system to analyze the expected behavior or performance of the system
- Steps of a Monte Carlo Analysis
 - 1) Evaluate the range for the variables being considered

- 2) Determine the probability distribution of each variable
- 3) For each variable, select a random value based on the probability distribution
- 4) Run a deterministic analysis or one pass through the model
- 5) Repeat steps 3 and 4 many times to obtain the probability distribution of the model's results.

11.3.4 Risk Response Planning

- part of **Planning** process group
- After identifying and quantifying risks, we must decide how to respond to them.
- It is the process of creating options to improve chances and cut down any obstacles to project goals, thus lowering total risk.
- Typical risk resolution actions are:
 - 1) Research and review
 - 2) Eliminate and review
 - 3) Reduce and protect
 - 4) Accept and protect

Strategies for Threats (Negative Risks)

- 1) **Risk avoidance**- eliminating a specific threat, usually by eliminating its causes.
- 2) **Risk acceptance**- accepting the consequences if a risk occurs
- 3) **Risk transference**- eg. shifting the consequences of a risk and responsibility for its management through outsourcing
- 4) **Risk mitigation**- reducing the impact of a risk event

Response Strategies for Positive Risks (Opportunities)

- 1) **Risk exploitation** - doing whatever we can to make sure the positive risk happens.
- 2) **Risk sharing** - allocating ownership of the risk to another party so that they can better realize the opportunity
- 3) **Risk enhancement** - changing the size of the opportunity by identifying, increasing probability and maximizing key drivers of the positive risk
- 4) **Risk acceptance** - the project team does not take any actions toward a risk
- 5) **Risk escalation** - notifying a higher level authority

Response Strategies for Both Positive and Negative Risks

1) Risk acceptance

- Allowing the consequences should a risk occur in a known and controlled manner
- Live with consequences

2) Contingent response strategies

- Develop contingency plan
- Create contingency allowance procedure
- Plan for alternative development

11.3.5 Risk monitoring and control process

- It is the process of keeping track of identified risks, monitoring residual risks, executing risk plans and evaluating the effectiveness of reducing risks.
- Monitoring risks involves knowing their status
- Controlling risks involves carrying out the risk management plan as risks occur

11.4 Terms

1) Residual Risks

- It is the risks that still remain after all of the response strategies have been implemented.
- E.g., even though a stable hardware product may have been used on a project, there may still be a risk that it fails to function properly.

2) Secondary Risks

- It is the direct result of implementing a risk response.
- E.g., using the more stable hardware may have caused a risk of peripheral devices failing to function properly.

11.5 Results of Good Project Risk Management

- Unlike crisis management, good project risk management often goes unnoticed
- Well-run projects appear to be almost effortless, but a lot of work goes into running a project well
- Project managers should strive to make their jobs look easy to reflect the results of well-run projects

Chapter 13:

Project Procurement Management (5 Marks)

13.1 Project Procurement

- The term "procurement" refers to buying goods and/or services from a third party.
- It's a group of processes required to purchase the products, services, goods or results needed from outside of the project team to perform the work.
- Most organizations do some form of outsourcing to meet their IT needs and spend most money within their own country
- Organizations or individuals who provide procurement services are referred to as suppliers, vendors, contractors, subcontractors, or sellers. Suppliers are the most widely used.

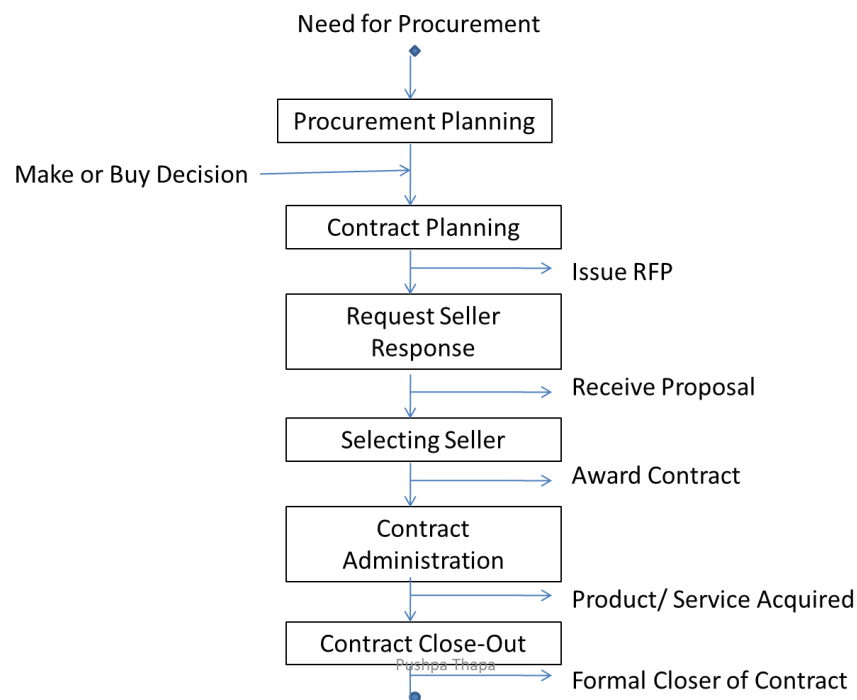
13.2 Project Procurement Management

- PPM includes the processes required to acquire goods and services for a project from outside the performing organization.
- Organizations can be either the buyer or the seller of products or services under a contract or other agreement.

Procurement Processes

- 1) **Planning purchases and acquisitions:** decide what to buy, when to buy it, and how
- 2) **Planning contracting:** Describe the requirements for the goods or services you're looking for, then find potential suppliers or dealers.
- 3) **Requesting seller responses:** obtain information, quotes, bids, offers, or proposals from sellers, as appropriate
- 4) **Selecting sellers:** choose the potential suppliers through a process of evaluating potential sellers and negotiate the contract
- 5) **Administering the contract:** manage the relationship with the selected seller
- 6) **Closing the contract:** complete and settle each contract, including resolving any open items

13.3 Process flow of Project Procurement Management



13.3.1 Plan Purchase and Acquisition Process

- It is the process that identifies the different project needs that could be purchased.
- This process involves considering what to acquire, how much to acquire, how to do so, and when to acquire. However, before doing so, we must decide whether or not to buy.
- It's a part of Project Planning.

Organizational Process Assets (OPA)

- OPA are valuable information, documents and knowledge tools that the organization accumulates over time.
- OPA are simply the lessons learned from the past.
- An organization gradually gains and develops more Organizational Process Assets as it gets involved in various projects.
- Includes Procurement (purchase) history of the organization

Enterprise Environmental Factors (EEF)

- Factors that bring the change in the business are known as internal and external environmental factors.
- Factors are : -
 - The conditions of the marketplace.
 - What products, services, goods or results are available in the market.
 - From whom could we get the goods?
 - Under what conditions.
- **Internal Factors**
 - Factors that occur inside the organization cause internal changes
- **External Factors**
 - factors that occur outside the organization but which can cause internal changes

Make or Buy Checklist

Reasons to Make

- Cheaper to Make
- Experience making it
- Idle production facility available
- Compatible and fits in production line
- Part is proprietary
- Not dependent on supplier
- High transportation costs

Reasons to Buy

- Cheaper to Buy
- No Production Facility
- Avoid fluctuating or seasonal demand
- Inexperience with making process
 - Available suppliers
- Maintain existing suppliers
- Higher reliability and quality

13.3.2 Plan Contracting Process

- The Procurement Management Plan describes how the procurement process will be managed from developing procurement documentation through contract closure.

Approach:

1) Request for Proposals:

- used to request bids from potential vendors
- A request for proposal (RFP) is a document that describes in detail specifically what product or service a customer wants to purchase and how bids will be evaluated.
- A vendor will create a proposal if there are several ways to satisfy the needs of the buyer.

2) Requests for Quotes: (bid)

- It is a method for requesting quotations or bids from potential suppliers
- The customer wants to compare prices between suppliers.
- It is a document created by sellers that lists prices for common items that the buyer has defined in detail.

Contract Statement of Work (SOW)

- A statement of work is a description of the work required for the procurement (purchase).
- If a SOW is used as part of a contract to describe only the work required for that particular contract, it is called a contract statement of work
- A SOW is a type of scope statement.
- A good SOW gives bidders a better understanding of the buyer's expectations.

Contract

- A contract is a mutually binding agreement that forces the seller to provide the specified products or services and forces the buyer to pay for them.
- Since contracts are legally binding, there is more responsibility to deliver the work as stated in the contract

Types of Contracts

1) Fixed Price (or lump sum)

- Contracts with a fixed total price for a clearly defined good or service are known as lump-sum or fixed-price contracts.
- Due to the fixed price in this scenario, the buyer faces less risk.
-

2) **Cost Reimbursable**

- Reimbursement means any money spent by the contractor will be given back during or after the project has been completed.
- A cost reimbursement contract is an agreement made between parties in the construction project that ensures the owner will pay the contractor's expenses while they are working on the project

Types

1) **Cost plus fixed fee (CPFF) contract:**

A contract in which the buyer pays the supplier for allowable costs plus a fixed fee payment that is usually based on a percentage of estimated costs

2) **Cost plus award fee (CPAF) contract**

A contract in which the buyer pays the supplier for allowable costs (as defined in the contract) plus an award fee based on the satisfaction of subjective performance criteria

3) **Cost plus percentage of costs (CPPC) contract:**

A contract in which the buyer pays the supplier for allowable costs (as defined in the contract) along with a predetermined percentage based on total costs

4) **Cost plus incentive fee (CPIF) contract:**

A contract in which the buyer pays the supplier for allowable costs (as defined in the contract) along with a predetermined fee and an incentive bonus

3) **Time and Material**

- Time and material (T&M) contracts are a hybrid of fixed-price and cost-reimbursable contracts

4) **Unit Price**

- Unit pricing can also be used in different kinds of contracts to define that the buyer must pay the supplier a specific sum for each unit of a good or service.
- The quantity required to accomplish the work determines the contract's overall cost.

Point of Total Assumption (PTA)

- The cost at which the contractor bears full responsibility for every extra dollar of contract costs is known as the Point of Total Assumption (PTA).
- Contractors have a financial motivation to limit cost overruns since they do not want to reach the point of total assumption.
- The PTA is calculated with the following formula:
$$PTA = (\text{ceiling price} - \text{target price}) / \text{government share} + \text{target cost}$$

Contract types versus risk

- Buyers have the least risk with organization fixed price contracts, because they know exactly what they must pay the supplier.
- Buyers have the most risk with cost plus percentage of costs (CPPC) contracts because they do not know the supplier's expenses in advance and the suppliers can be encouraged to keep raising prices.
- From the supplier's perspective, a CPPC contract carries the least risk and an organization fixed price contract carries the most risk.

Standard Forms

It includes standard contracts, standard descriptions of procurement(purchase) items, non-disclosure agreements, proposal evaluation criteria checklists, standardized versions of all parts of the needed bid documents.

Evaluation Criteria

Evaluation Criteria are the borderlines that are developed and used to rate or score proposals. Evaluation Criteria are often included as a part of procurement documents.

- Organizations prepare evaluation criteria for source selection, normally before publishing formal RFP.
- Organizations use criteria to rate/score proposals, and assign a weight to each criterion to indicate its importance.
- Example criteria and weights: the technical approach (30 percent weight), management approach (30 percent weight), past performance (20 percent weight), and price (20 percent weight).

13.3.3 Request Seller Responses Process

- The request seller responses process gathers information from potential sellers on how to meet project needs in the form of bids, quotes, and proposals.
- Organizations have various options for advertising to obtain goods and services.
 - Approaching the preferred vendor
 - Approaching several vendors
 - Advertising to anyone interested
- The buyer's expectations can be explained during a bidders' conference.
- Request Seller Responses process is a part of "Project Planning Phase"

Requesting Seller Responses Overview		
INPUTS	TOOL & TECHNIQUES	OUTPUT
1. Organizational Process Assets	1. Bidder Conference	1. Qualified Seller List
2. Procurement Management Plan	2. Advertising	2. Procurement Document
3. Procurement Documents	3. Develop Qualified Seller List	Package
		3. Proposals

Bidder Conferences

- These are the meetings with vendors.
- These gatherings take place before a bid or proposal is prepared.
- They are used to ensure that all prospective sellers have a clear common understanding of procurement.
- Its goals are to provide equal chances to all potential sellers, treat people with consideration and respect, understand what motivates them, and communicate carefully with them.

Advertising

- Advertising in newspapers or specialized media like professional journals can frequently increase the size of already-existing lists of potential vendors.
- While designing and producing an advertisement, advertisement writing style can be defined in order to achieve the following objectives:
 - Get attention, Gain interest, Create desire, and Get action

Source Selection Criteria

- It's important to develop some sort of evaluation standards, especially before releasing a formal RFP or RFQ.
- Avoid accepting ideas just because they appear good on paper; instead, carefully consider things like previous performance and management style.

Conducting Procurements

- Deciding whom to ask to do the work
- Sending appropriate documentation to potential sellers
- Obtaining proposals or bids
- Selecting a seller
- Awarding a contract

13.3.4 Seller Selection Process

- It is a part of the Project Execution Phase.
- Organizations often do an initial evaluation of all proposals and bids and then develop a short list of potential sellers for further evaluation
- Sellers on the shortlist often prepare a best and final offer (BAFO)
- Final output is a contract signed by the buyer and the selected seller.

Select Seller Process Overview					
INPUTS		TOOL & TECHNIQUES		OUTPUT	
1.	Organizational Process Assets	1.	Weighting System	1.	Selected Seller
2.	Procurement Management Plan	2.	Independent Estimates	2.	Contract
3.	Evaluation Criteria	3.	Screening System	3.	Contract Management Plan
4.	Procurement Document	4.	Contract Negotiation	4.	Resource Availability
	Package	5.	Seller Rating System	5.	Update Procurement
5.	Proposals	6.	Expert Judgment		Management Plan
6.	Qualified Seller List	7.	Proposal Evaluation Technique	6.	Requested Changes
7.	Project Management Plan				

Contract Negotiation

- It makes the contract's requirements and organizational structure clear so that both parties can agree on them before the contract is signed.

- **Five Stages of Contract Negotiation:**
 - 1) **Protocol**- Learn about one another.
 - 2) **Probing**- Everybody mentions their main priorities.
 - 3) **Scratch bargaining**- There is bargaining, and adjustments are made.
 - 4) **Closure**- Final agreements are made after summarizing the two positions.
 - 5) **Agreement**- The agreements are equally understood by both parties.
- Contract is awarded to each selected seller. It is the key output of the select seller process.
- Project contracts should be treated as “two-way” contracts.

13.3.5 Contract Administration Process

- It makes sure that the buyer agrees with the terms of the contract and that the sellers' performance satisfies contract agreements.
- The contract is managed by both the buyer and the seller for comparable reasons.
- Each party makes sure their rights are respected and that they fulfill their commitments.
- It's a part of the project controlling phase.

Buyer Conducted Sellers' Performance Review

- An organized evaluation of the seller's progress in providing the project scope and quality on time, within budget, and according to the contract is known as a procurement performance review.
- It includes an evaluation of the paperwork created by the seller, a buyer evaluation, and quality testing conducted while the seller was carrying out the service.
- This document suggests:
 - Completeness of contractual agreement
 - Overall performance or fulfill contacts requirements
 - Payment direction as per contract

Inspections and Audit

- It is possible to conduct inspections and audits during project execution to find any flaws in the seller's work processes or deliverables, as defined in the contract documents and requested by the buyer and supported by the seller.

13.3.6 Closing Procurements

- Part of Project Closing Phase
- The project team should:
 - Determine if all work was completed correctly and efficiently
 - Update records to reflect final results
 - Keeping records will help in the future.
- The formal approval and closing requirements should be specified in the contract itself.
- Early termination of a contract is a unique case of contract closure that may be brought about by the parties' agreement or by one of the parties' defaults.

Tools to Assist in Contract Closure

- Procurement audits identify lessons learned in the procurement process
- Negotiated agreements help to close contracts more smoothly
- A records management system provides the ability to easily organize, find, and archive procurement-related documents

13.4 Controlling Procurements

- Ensures that the seller performs according to the terms of the contract
- Since contracts are legal agreements, it is important that legal and contract experts participate in their formation and management.
- Project managers and team members must keep an eye out for positive change orders.

Suggestions for Change Control in Contracts

- Evaluation of any change should include an impact analysis.
- How will the change affect the scope, time, cost, and quality of the goods or services being provided?
- Changes must be documented in writing.
- Members of the project team should also keep records of all significant meetings and phone calls.

Procurement Audits

- The goal of a procurement audit is to increase productivity by regularly evaluating your contracts, processes, and history with vendors to guarantee accuracy and stick to the requirements of your contracts.

Closed Contracts

- The buyer officially informs the seller in writing that the contract has been fulfilled, usually through its approved contract administrator.
- Generally, formal contract closing requirements are specified in the contract's terms and would be covered by the contract management plan.

13.5 Using Software to Assist in Project Procurement Management

- Several types of software can help in project procurement management.
- E-procurement software helps organizations save money in procuring various goods and services.
 - Web-based ERP (Electronic Resource Planning)
 - E-MRO (Maintenance, Repair, and Overhaul)
 - E-sourcing (Identifying new suppliers)
 - E-tendering o etc.
- Organizations can also use the web, industry publications, and discussion groups to research and compare various suppliers

Chapter 14: Developing custom Processes for IT projects (5 Marks)

14.1 Developing IT Project Management Methodology

- Some organizations invest a lot of time and money in training project managers in general skills, yet even after the training, project managers might not be aware of how to adapt their methods to the organization's specific needs.
- Because of this problem, some organizations develop their own internal IT project management methodologies.
- The phases used are:
 - Initiation
 - Planning
 - Executing/Production
 - Monitoring and Control
 - Closing

14.1.1 PRINCE2

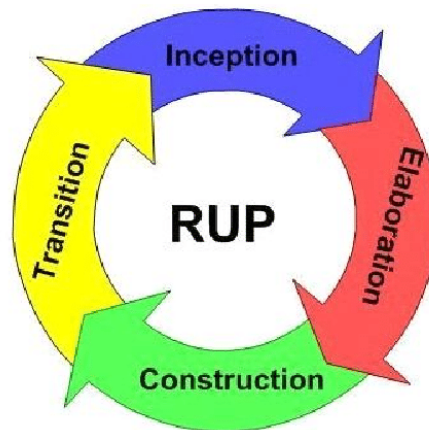
- Projects in Controlled Environments was originally developed for IT projects.
- PRINCE2 has eight process groups:
 - 1) Starting up a project
 - 2) Planning
 - 3) Initiating a project
 - 4) Directing a project
 - 5) Controlling a stage
 - 6) Managing product delivery
 - 7) Managing stage boundaries
 - 8) Closing a project

Some of the methods

- 1) Agile methods: (explained in chapter 4)
- 2) Rational Unified Process (RUP) Framework
- 3) Six Sigma methodologies

14.1.2 Rational Unified Process (RUP) Framework

RUP is an iterative software development process that focuses on team productivity and delivers software best practices to all team members. Rational Unified Process (RUP) is a software development process for object-oriented models. It is also known as the Unified Process Model.



Three reason behind using UP are:

- 1) The UP is an iterative process.
- 2) UP practices provide an example structure for how to do and thus how to explain OOA/D.
- 3) The UP is flexible, and can be applied in a lightweight and agile approach.

Inception

- Requirements are gathered.
- Feasibility study and scope of project is determined.
- Actors and their interactions are analyzed.

Elaboration

- Project plan is developed.
- Risk assessment is performed.
- Non-functional requirements are elaborated.
- Software architecture is described.
- Use case model is completed.

Construction

- All the components are developed and integrated.

- All features are tested.
- In each iteration, refactoring (clarifying and simplifying the design of existing code, without changing its behavior) is done.
- Stable product should be released.

Transition

- Software product is launched to the user.
- Deployment baseline should be complete.
- Final product should be released.

14.1.3 Six Sigma methodologies

- It's a methodology that is used to improve the output quality in a process.
- It does this by first identifying, and then removing, the causes of defects.
- This is achieved via a set of quality management methods that feature both empirical and statistical approaches.
- A staff member with Six Sigma expertise is also usually hired to monitor the process.
- Many organizations have projects that use Six Sigma methodologies.
- Two main methodologies are used on Six Sigma projects:
 - **DMAIC** (Define, Measure, Analyze, Improve, and Control) is used to improve an existing business process
 - **DMADV** (Define, Measure, Analyze, Design, and Verify) is used to create new product or process designs to achieve predictable, defect-free performance.

DMAIC methodology

1) Define :

- This step is used to figure out the business problem, what the goal is, the potential resources, the project scope and the timeline for completion.

2) Measure :

- The team will then make the determination on what will be measured and how.
- The better the data, the better this system performs.

3) Analyze :

- Next comes identifying what's preventing the progress of the process so it can be eliminated.
- Again, you'll use a data collection plan until you discover the root causes of the problem.

4) Improve :

- Test and implement solutions to the identified problems using creative solutions to eliminate root causes and prevent them from coming up again.
- This is done with various brainstorming techniques and other problem-solving methods.

5) Control :

- To ensure that these improvements are sustainable, monitor them by creating a control plan.
- Be sure to update that plan as needed.

DMADV methodology

1) Define : Define the process and design goals of your project.

2) Measure : Next, identify and measure the critical-to-quality characteristics of the product, service or process. Be sure you're not neglecting risk and production capabilities.

3) Analyze : Here's where you choose the best design after analyzing the data you collected in the first two steps.

4) Design : Once you have a design, start building and testing the product, service or process.

5) Verify : You need verification and validation to ensure your design output meets the design input requirements and that your product is performing as designed in a real-world environment

14.2 A customized SDLC Model for IT Projects

- To design and develop a “customized +tailored +tuned”, project management methodology a “multi-pass model”
- Multi-pass model need not be a full fledged development methodology, rather it encompasses both the “iterative & incremental model” and the agile model in a “merge and fit” manner.
- Key logical phases of multi-pass model are: •
 - Outline initial requirements, focus on mission –
 - Sort out major requirements, prioritize according to business need and develop an initial stable solution – Iterative & Incremental way.
 - Increase client’s involvement, develop more functional and relatively more sensible solutions – Agile way.
 - Finalize requirements, enhance design, develop complete solution, test rigorously – iterative & incremental way

14.3 IT Project Process Tailoring

Tailoring provides necessary adjustment of project activities to reflect the uniqueness of the project while keeping the project's goal in mind.

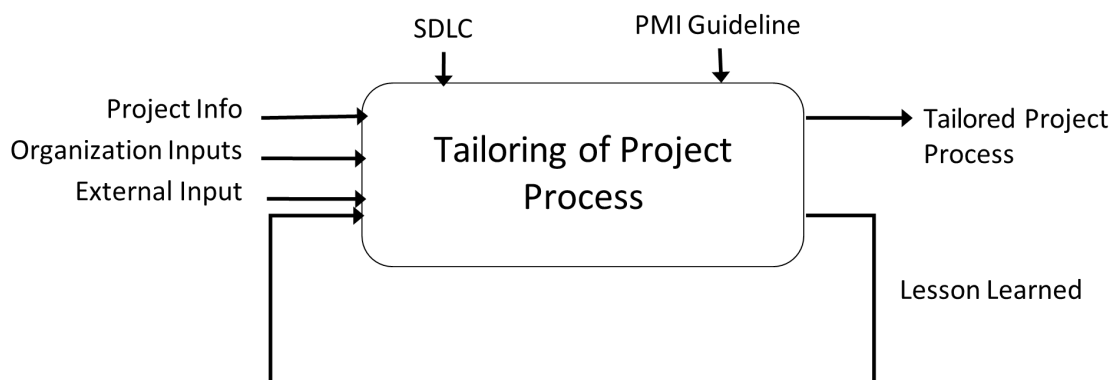


Figure: Tailoring of Project Process

14.4 Barriers to Implementation of Tailored Project Management

- 1) Lack of sustained top management commitment
- 2) Insufficient training on integrated project management
- 3) Lack of planning
- 4) Lack of resources
- 5) Lessons learned and good practices not shared across projects
- 6) Not invented here syndrome
- 7) Poor incentives / award criteria
- 8) Over-extended reviews

14.5 Moving forward with customized management processes

Each project is unique, it is important to understand that an organization's project management processes will likely need to be tailored in order to ensure project success.

Why Custom Processes for IT Projects?

- 1) Customer's role is dominant
- 2) Environment is more dynamic
- 3) Tasks are mental, unique, and complex
- 4) Society is more democratic and educated
- 5) Individuals have higher aspirations and expectations
- 6) Government's role is less clear, and its performance more closely scrutinize
- 7) Sources of knowledge are different (tacit knowledge, the practitioner)

Outsourcing and Offshoring

- **Outsourcing** occurs when a company contracts a specific process out to a third party, finding someone who specializes in whatever needs to be done.
- **Offshoring** occurs when businesses send in-house jobs overseas.
 - Offshoring sends jobs out of the country.
- Both saves organizational finance.

14.6 Preventive measures of office computer system

1) General Issues

- Working environment : Heat, Humidity, Air flow, etc.
- Conditioning power supply
- Keep apart from strong magnetic thing
- Maintain normal shutoff procedure

2) Daily Issues

- Automated virus scan
- Differential or incremental backup

3) Weekly Issues

- Clear temporary files and folders and recycle bin
- Update antivirus software
- Defrag hard disk
- Clean the printer

4) Monthly Issues

- clean machines, keyboard, mouse, external drive, etc

5) Yearly Issues

- clean the inside of the machine

14.6 Certified Associate in Project Management

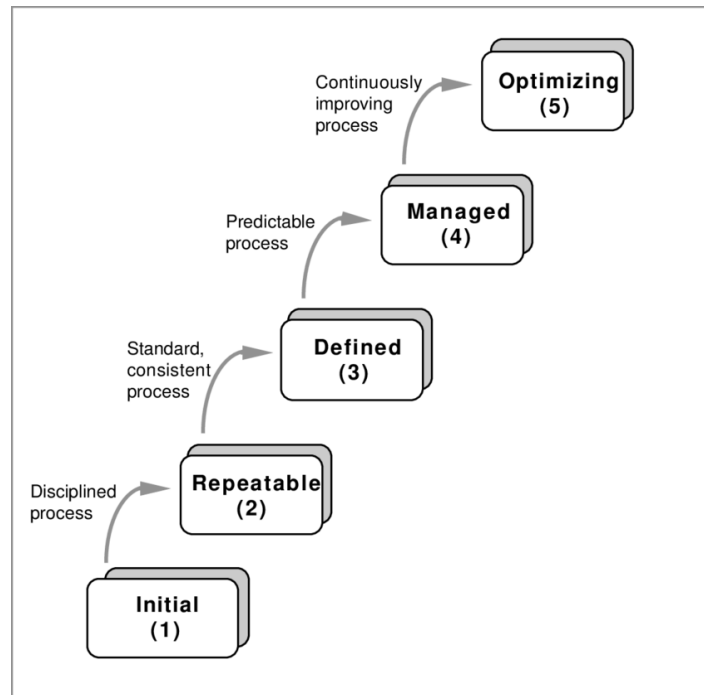
- PMI's Certified Associate in Project Management (CAPM) is an entry-level certification for project practitioners, designed for those with less experience.
- PMI provides certification as a Project Management Professional (PMP)
- A PMP has documented sufficient project experience, agreed to follow a code of ethics, and passed the PMP exam
- The number of people earning PMP certification is increasing
- The certification program includes: -
 - 1) Certified Associate in Project Management (CAPM) certification
 - 2) PMI Agile Certified Practitioner (PMI-ACP) certification
 - 3) PMI Risk Management Professional (PMI-RMP) certification
 - 4) PMI Scheduling Professional (PMI-SP) certification
 - 5) Portfolio Management Professional (PfMP) certification
 - 6) Program Management Professional (PgMP) certification

- 7) Project Management Professional (PMP) certification
- 8) PMI Professional in Business Analysis (PMI-PBA) certification

14.7 Project Management Maturity

- It is a formal tool used to measure an organizational project management maturity.
- It provides a path to take advancement of PM maturity once the initial level of maturity and areas for improvement are identified.
- It provides a logical path for progressive development.
- It follows software engineering institute's Capability Maturity Model's five evolutionary maturity levels and examines maturity development across ten knowledge areas.

Five Maturity Level



1) Initial

- At maturity level 1, processes are usually ad hoc and chaotic.
- The organization usually does not provide a stable environment.
- Maturity level 1 organizations often produce products and services that work; however, they frequently exceed the budget and schedule of their projects.

2) Managed (Repeatable)

- The projects of the organization have ensured that requirements are managed and that processes are planned, performed, measured, and controlled.
- At maturity level 2, requirements, processes, work products, and services are managed

3) Defined

- At maturity level 3, an organization has achieved all the specific and generic goals of the process areas assigned to maturity levels 2 and 3.
- At maturity level 3, processes are well characterized and understood, and are described in standards, procedures, tools, and methods.

4) Quantitatively Managed

- At maturity level 4, an organization has achieved all the specific goals of the process areas assigned to maturity levels 2, 3, and 4 and the generic goals assigned to maturity levels 2 and 3.
- At maturity level 4 Subprocesses are selected that significantly contribute to overall process performance.
- These selected subprocesses are controlled using statistical and other quantitative techniques.

5) Optimizing

- At maturity level 5, an organization has achieved all the specific goals of the process areas assigned to maturity levels 2, 3, 4, and 5 and the generic goals assigned to maturity levels 2 and 3.
- Processes are continually improved based on a quantitative understanding of the common causes of variation inherent in processes.

14.8 Future Trends

- Greater emphasis on leadership and communication skills.
- Talent retention
- Agile framework
- Increasing complexity of IT projects
- Remote project works

14.8 Code of Ethics

- Ethics is a set of principles that guides decision making based on personal values of what is considered right and wrong.
- Making ethical decisions is an important part of project managers' personal and professional lives
- Code of Ethics are:
 - Responsibility (towards community)
 - Respect (relationship between employers and clients)
 - Fairness (work performance)
 - Honesty (personal performance)

How to Create a Code of Ethics

- 1) Decide why you are writing your code of ethics. Is it to inspire your employees? Is it to spell out the expected behavior?
- 2) Begin with an introduction that explains the purpose of the code of ethics and what you hope to achieve by instituting such a code.
- 3) Add the items to your code of ethics. Cover the issues like interpersonal relationships, behavior expected around customers and clients and other items specific to your company/industry.
- 4) Decide how you will implement a code of ethics.

14.9 Promoting Project Excellency Through Awards and Assessment

- Excellent projects respect and appreciate their people.
- They recognize their potential and skills, as well as promote their development.
- They create a culture where they are enabled and motivated to apply their knowledge and abilities for the success of the project.

Dealing responsibly with people?

- The skills and abilities that are necessary to fulfill the project objectives are identified. and People are deployed accordingly.
- The dialogue with people takes place systematically and transparently.
- Legal and moral responsibilities for people, especially work safety, are carried out consciously.

Enablement

- The knowledge and abilities of people are promoted and further developed.
- In the process, personal interests, career development and self-development are supported.

Empowerment

- People work in an environment that allows them and motivates them to apply their skills, talents and creativity for the success of the project.
- They act in coordination, are involved and empowered towards taking initiative.

Chapter 15: Balanced Scorecard and ICT project

Management (2 Marks)

Balanced Scorecard (BSC)

- A balanced scorecard is a performance metric used to identify, improve, and control a business's various functions and resulting outcomes.
- is a strategy map for balancing today and tomorrow
- BSCs were originally developed for for-profit companies but were later adapted for use by nonprofits and government agencies.
- The balanced scorecard involves measuring four main aspects of a business:
 - Learning and growth,
 - business processes,
 - customers perspective, and
 - Financial Data.

1) Learning and Growth

- Learning and growth are analyzed through the investigation of training and knowledge resources.
- handles how well information is captured and how effectively employees use that information to convert it to a competitive advantage within the industry.

2) Business Processes

- Business processes are evaluated by investigating how well products are manufactured.
- Operational management is analyzed to track any gaps, delays, bottlenecks, shortages, or waste.

3) Customers Perspective

- Customer perspectives are collected to gauge customer satisfaction with the quality, price, and availability of products or services.
- Customers provide feedback about their satisfaction with current products.

4) Financial Data

- Financial data, such as sales, expenditures, and income are used to understand financial performance.
- These financial metrics may include dollar amounts, financial ratios, budget variances, or income targets.

Main Role of Balanced Scorecard:

The balanced scorecard directs you to develop metrics, set objectives, and gather and analyze data for each of the viewpoints.

As a result, scorecard provides an effective method for measuring and analyzing the implementation of a plan ensuring the better quality of the project.

3.8.1 Examples of a Balanced Scorecard (BSC)

Steps for BSC

- 1) Establish a clear vision of the future
- 2) Define strategic objectives
- 3) Determine the critical resources factors
- 4) Choose indicators to measure and monitor performance
- 5) Set goals, action plans and initiatives

Strategic Maps of an Online Shopping System Business

	Objectives	Goals	Indicators	Initiatives
Financial	↑ Sales ↓ Cost	↑ 15% Net Sales ↓ 10% operational Cost	Financial Statements	Negotiate with suppliers
Customers	To be a reference for variety of products	↑ Product launched every quarter by 15%	No of new products launched quartely	Create an innovation committee
Internal Process	To be able to develop new products constantly	Start min of 5 new dev projects every month	Project innovation report	Acquare specific software for managing products
Learning and Growth	To have knowledgeable staffs in product development	Have at least 2 professionals with master degree in product development	No of product development professionals with master degree	• Hire employees with master degree • Send employess to get master degree

Some Extra Questions

1) Cocomo Model

Cocomo (Constructive Cost Model) is a regression model based on LOC, i.e number of Lines of Code. COCOMO predicts the efforts and schedule of a software product based on the size of the software.

The necessary steps in this model are:

- 1) Get an initial estimate of the development effort from evaluation of thousands of delivered lines of source code (KDLOC).
- 2) Determine a set of 15 multiplying factors from various attributes of the project.
- 3) Calculate the effort estimate by multiplying the initial estimate with all the multiplying factors i.e., multiply the values in step1 and step2.

In COCOMO, projects are categorized into three types:

- 1) **Organic** : A Software project is said to be an organic type if the team size required is adequately small, the problem is well understood and has been solved in the past and also the team members have a nominal experience regarding the problem.
- 2) **Semi Detached** : Software project is said to be a Semi-detached type if the vital characteristics such as team size, experience, knowledge of the various programming environment lie in between that of organic and Embedded

3) Embedded

A software project requiring the highest level of complexity, creativity, and experience requirement fall under this category.

Such software requires a larger team size than the other two models and also the developers need to be sufficiently experienced and creative to develop such complex models.

- a) Basic COCOMO Model
- b) Intermediate COCOMO Model
- c) Detailed COCOMO Model

2) Tornado Analysis

Types of simulation modeling

There are different types of charts used in project management. The tornado diagram is a special bar chart that is used in sensitivity analysis.

The sensitivity analysis is a modeling technique that determines which risks have the most impact on the project. The tornado diagram is used to compare the importance (relative) of different variables.

- The tornado diagram is one of the methods used to display the sensitivity analysis.
- It is used to compare the relative importance and impact of variables with a high degree of uncertainty and stability.

This particular bar chart has a tornado-like form because the risk data categories are arranged vertically, from largest to smallest, on horizontal bars.

For this particular chart, the Y-axis contains uncertain variables at their base values while The correlation between known and uncertain results is shown on the X-axis.

Alternatives of tornado diagrams are decision trees.

